



Managing Risk Under Climate Extremes: How Hazards Propagate Through Natural, Engineered, and Financial Systems

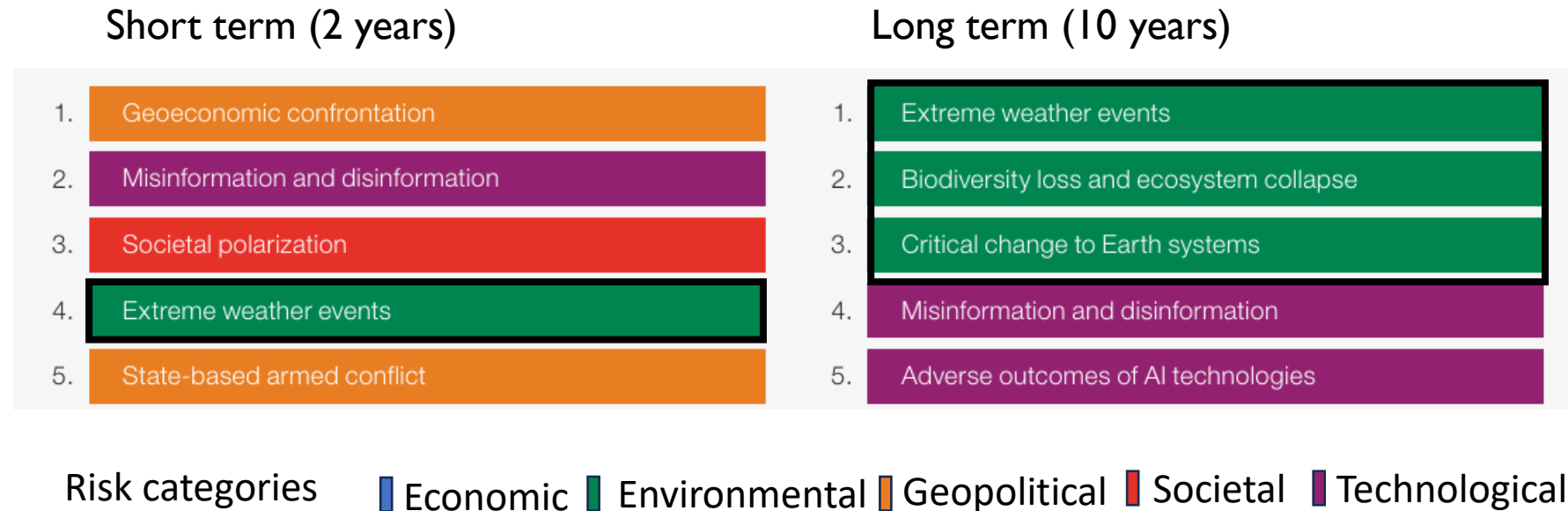
Dan Li, Ph.D.

Institute for Risk Management and Insurance Innovation
University of North Carolina at Chapel Hill

May 14, 2026

2026 Global Risks Ranking by Severity

as ranked by participants at the World Economic Forum in Davos



Climate-related risks rank among the **top global threats** over both short and long horizons.

Environmental Risk Is Now a Major Financial Concern



BUSINESS

Mind the adaptation gap: Despite rising climate costs, few companies have adaptation plans

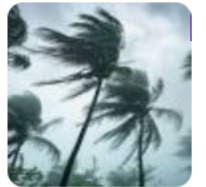
Sep 15, 2025



IB Insurance Business

Environmental risk tops insurance CEO concerns

Nine insurance CEOs from the US, Australia, Japan, the UK, Slovenia, Germany, Switzerland and France identify environmental risk as the...



KPMG

Banks— managing the financial risks of climate and environmental change

Regulatory expectations around climate-related financial risk continue to be a significant concern for banks. There are already challenging requirements...

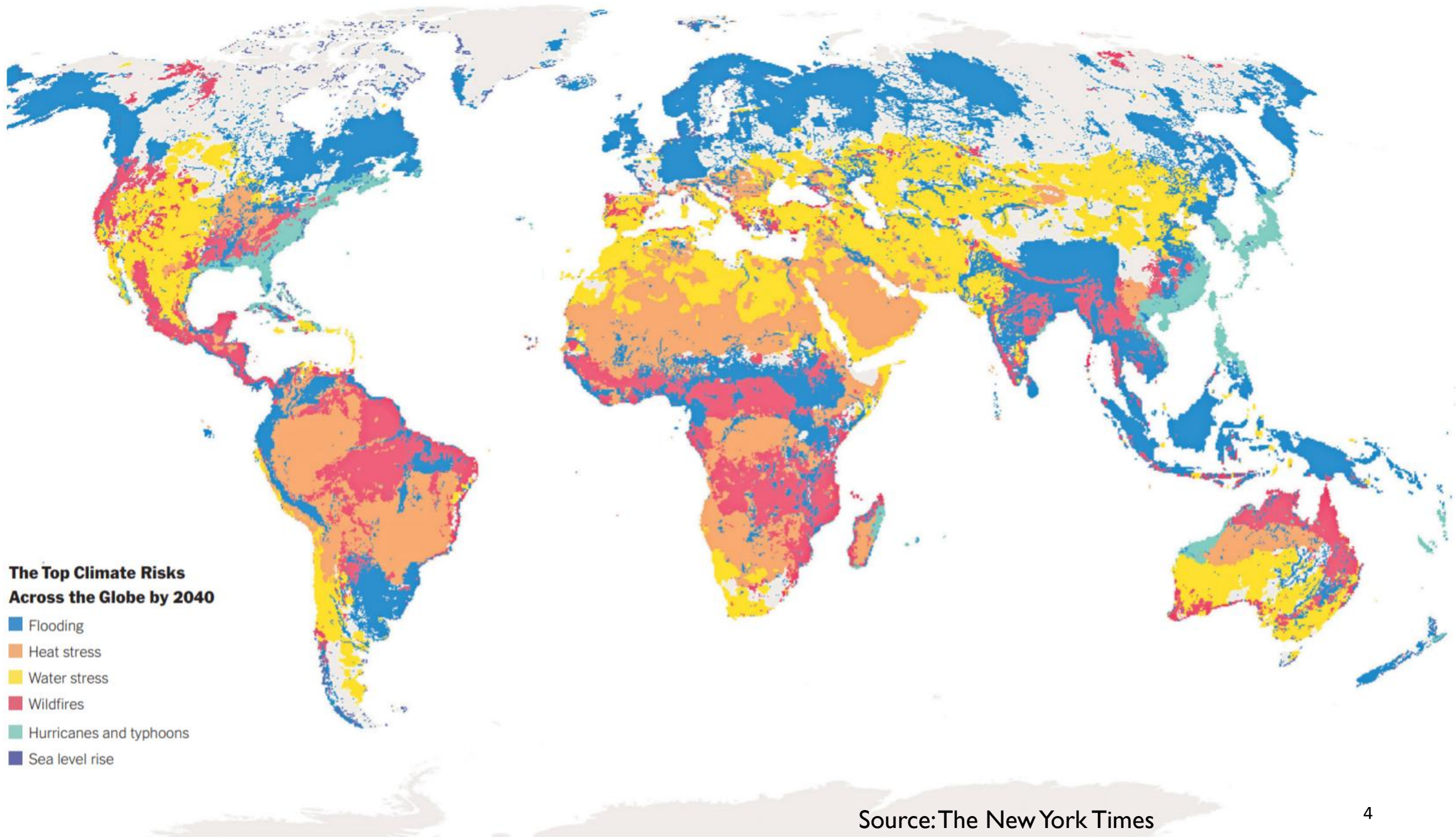
 World Bank

When green meets green: Environmental risk-pricing by banks

The “after the Paris Agreement” subsample shows that green banks offer green firms a discount of about 60–69 basis points relative to brown firms.

Mar 20, 2024





Source: The New York Times

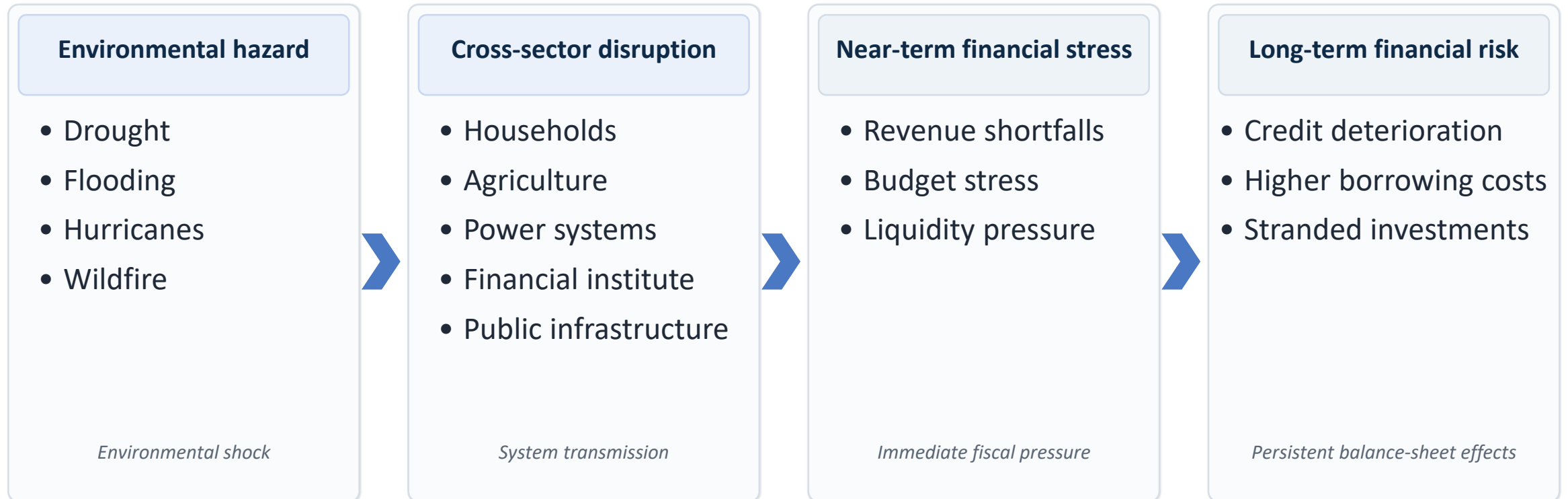
Environmental Risks:

What? Where? When? How?

**The Top Climate Risks
Across the Globe by 2040**

- Flooding
- Heat stress
- Water stress
- Wildfires
- Hurricanes and typhoons
- Sea level rise

How Environmental Risks Became Financial Impacts

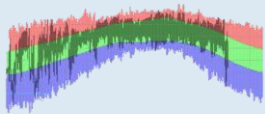


Research Roadmap

Linking Large-Scale Measures to System-Based Risk Mitigation
(Hazard → Exposure → Vulnerability)

Evidence

Measure risk: How often? How severe? Who is exposed?



Large-scale hazard characterization

- data-driven econometrics
- machine learning
- remote sensing

Scale: national / regional / multi-decadal

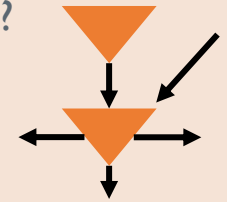
Mechanisms

- Who/what is exposed?
- How do hazards become operational and financial risk?



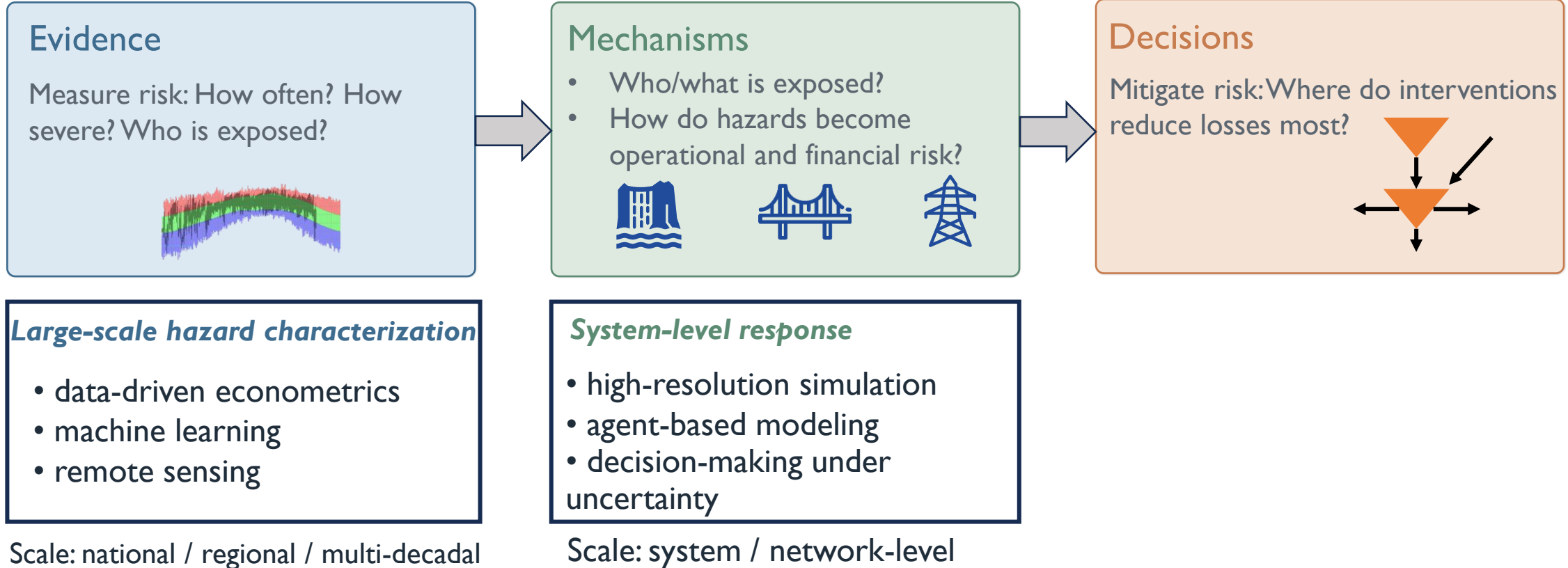
Decisions

Mitigate risk: Where do interventions reduce losses most?



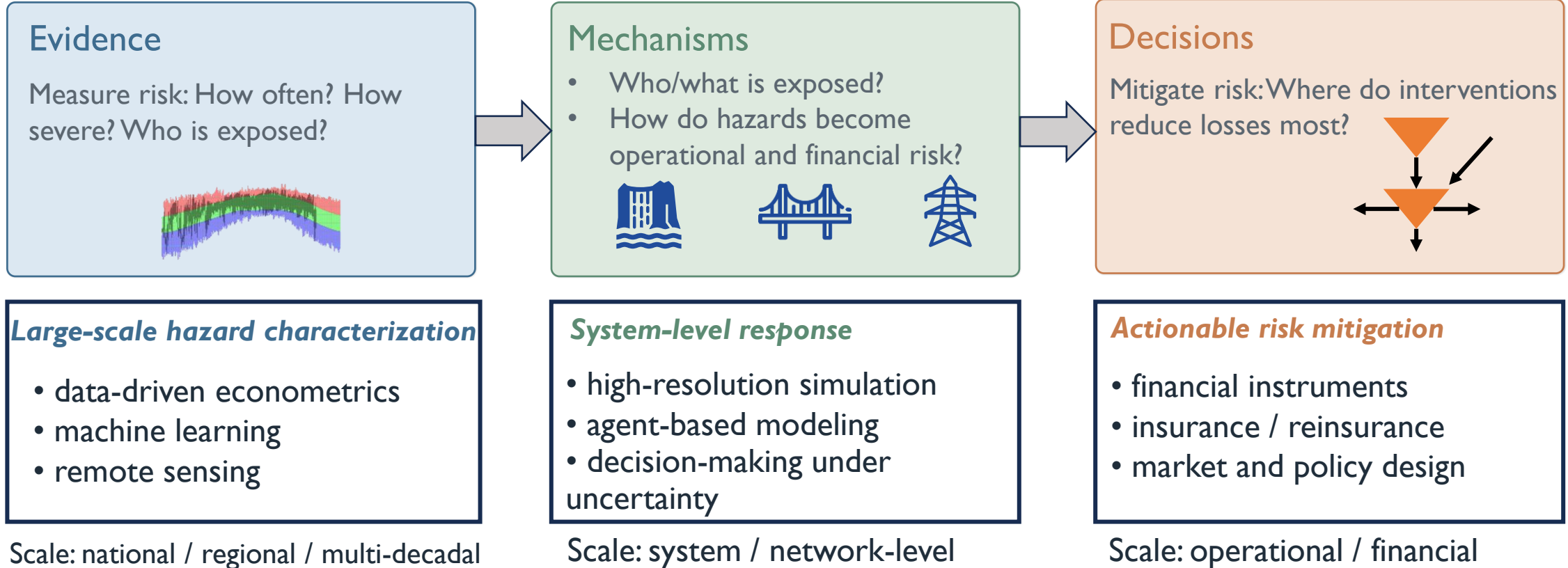
Research Roadmap

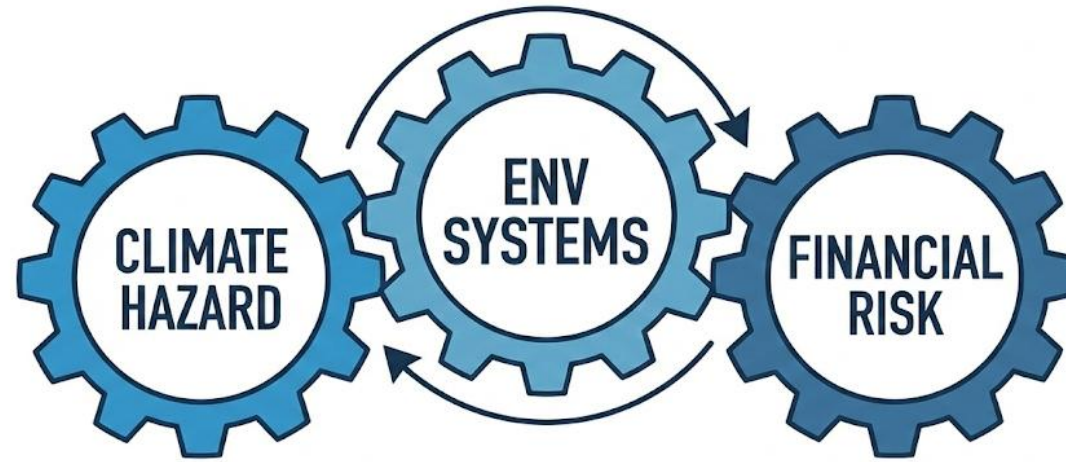
Linking Large-Scale Measures to System-Based Risk Mitigation
(Hazard → Exposure → Vulnerability)



Research Roadmap

Linking Large-Scale Measures to System-Based Risk Mitigation
(Hazard → Exposure → Vulnerability)





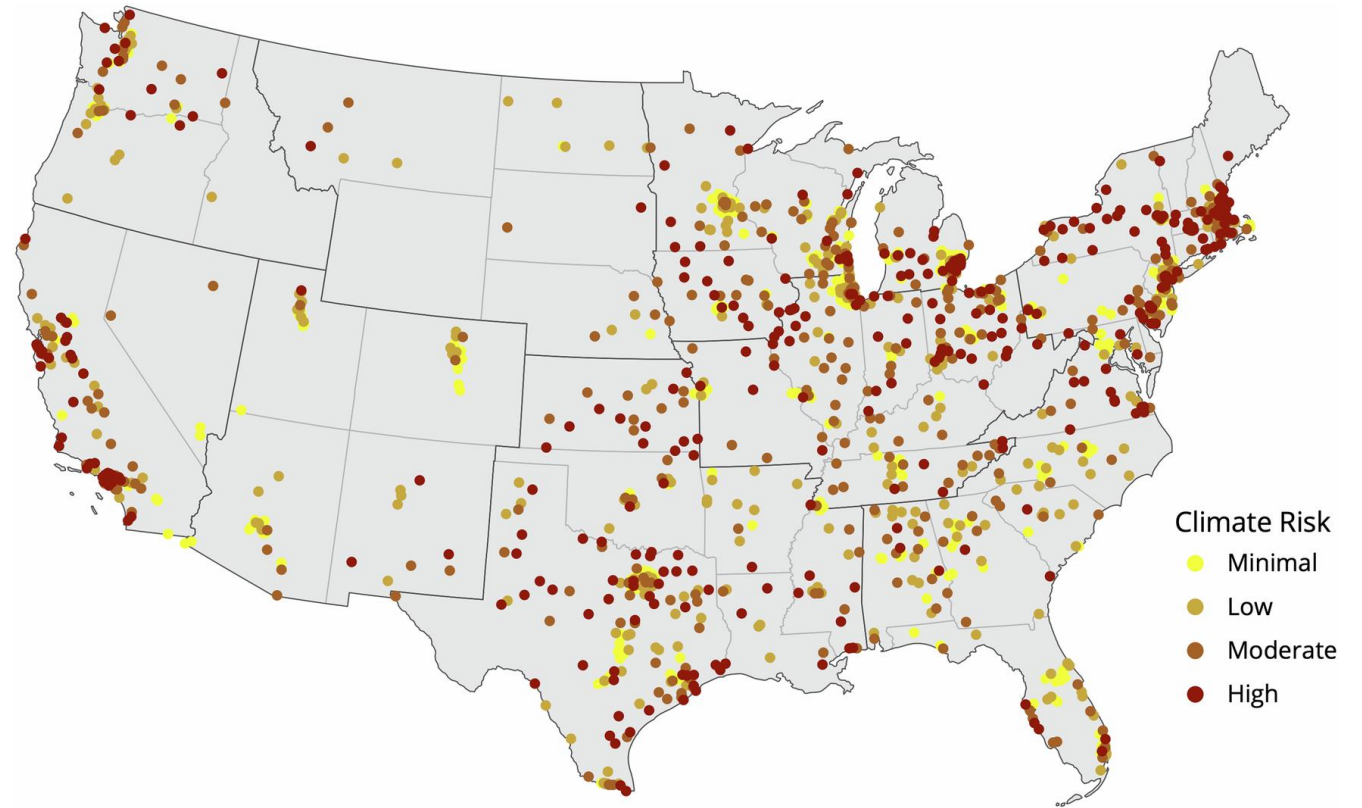
Scalable, cross-disciplinary framework

linking hazard signals to decision-making for proactive risk management.

Water Utility's Unique Drought Exposure

- 15,000 utilities, serving 90% of population
- Annual revenue: \$42.1 billion
- Annual investment needed: \$63 billion

Map of climate risk for US drinking water utilities.



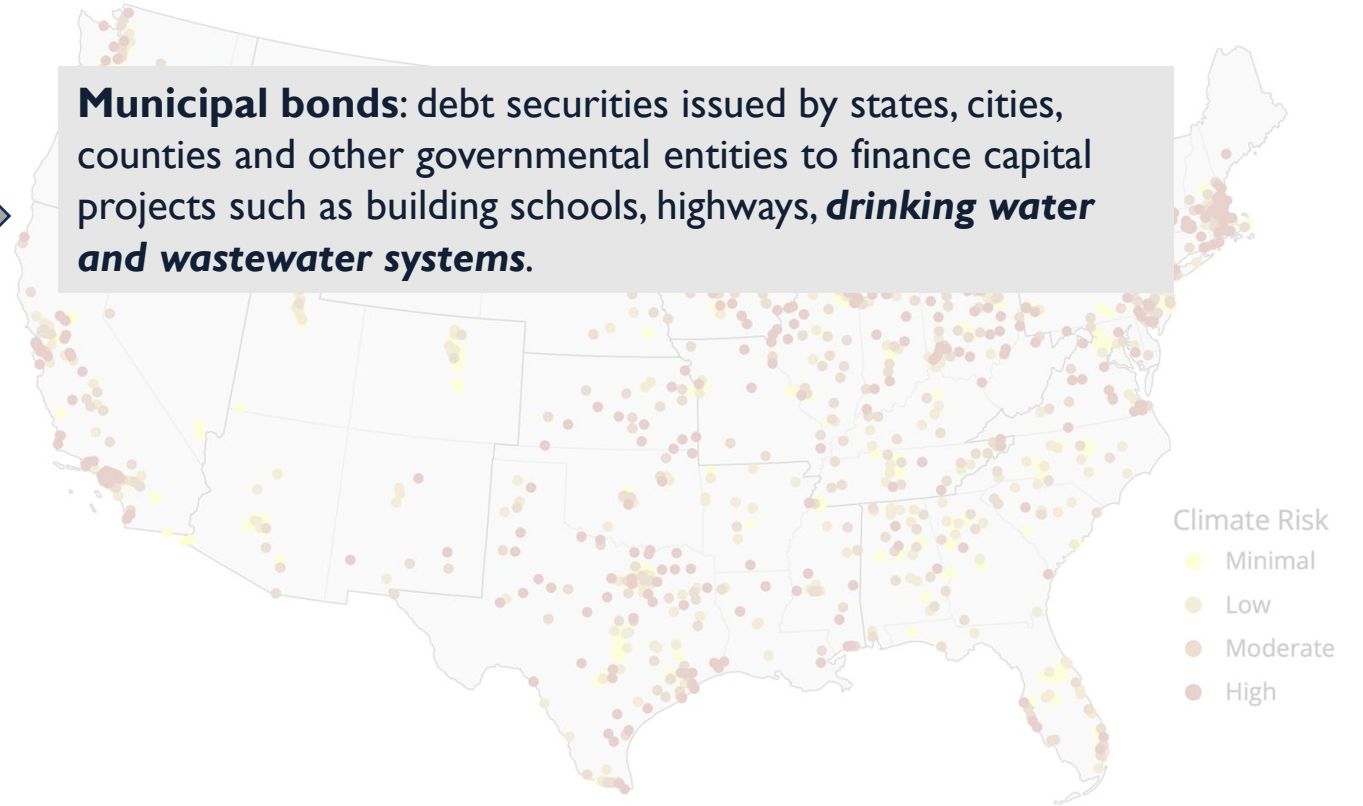
Water Utility's Unique Drought Exposure

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Funded through



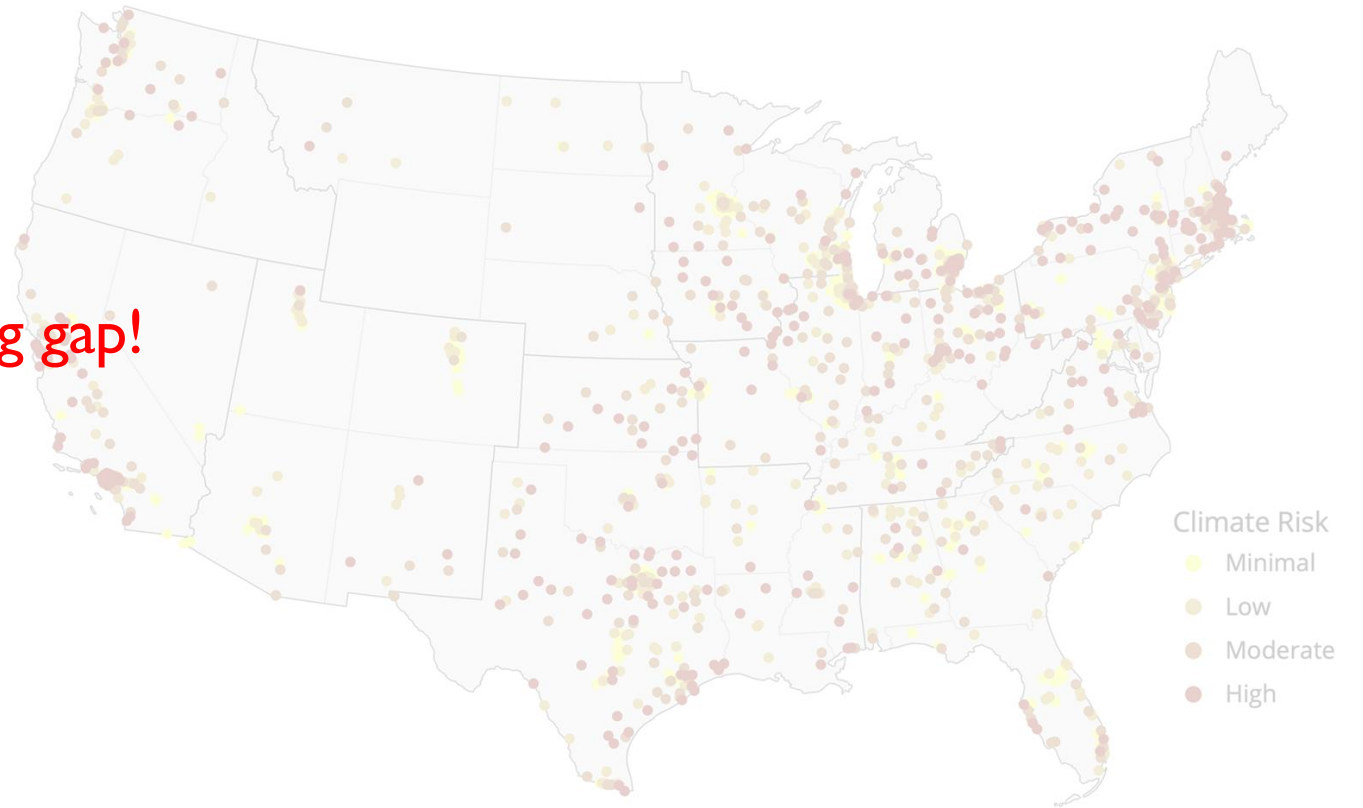
Municipal bonds: debt securities issued by states, cities, counties and other governmental entities to finance capital projects such as building schools, highways, ***drinking water and wastewater systems.***



Water Utility's Unique Drought Exposure

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- Annual revenue: \$42.1 billion
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Funding gap!

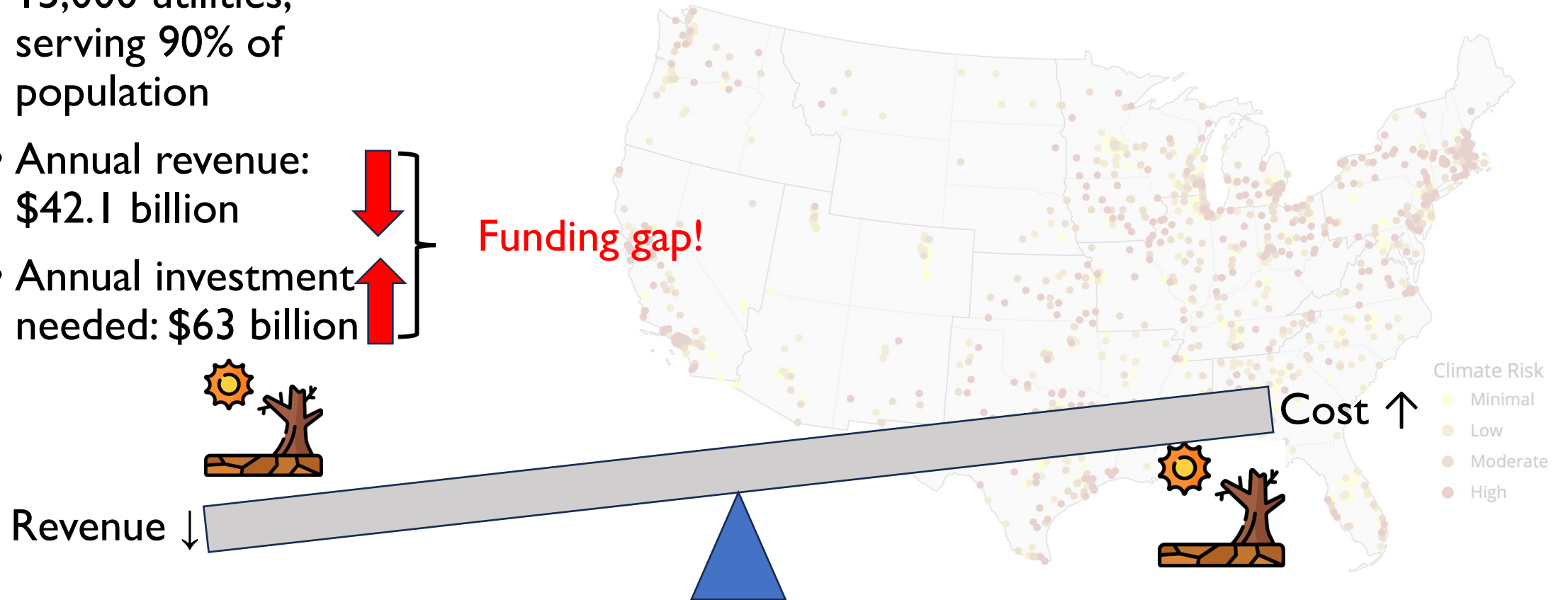


Climate Extremes Deepen the Funding Gap

- 15,000 utilities, serving 90% of population

- Annual revenue: \$42.1 billion

- Annual investment needed: \$63 billion



Climate shocks **disrupt water supply and demand**, then ripple into **operational and financial instability**.

Attention To These Risks Has Grown

S&P Global

29-Oct-2021 | 13:20 EDT

20 California Irrigation District Rating Outlooks Revised To Negative From Stable On Rising Drought Severity, Pressure

yahoo!news

Residents of this California community just voted to hike their own water bills by 300% — from \$145 to \$568

Cory Santos

Thu, July 3, 2025 at 4:20 PM EDT

4 min read



132

Fitch Revises Outlook on Corpus Christi, TX's Utility System Senior Lien Revs to Negative

Fri 31 Oct, 2025 - 5:04 PM ET

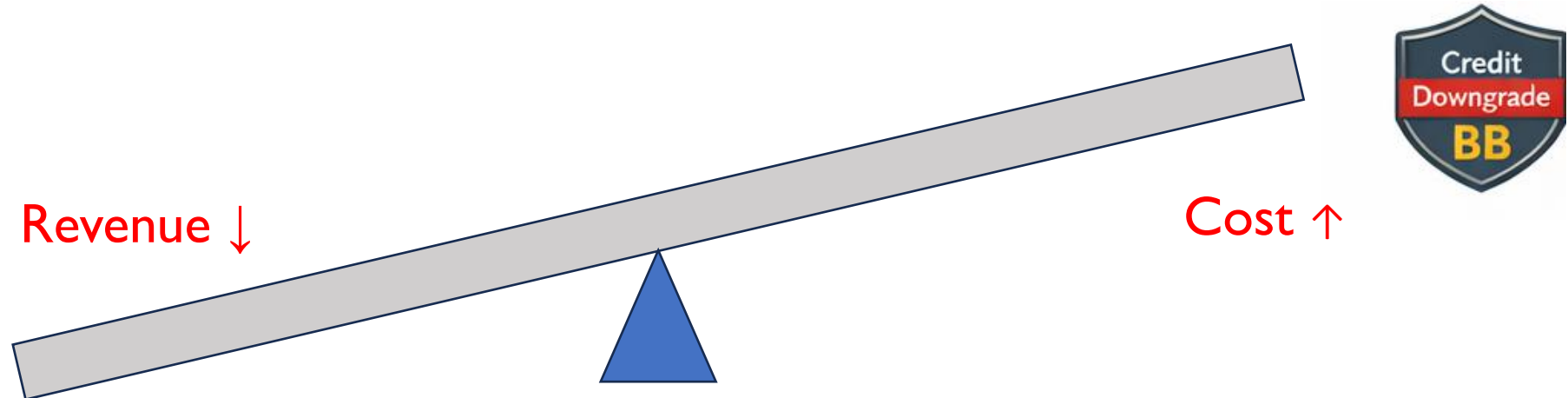
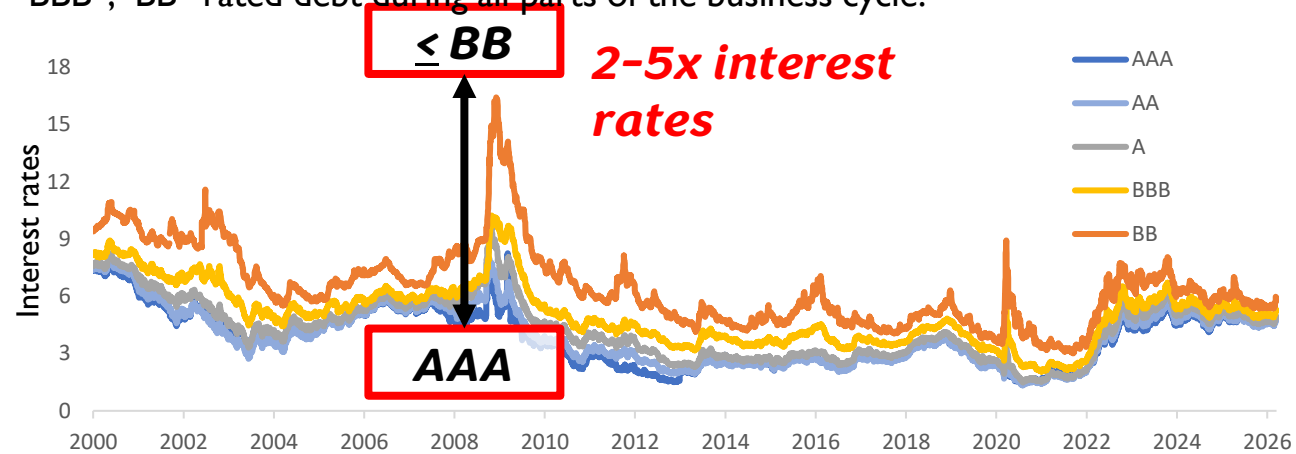
Fitch Revises Denton, TX's Utility Outlook to Negative; Affirms Rev Bonds at 'A'

Tue 31 Mar, 2026 - 12:38 PM ET

Damage Is Not Confined to the Physical Environment

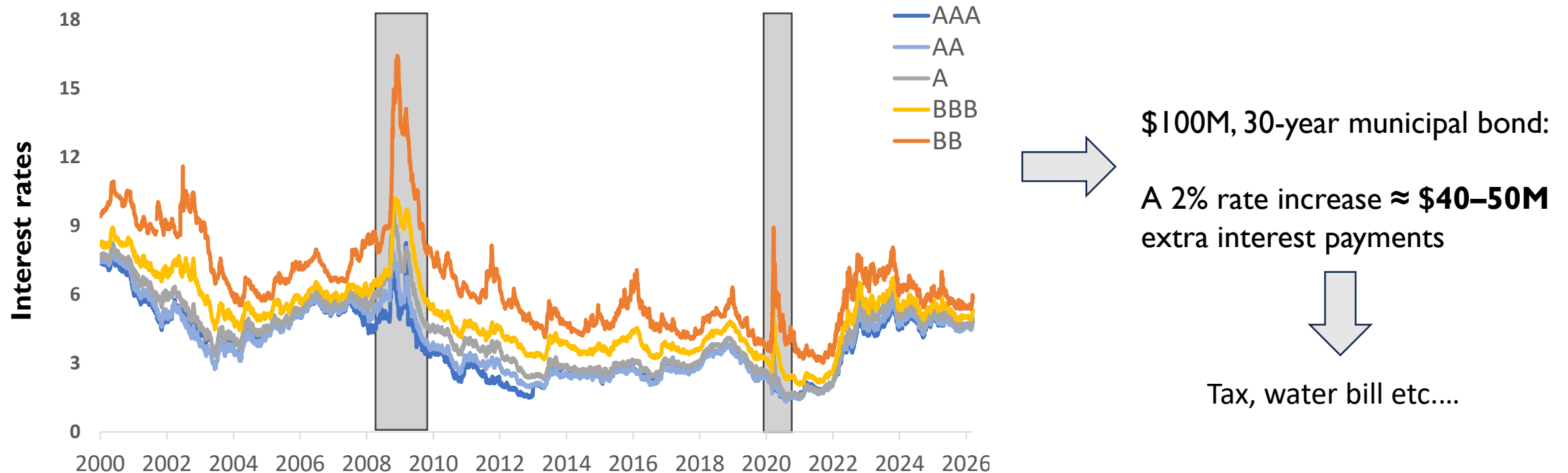
- **Revenues** ↓ as mandatory conservation reduces water use
- **Credit risk** ↑ raises borrowing and debt-service costs

“AAA”, “AA”, “A” rated long-term debt is noticeably more stable than “BBB”, “BB” rated debt during all parts of the business cycle.



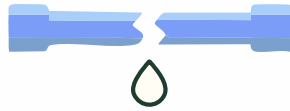
Short-Term Climate Shocks, Long-Term Cost Increase

Climate change does not stop during financial stress!



Infrastructure systems are financed over **30-50 year horizons**. Short climate shocks create long shadows.

Cascading Impacts of Extreme Weather



Slower
Infrastructure
Investment



Higher
Household
Bills



Burden on
Communities

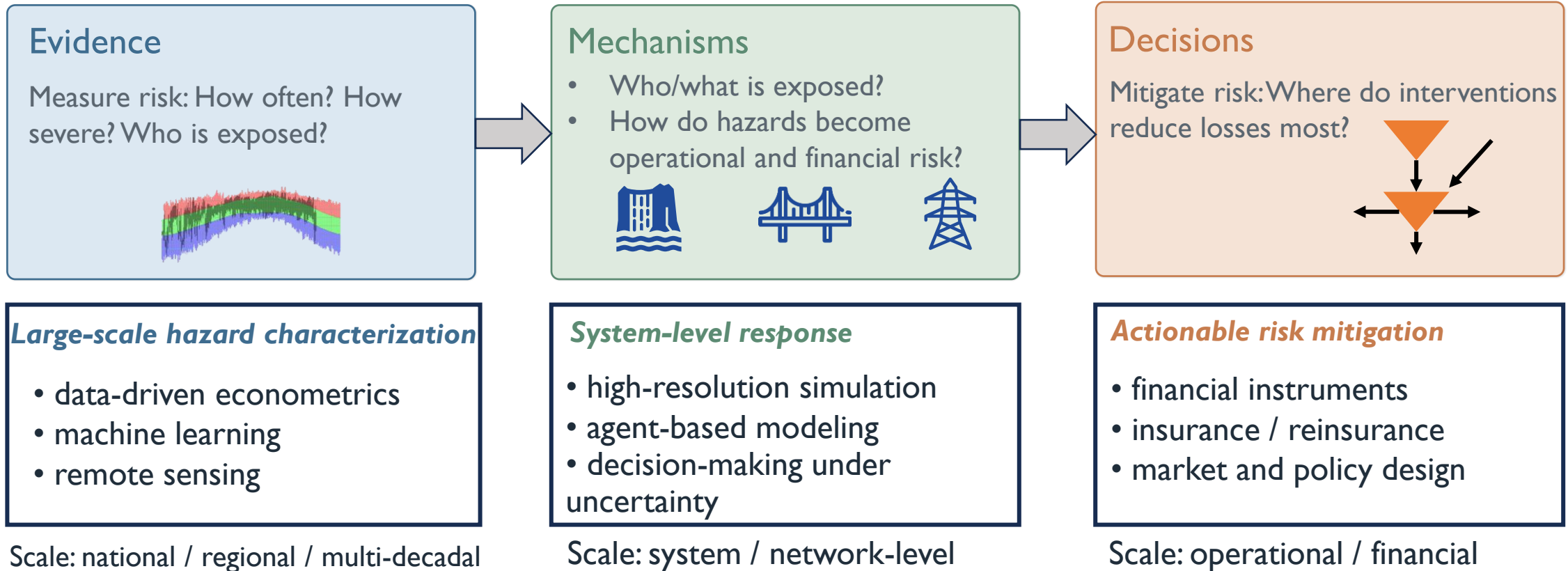
Higher borrowing cost → delayed infrastructure investment, higher water bills, and greater burdens on communities, making future adaptation increasingly difficult.

My Research Contribution:

Quantifying and *Managing* Climate-Driven
Financial Risk in Environmental Systems

Research Roadmap

Linking Large-Scale Measures to System-Based Risk Mitigation



Goal: Advancing risk management of climate extremes across natural, engineered, and financial systems.

1 Evidence

Where and how are hazards changing, and who or what is exposed?

2 Mechanisms

How do hazards propagate through coupled systems to become operational and financial risk?

3 Decisions

Which interventions, market tools, or policies reduce those risks most effectively?

Evidence

Hazard patterns and exposure

Mechanisms

Propagation through systems

Decisions

Mitigation and financial tools

National

Green bond issuance and carbon emissions

Li & Adriaens (2025) · ES&T

Water utility bond costs under flood and drought risk

Li, Saigal & Adriaens (2024) · ES&T

ESG impact on corporate bond pricing

Li & Adriaens (2023) · JME

State / local

ESG performance and disclosure for bond premium

Li & Adriaens (2024) · JSFI

Deep Learning Model for Wildfire Spread Prediction

Li, Zeff & Charackllis. In preparation

Parametric insurance for managing drought financial risk

Li, Zeff & Characklis · under review

Managing drought financial risk with water futures

Li, Zeff & Characklis · under review

Balancing supply and financial risk for water utilities

Petagna, Li, et al. (2026) · JWRPM

Evidence

Hazard patterns and exposure

Mechanisms

Propagation through systems

Decisions

Mitigation and financial tools

National

State / local

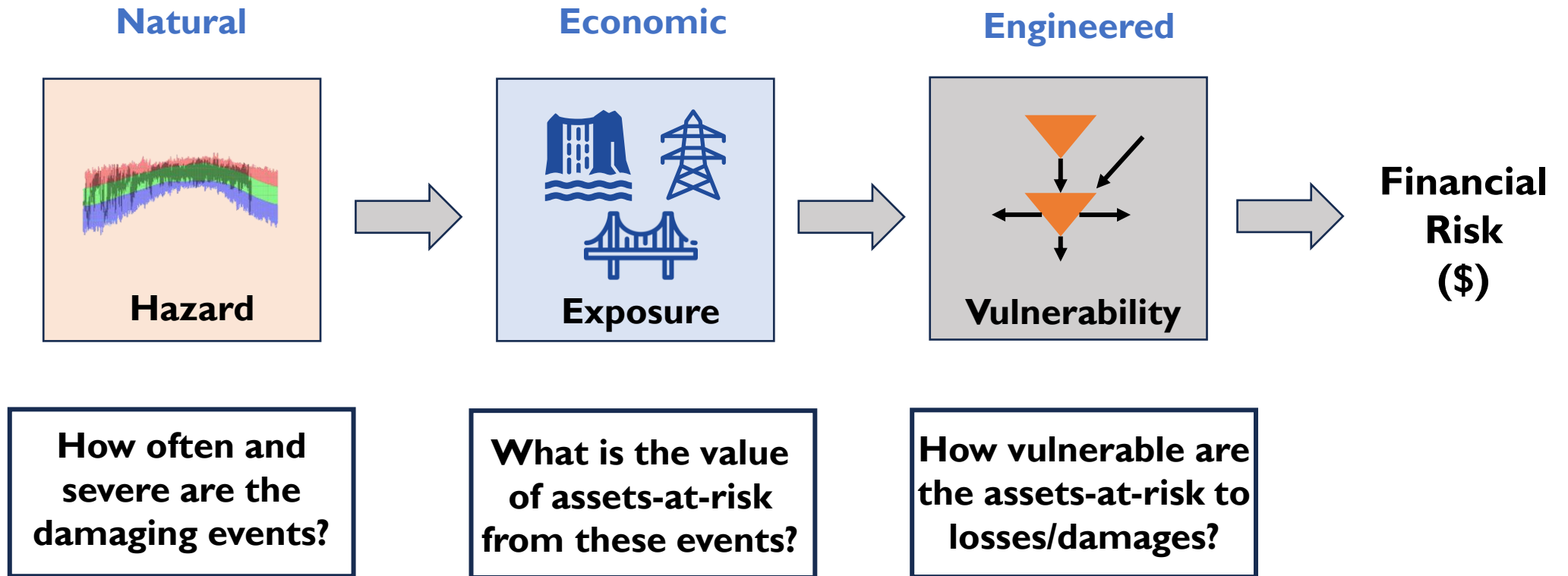
Managing drought financial risk with water futures

Li, Zeff & Characklis · under review

Balancing supply and financial risk for water utilities

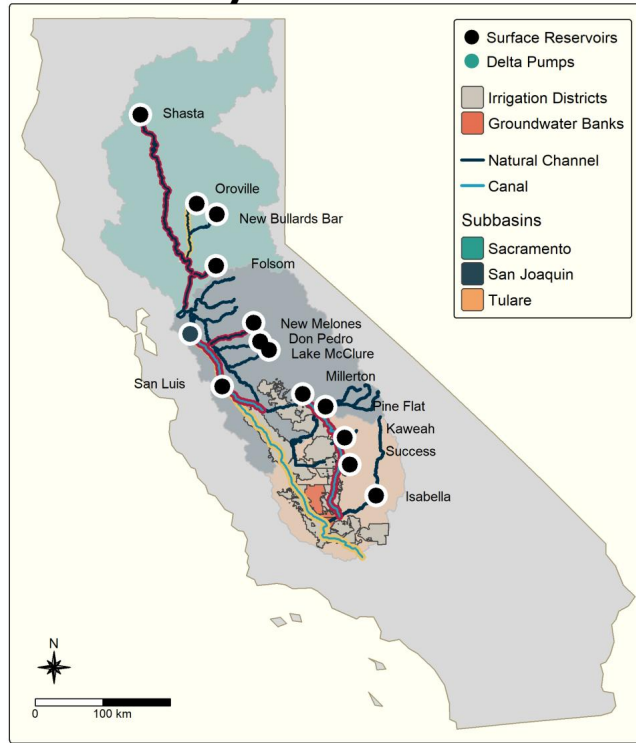
Petagna, Li, et al. (2026) · JWRPM

Financial Risk is a *Function* of Linked Systems

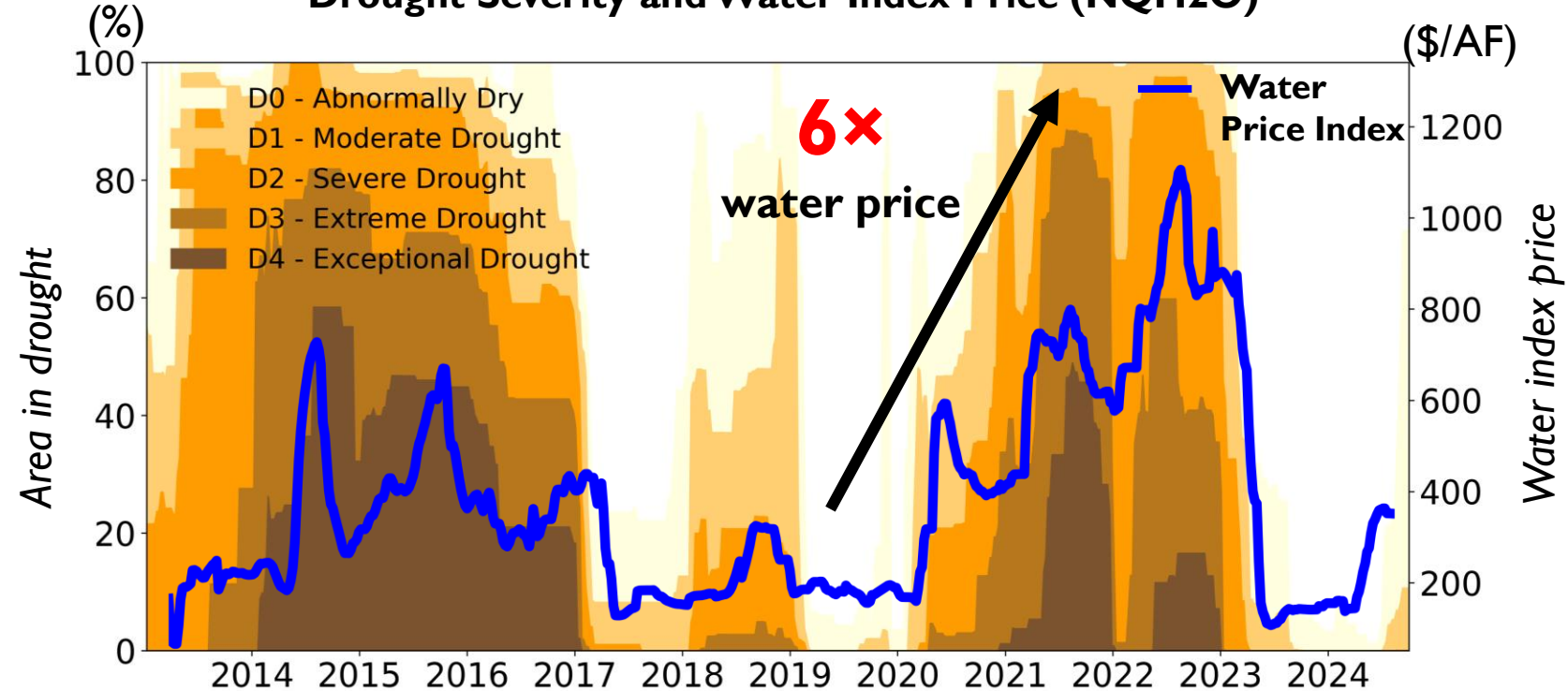


Managing Financial Risk in Water System

Water system in California

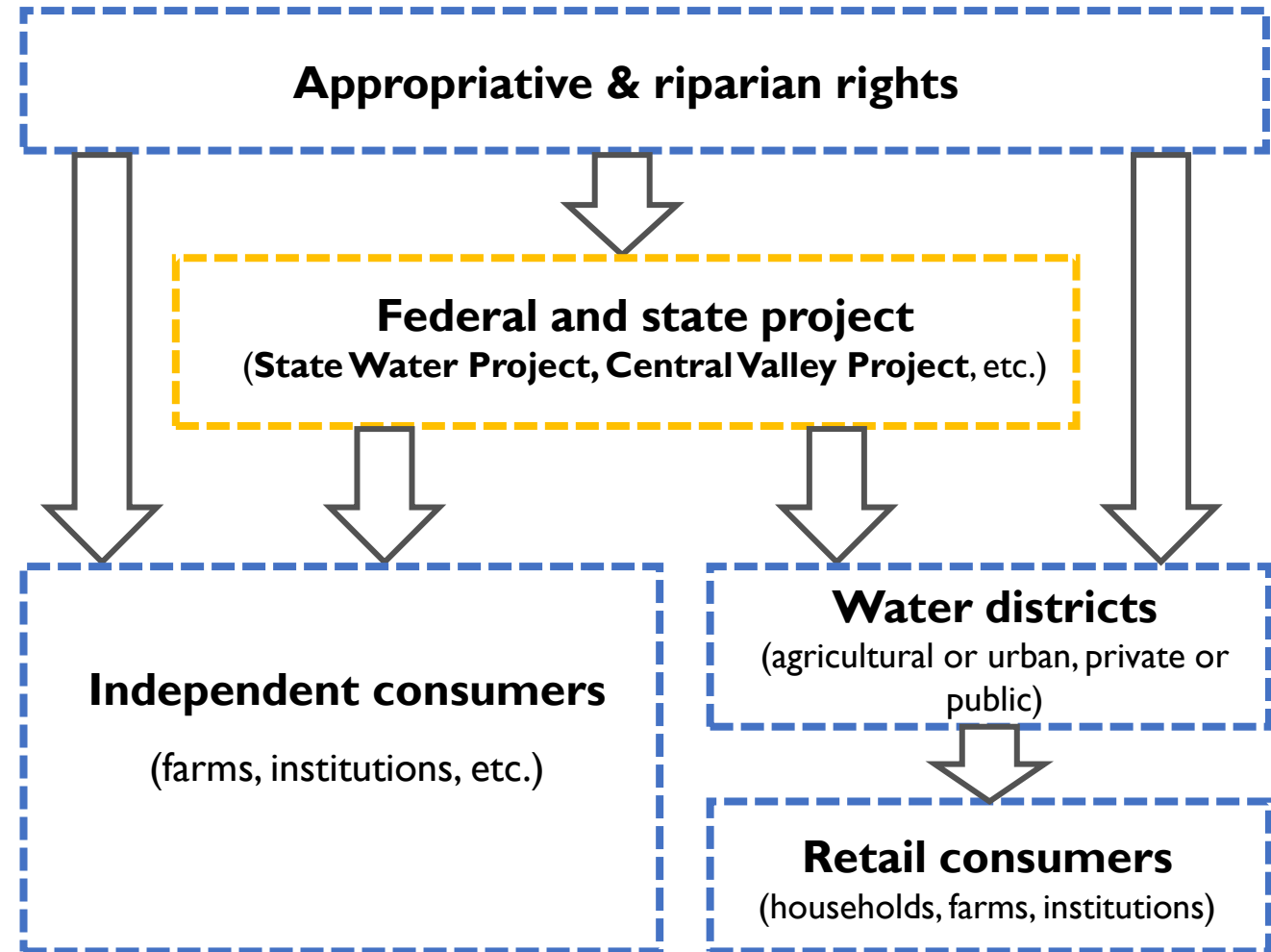
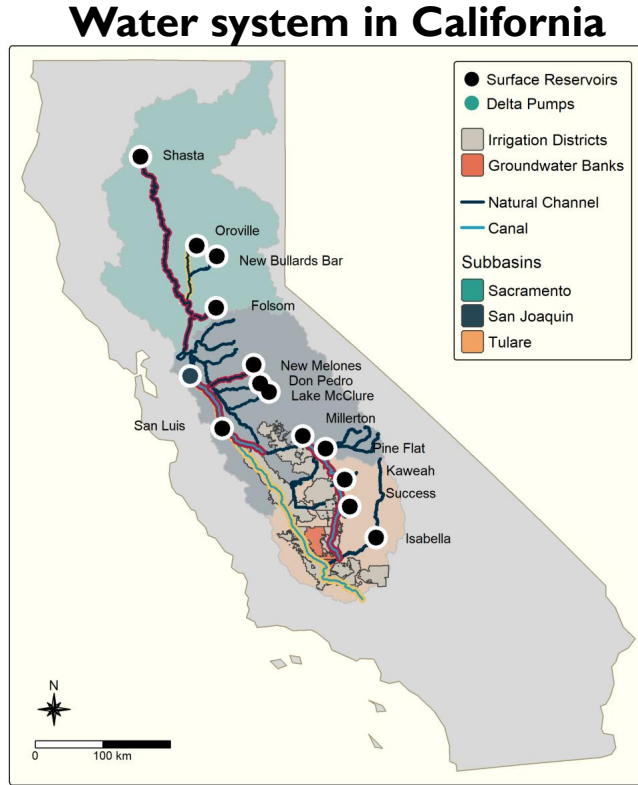


Drought Severity and Water Index Price (NQH2O)



Water price volatility is a major source of financial instability for water-dependent sectors.

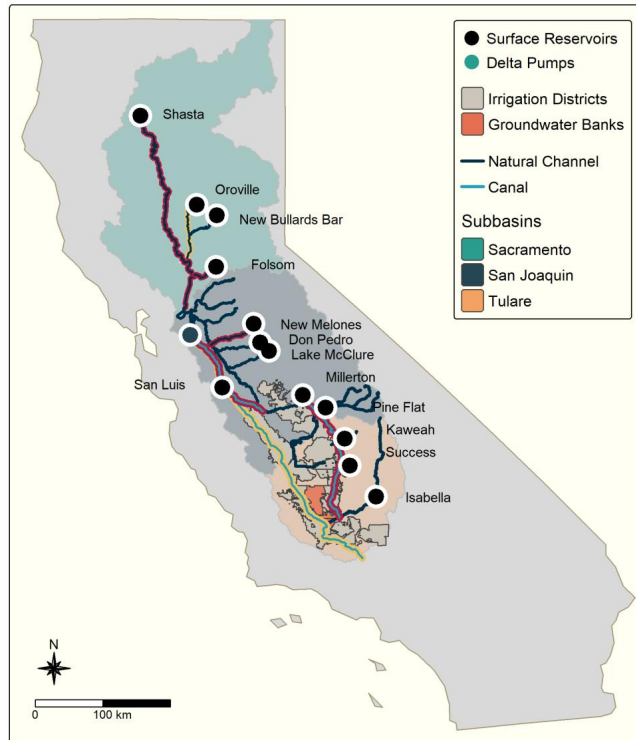
Water Allocation Is Governed By Sophisticated Rules



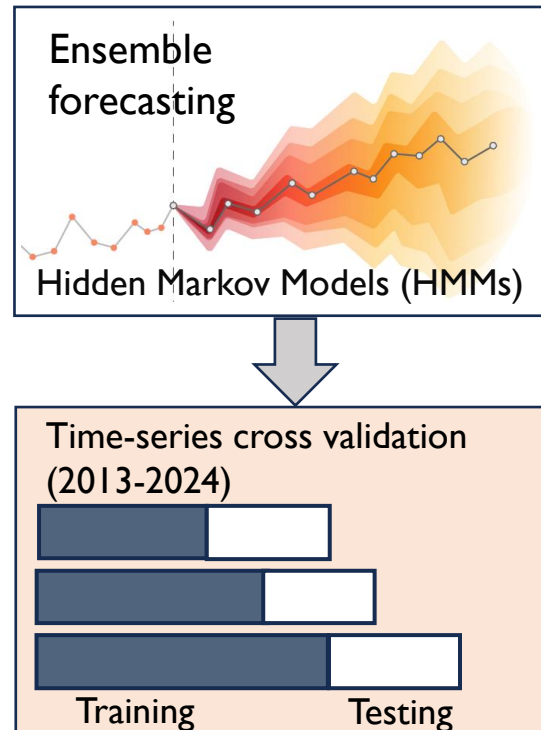
Allocation rules determine who loses water & who loses revenue, during drought.

Managing Financial Risk in Water System

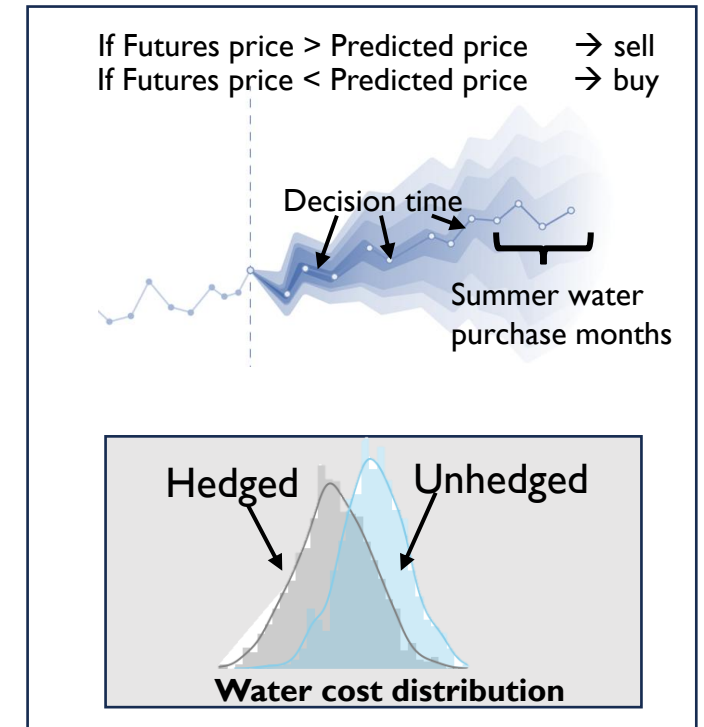
Hydrologic Modeling



Stochastic Forecasting



Dynamic Hedging Strategies



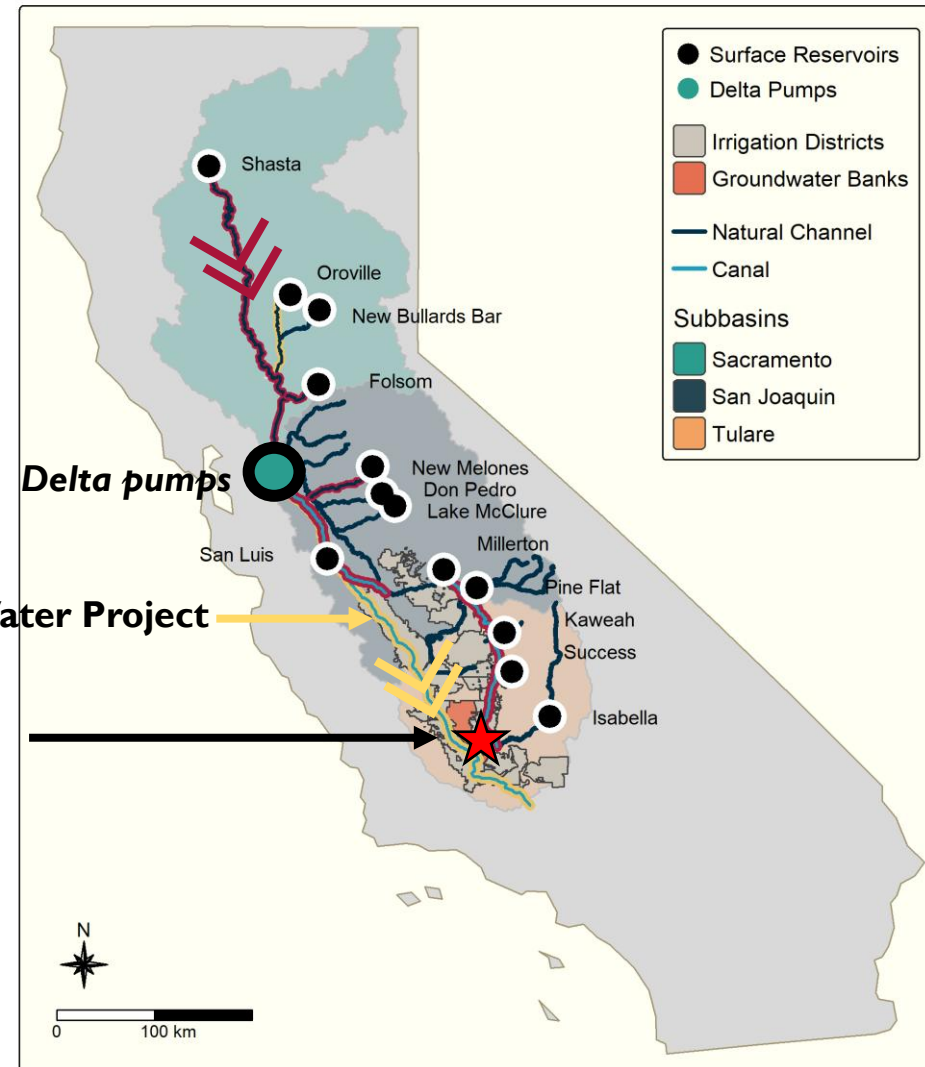
Managing Drought Risk Requires a **Systems Perspective**: Hydrology → Prices prediction → Hedging decision

Study Area

Industrial farms in Kern County:
financial risks caused by fluctuating water prices

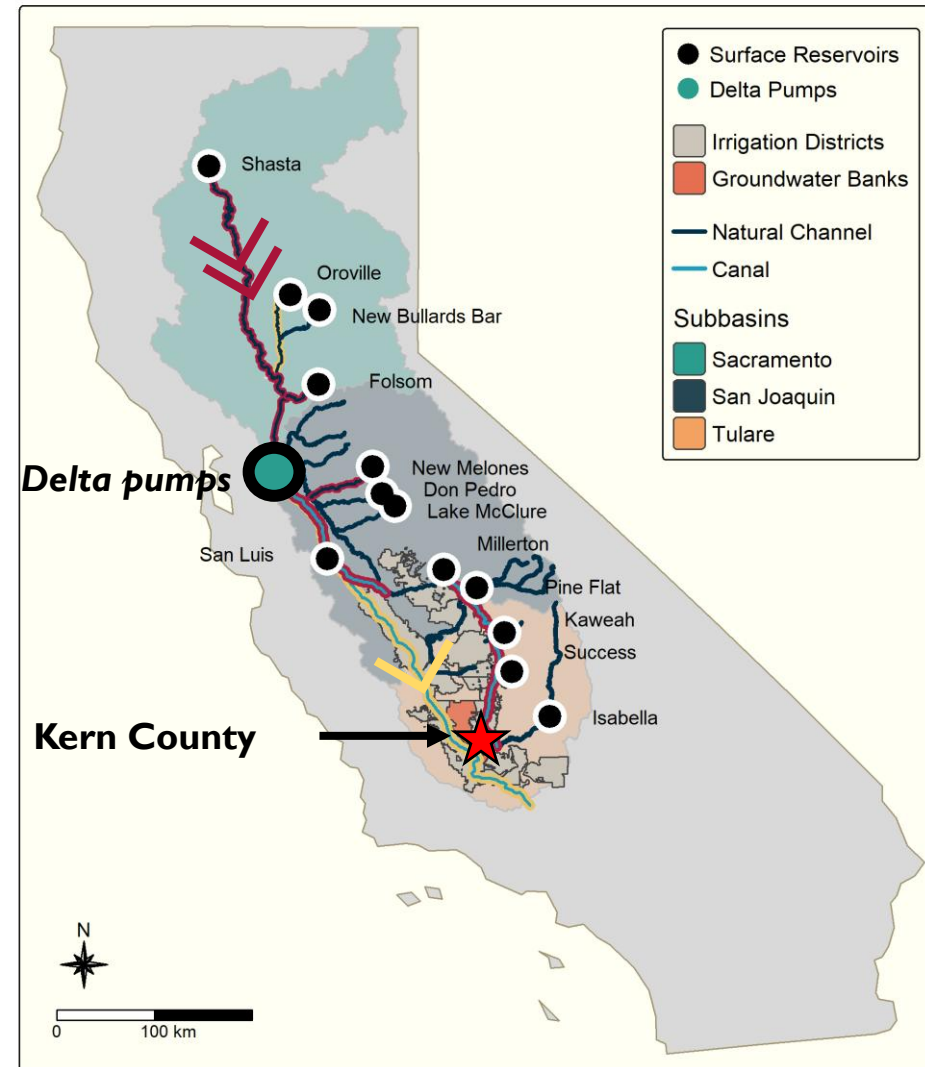


State Water Project
↓
Kern County



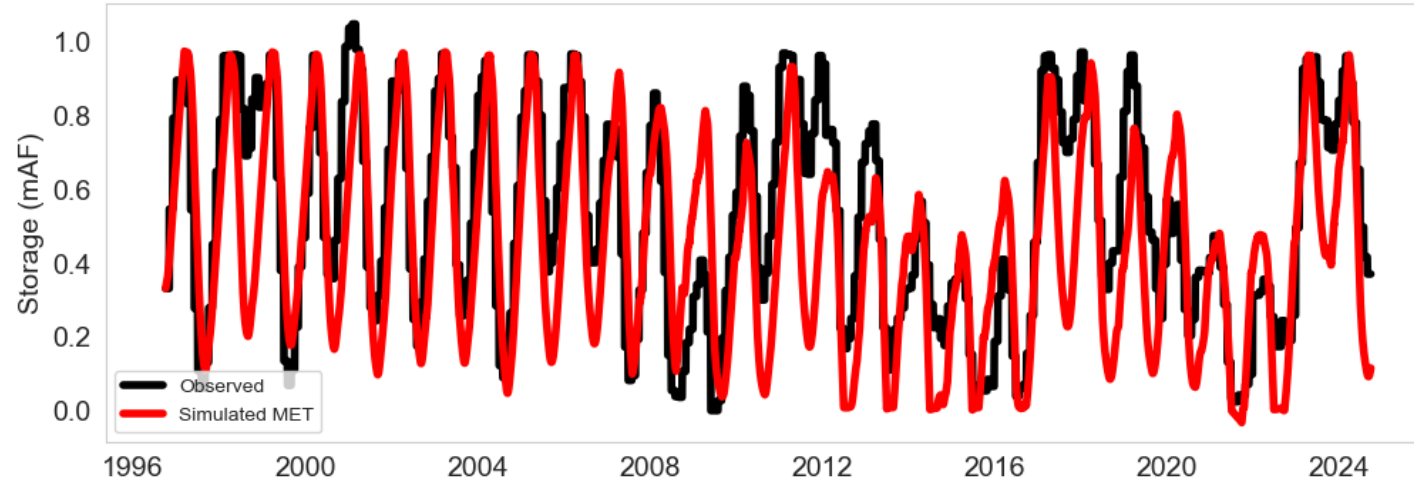
Simulation Model - CALFEWS

- Daily timestep
- The California Food-Energy-Water System (CALFEWS) is a rules-based simulation model.
- CALFEWS represents CA's coordinated water resources operations across institutionally complex system, including:
 - Surface water reservoirs
 - Delta pumping stations
 - Groundwater banks
 - Irrigation district

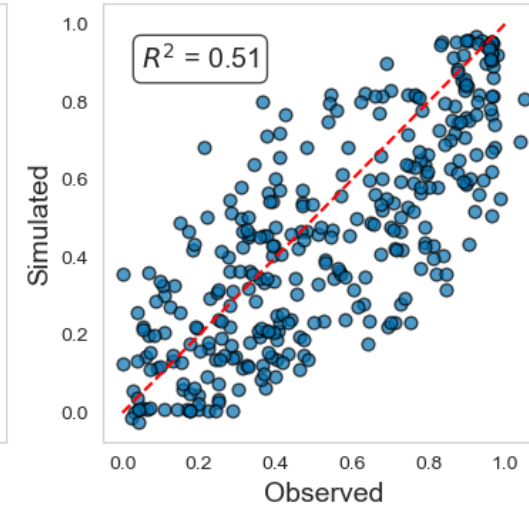


Simulation Model - CALFEWS

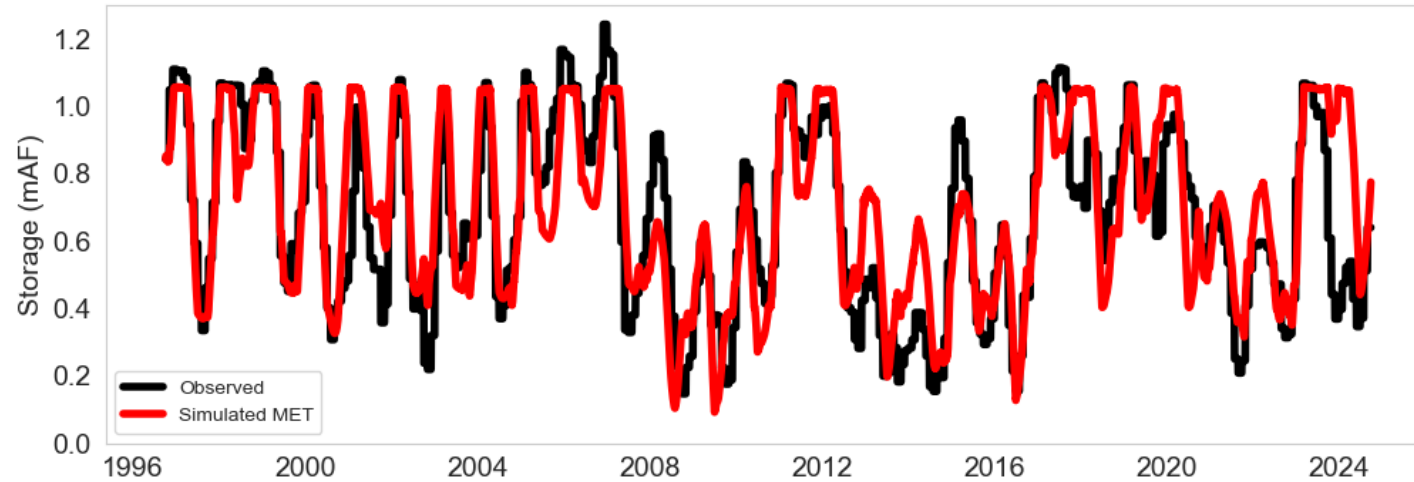
Federal (CVP) Portion, San Luis Reservoir



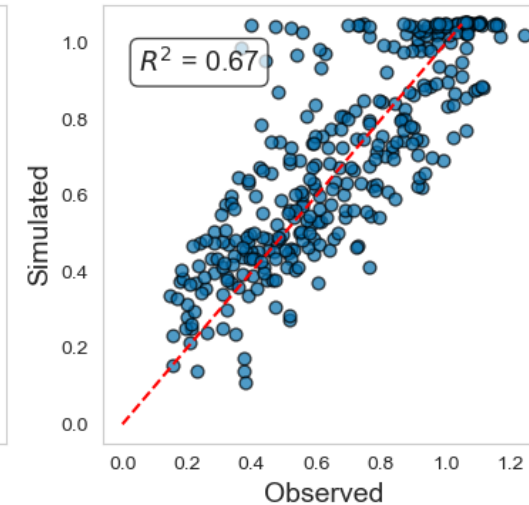
Observed vs. Simulated



State (SWP) Portion, San Luis Reservoir

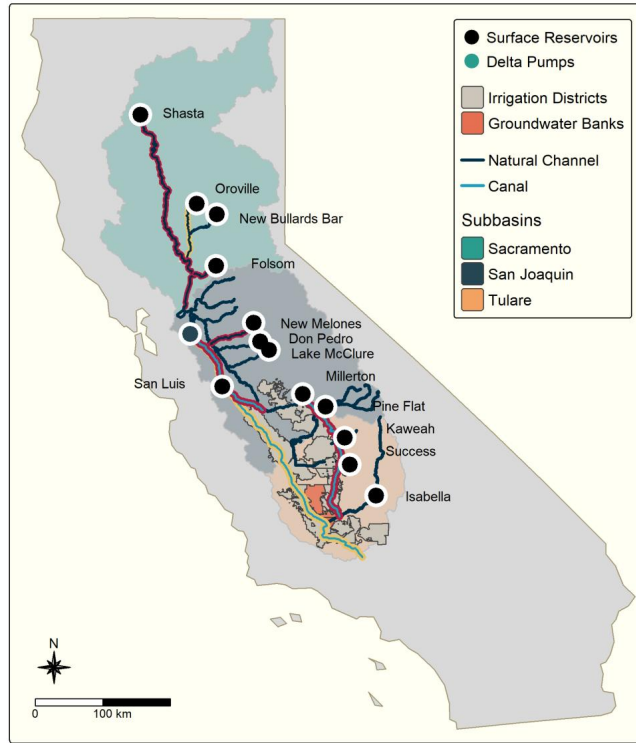


Observed vs. Simulated

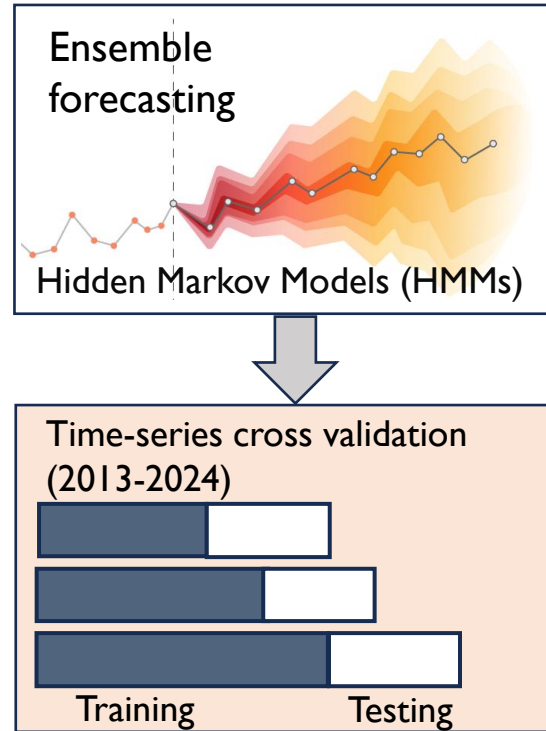


Managing Financial Risk in Water System

Hydrologic Modeling



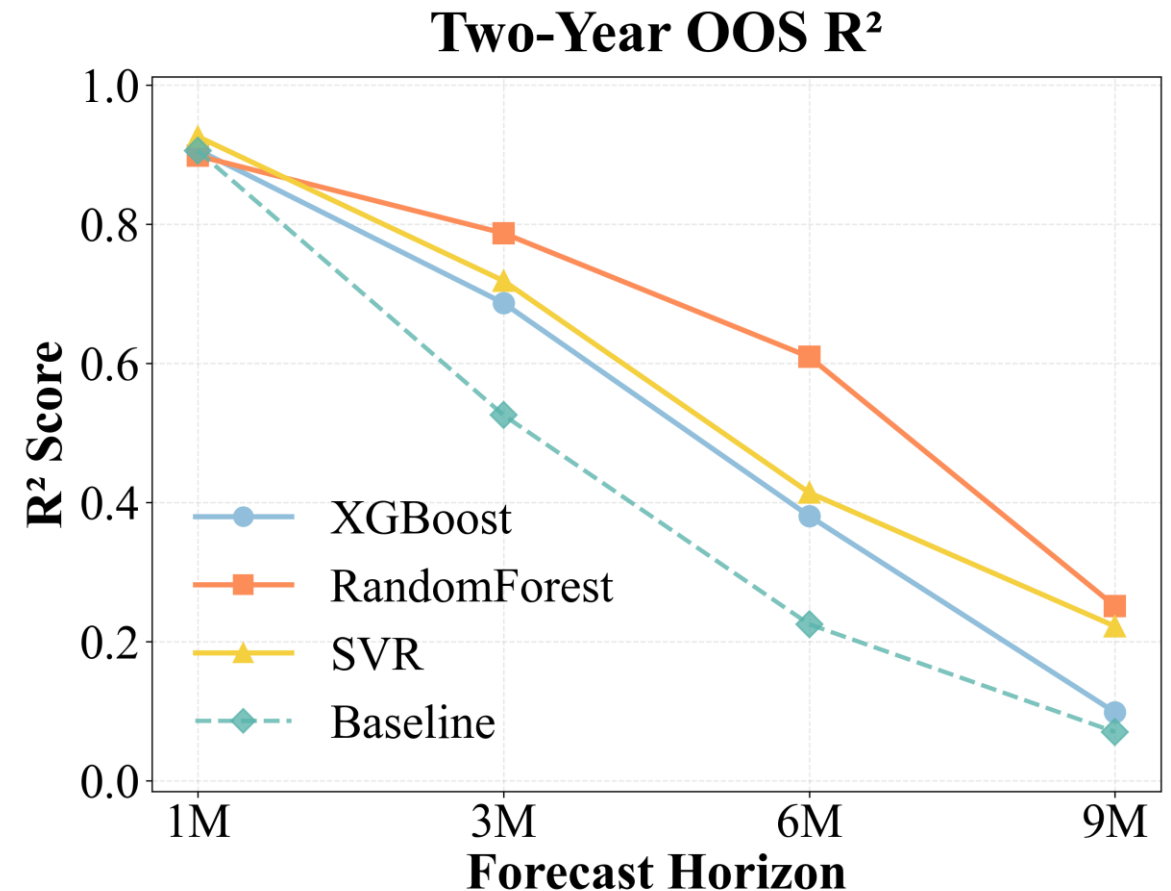
Stochastic Forecasting



Managing Drought Risk Requires a **Systems Perspective**: Hydrology → Prices prediction → Hedging decision

2-Year Out-of-Sample Test

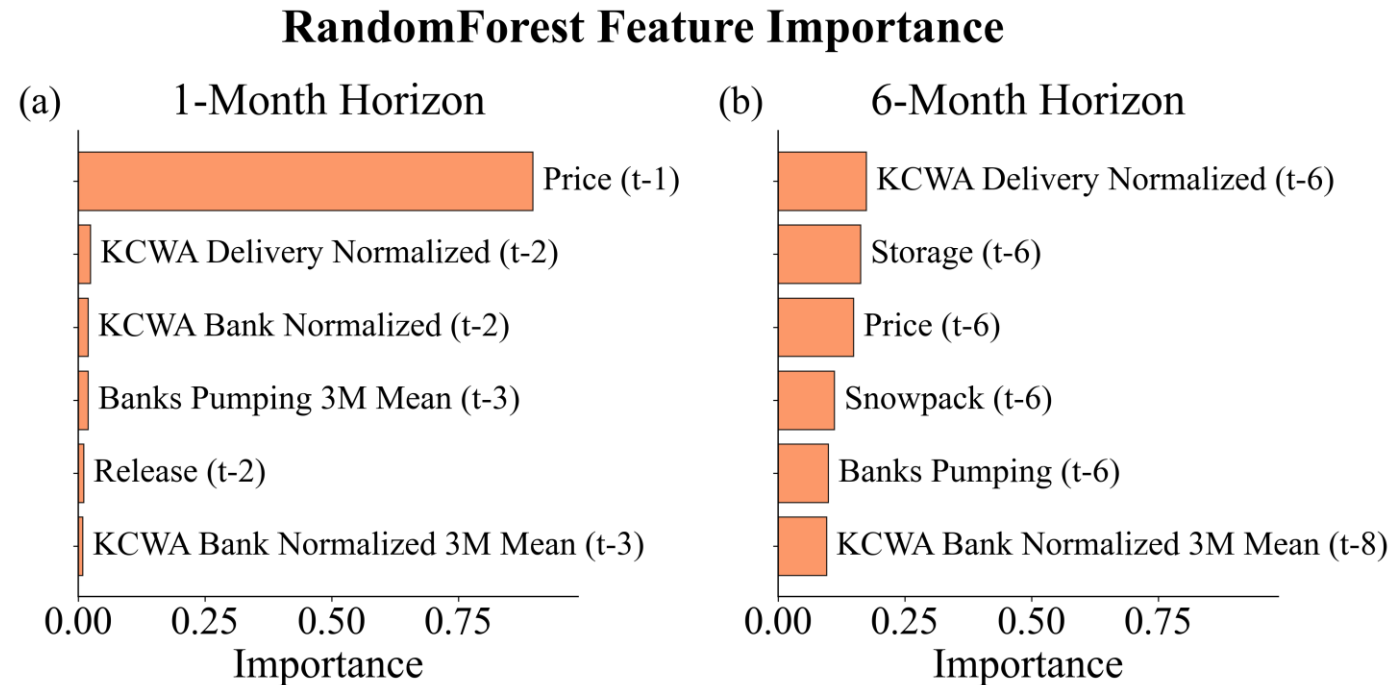
- Training: 2013–2022
Testing: 2022–2024
- The ML models show high skill at short lead times (1-M).
- By the 6-month forecast horizon, predictive power declines.
- **Forecast skill drops sharply beyond 6 months** and is very limited by 9 months.



→ **temporal limits** of hydrologic persistence and the **stochastic nature** of financial market dynamics.

How Predictors Change Across Forecast Horizons

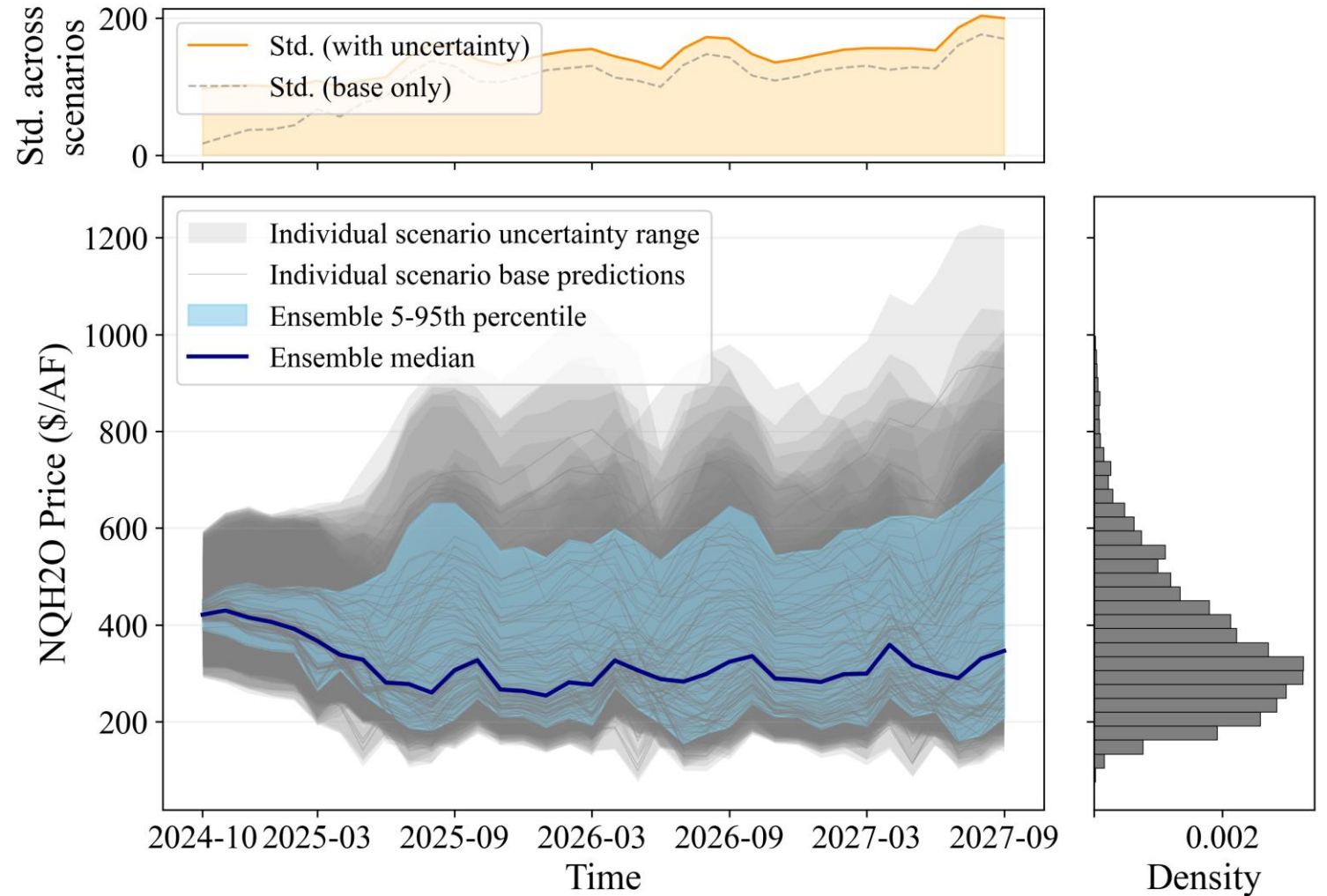
- **1 month ahead:** lagged prices dominate, showing strong short-term price persistence.
- **6 months ahead: hydrologic conditions** such as snowpack, storage, and Delta pumping become the main drivers.



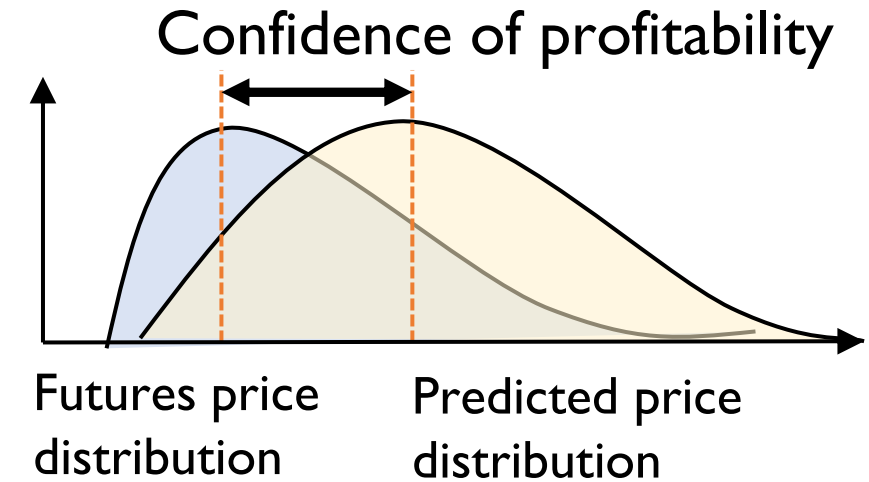
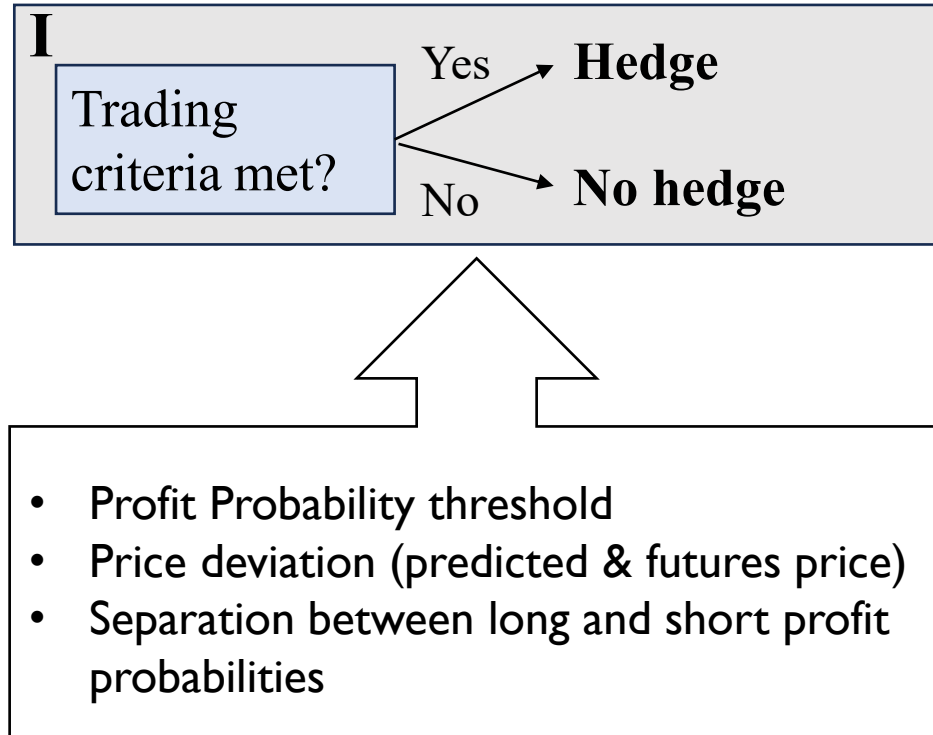
Water Price Forecasting

Modeled NQH2O price variability across 100 hydroclimatic ensembles (24 – 27)

→ a realistic range of prices under diverse conditions.



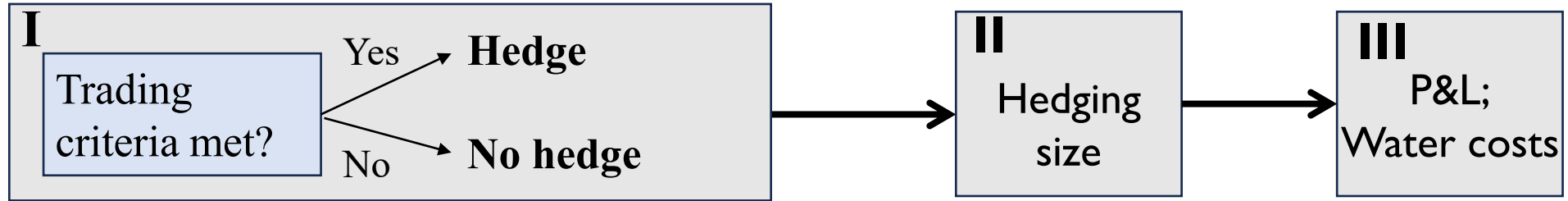
Dynamic Hedging Strategies



If Futures price < Predicted price → buy

If Futures price > Predicted price → sell

Dynamic Hedging Strategies



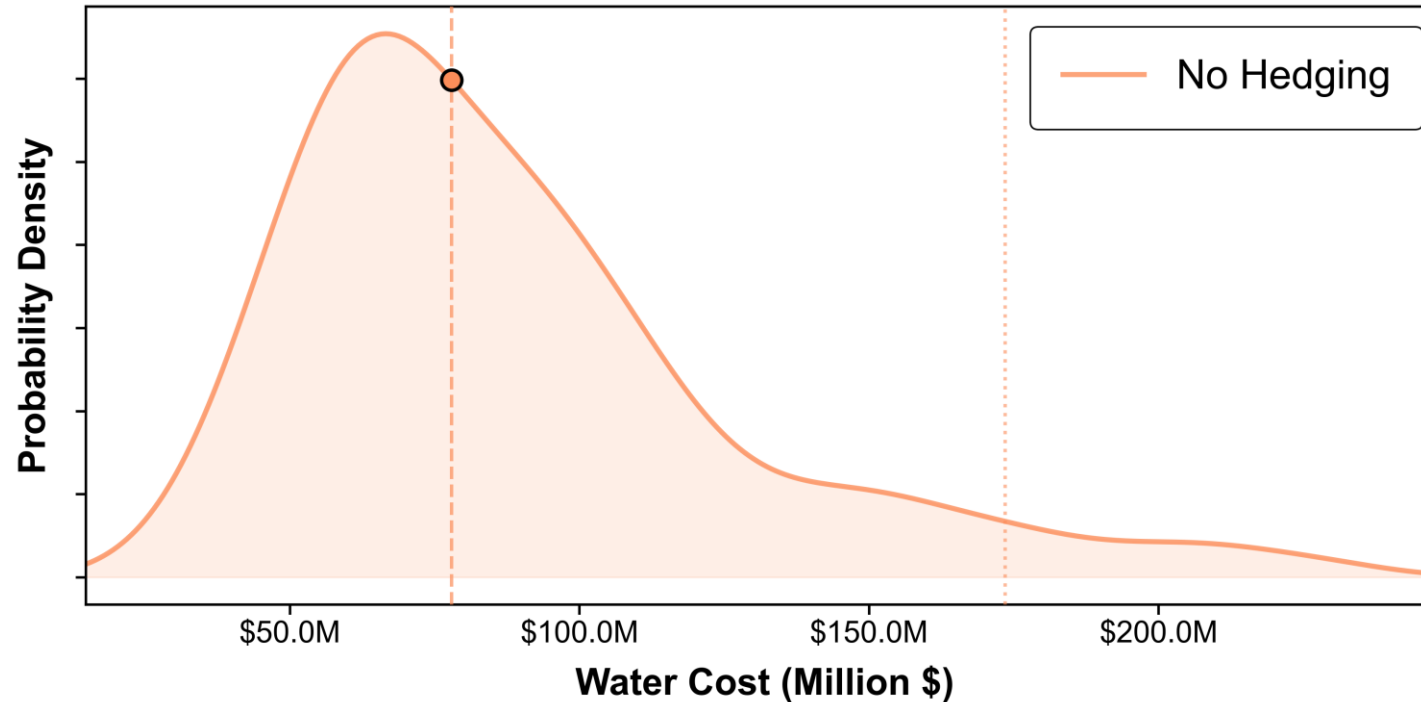
forecast uncertainty  hedge ratio 

→ limiting exposure while still allowing the strategy to act on high-confidence signals.

Hedging with Water Futures Reduces Water Costs

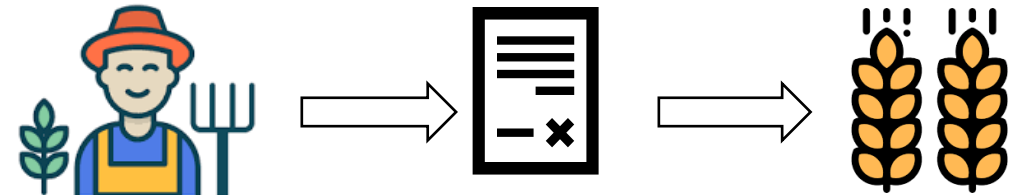


Water Cost Distribution Across the Ensemble (No Hedging)
= shortage $\times$ spot price



How Futures Contracts Work

- Futures contracts are standardized exchange-traded contracts used to hedge future price risk.
- Widely used in energy and agricultural markets for hedging.
- Water futures are cash-settled against the NQH2O index (no water delivery).



Farmers faces
harvest price risk

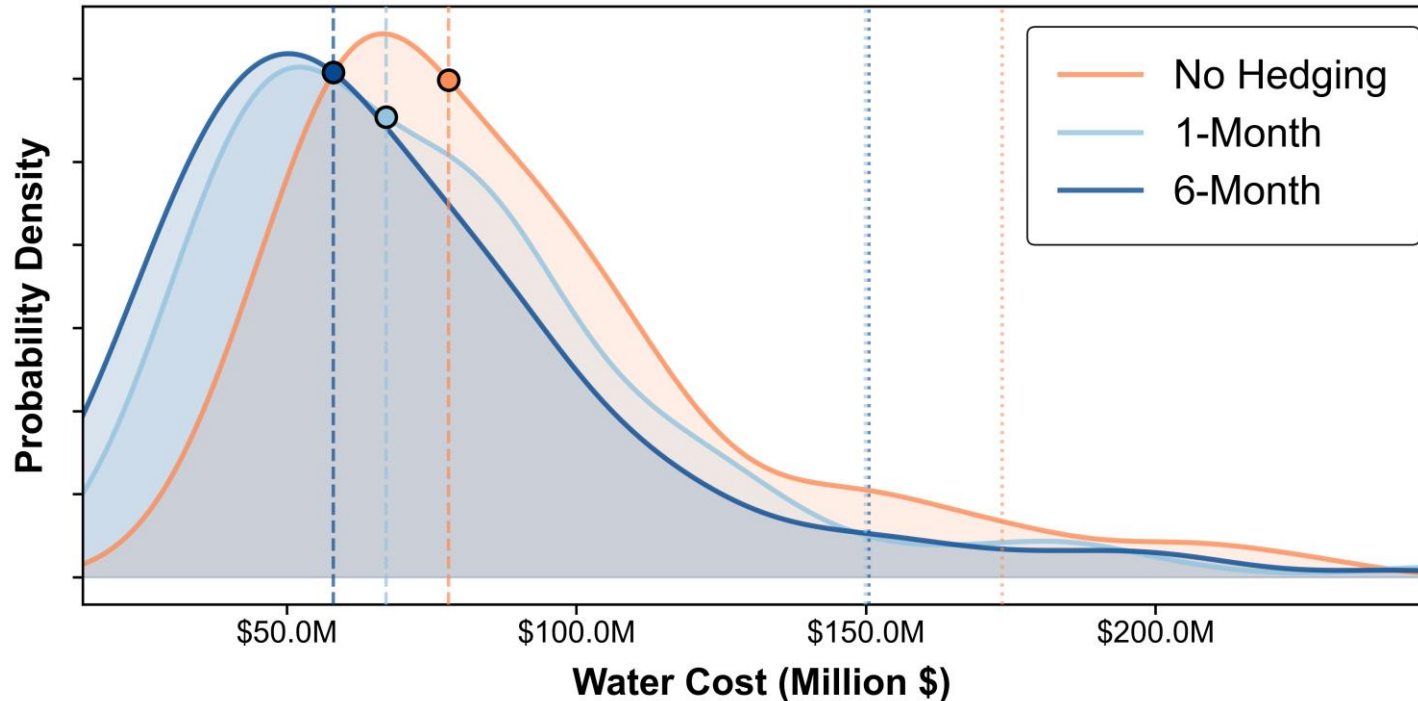
- Spot price ↓ → futures gain
- Spot price ↑ → futures loss

Growing season
(Enter contract)

Harvest
(Futures settlement)

Hedging with Water Futures Reduces Water Costs

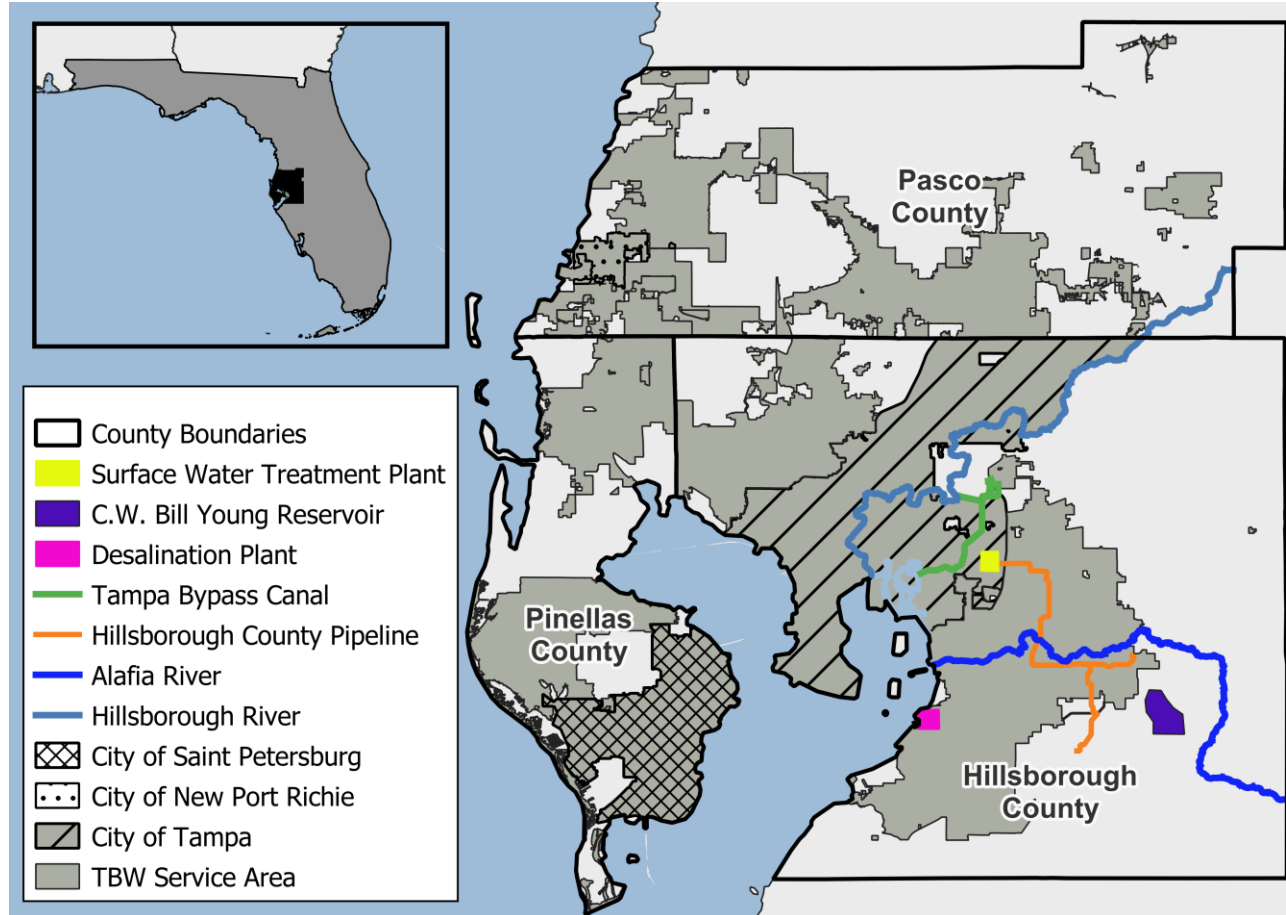
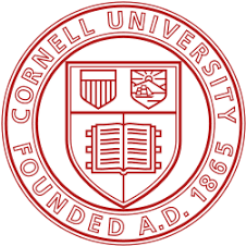
Water Cost Distribution Across the Ensemble (With Hedging)
= shortage $\times$ spot price



- A 6-month hedge cuts median costs by **25.5%** and reduces worst-5%-outcome by **13.3%** (in theory).
- Price-prediction tool could improve price discovery and **reduce information frictions**.

***How can utilities balance **supply reliability**
and **financial stability** under climate shocks?***

Tampa Bay Water Authority (TBW)



- Wholesale water provider to six member governments
- Diversified supply sources: groundwater, surface water, and desalination
- Credit Rating: AA+
- Issue **\$600 million new debt** in 2028 to expand supply

The central planning problem...

TBWV has to hit two goals at once:

Supply reliability

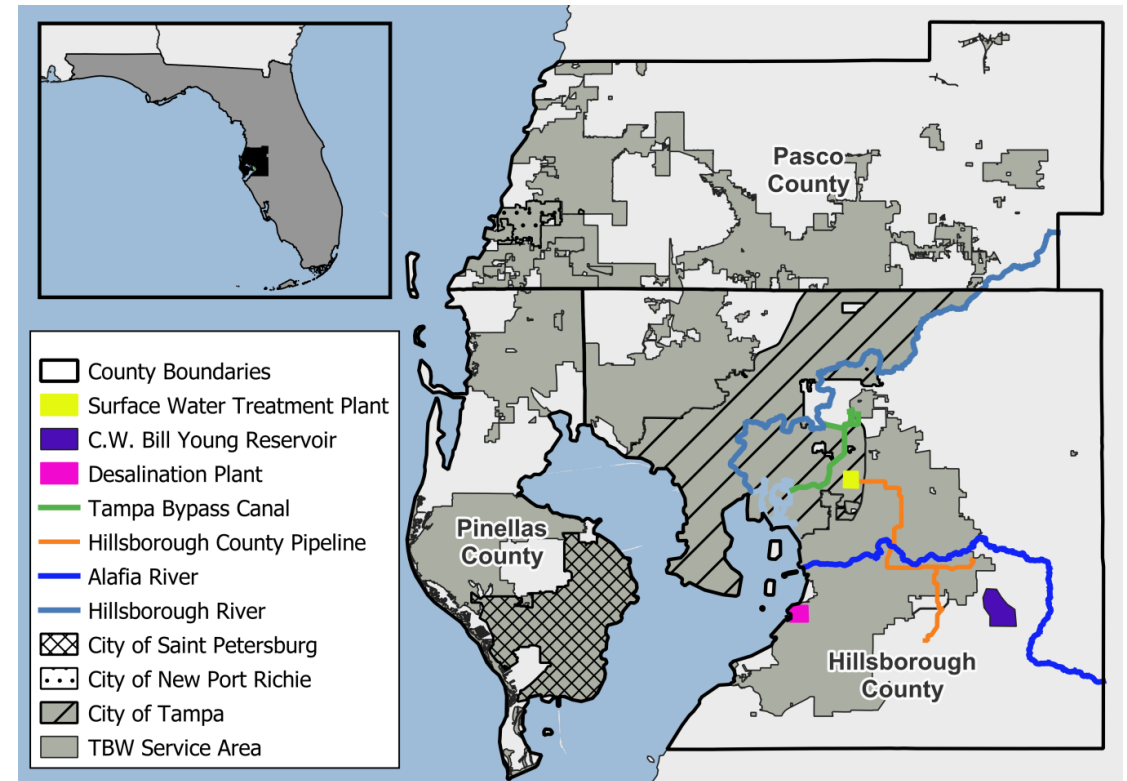
avoid critical shortfalls

Financial stability

meet debt and rate covenants

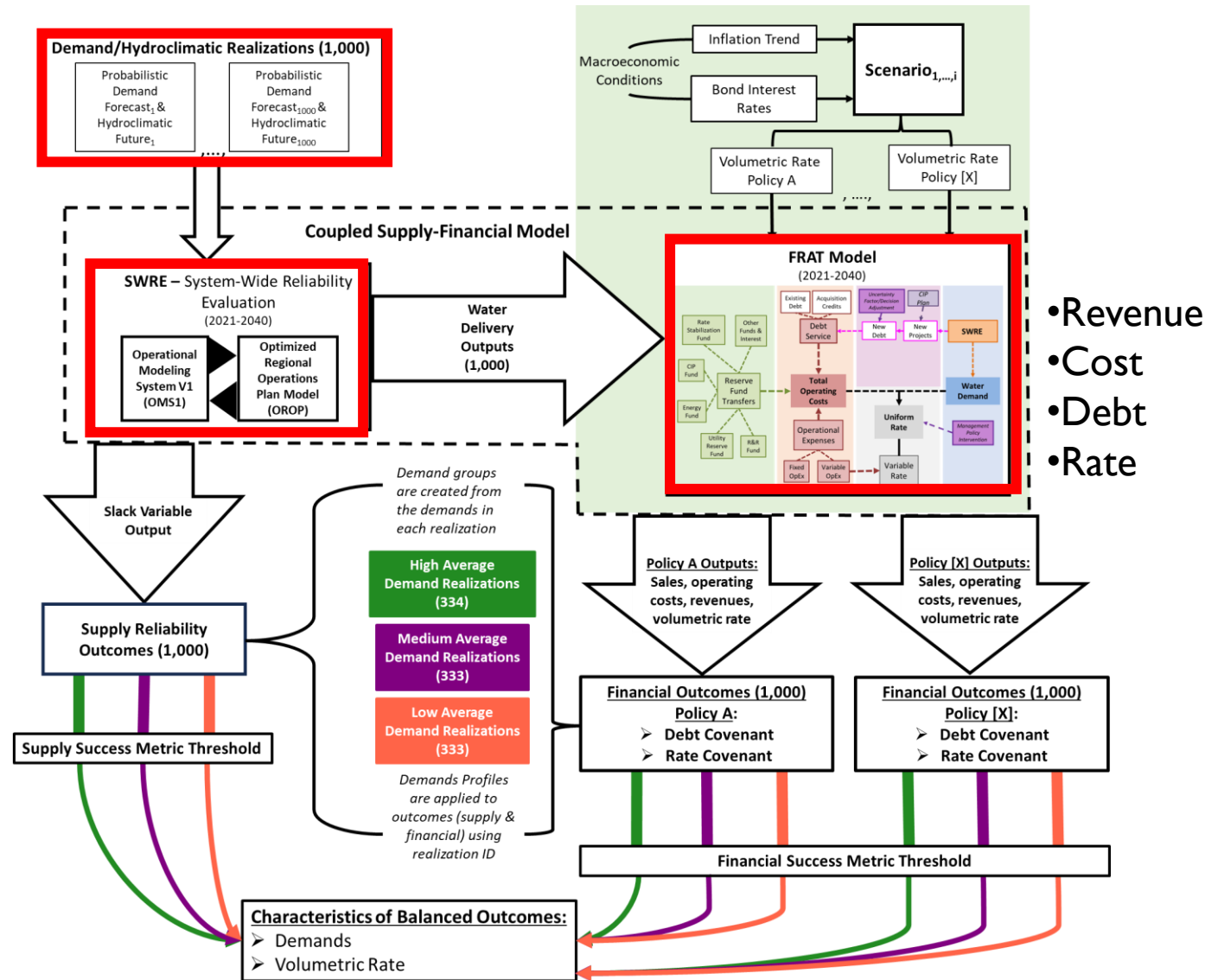
as climate and demand uncertainty grow...

- New supply projects improve reliability but raise debt service.
- Low demand reduces revenue & weakens financial stability
- Raising rates can create affordability issues.
- So what is the right mix of rate design and demand management to balance financial stability and supply reliability?



Study context: Tampa Bay Water serves **>2.6 million** people in west-central Florida.

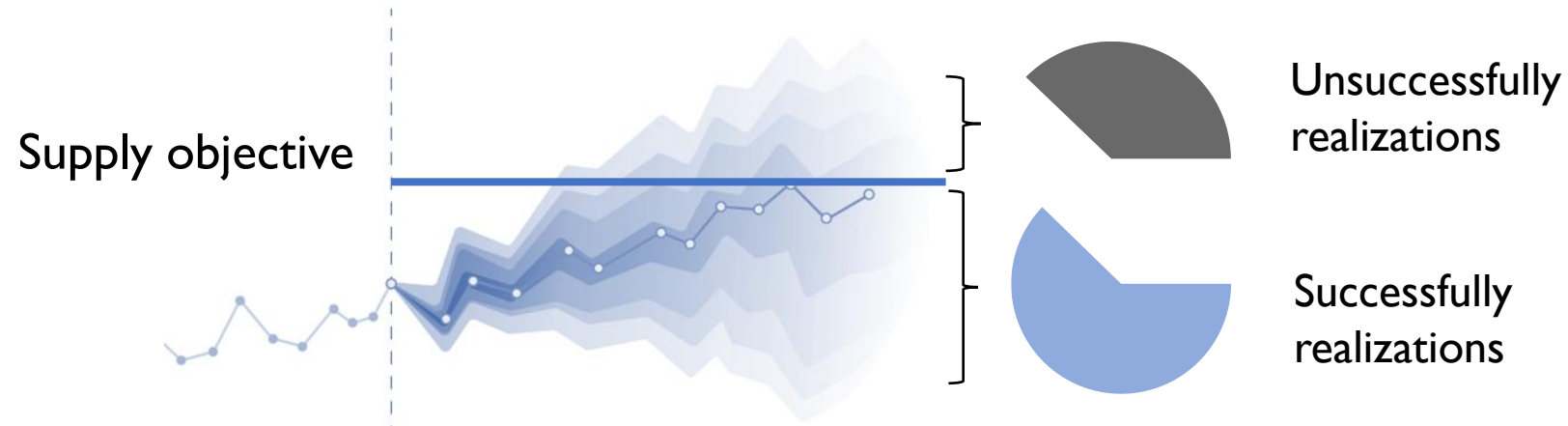
Coupled Supply-Financial Model



- Couples TBW's water supply and financial planning models (FRAT)
- Evaluates realizations meeting both supply and financial goals
- The integrated model connects hydrology, demand, operations, budgets, and credit risk.

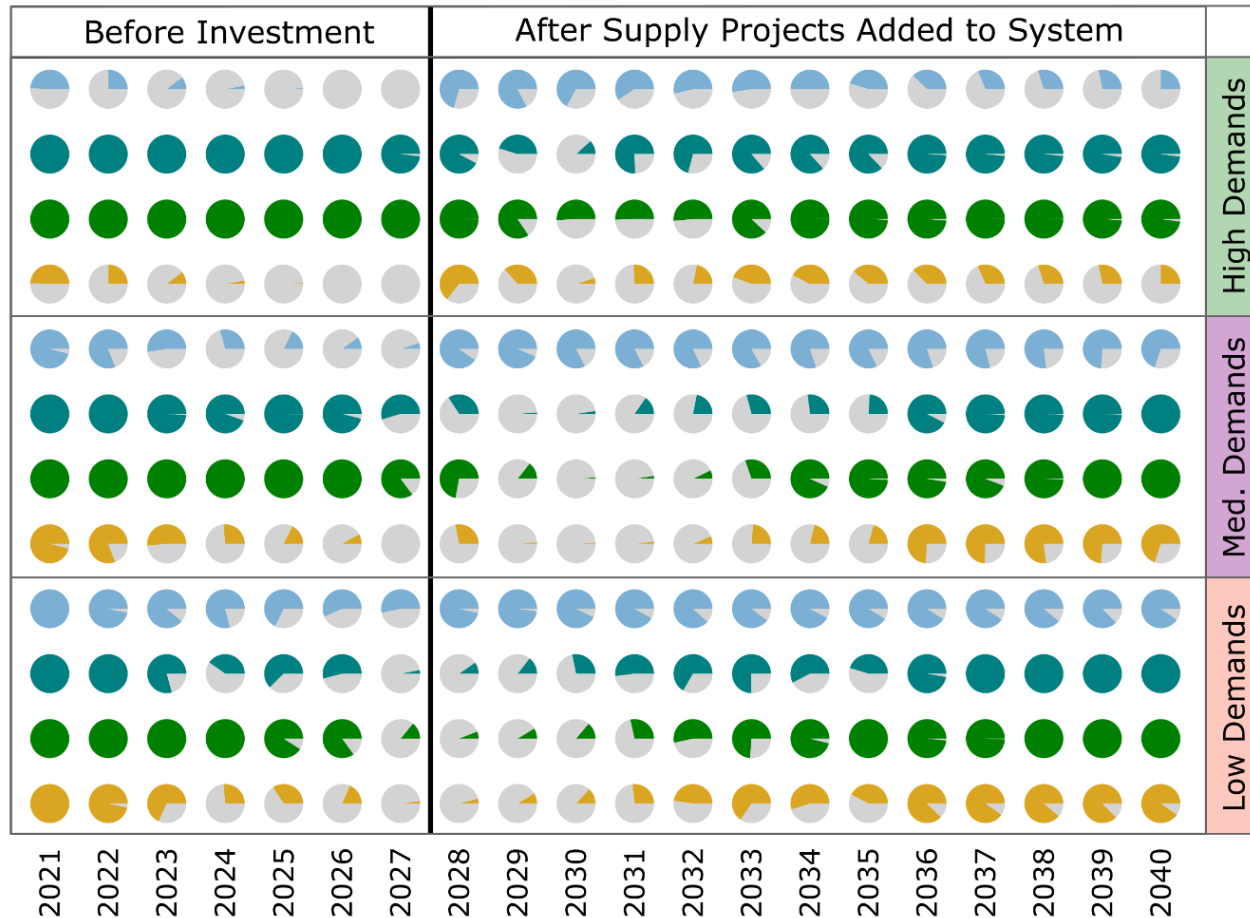
Where is our Goldilocks Zone?

Each pie shows the share of simulated outcomes that satisfy an objective in a given year.

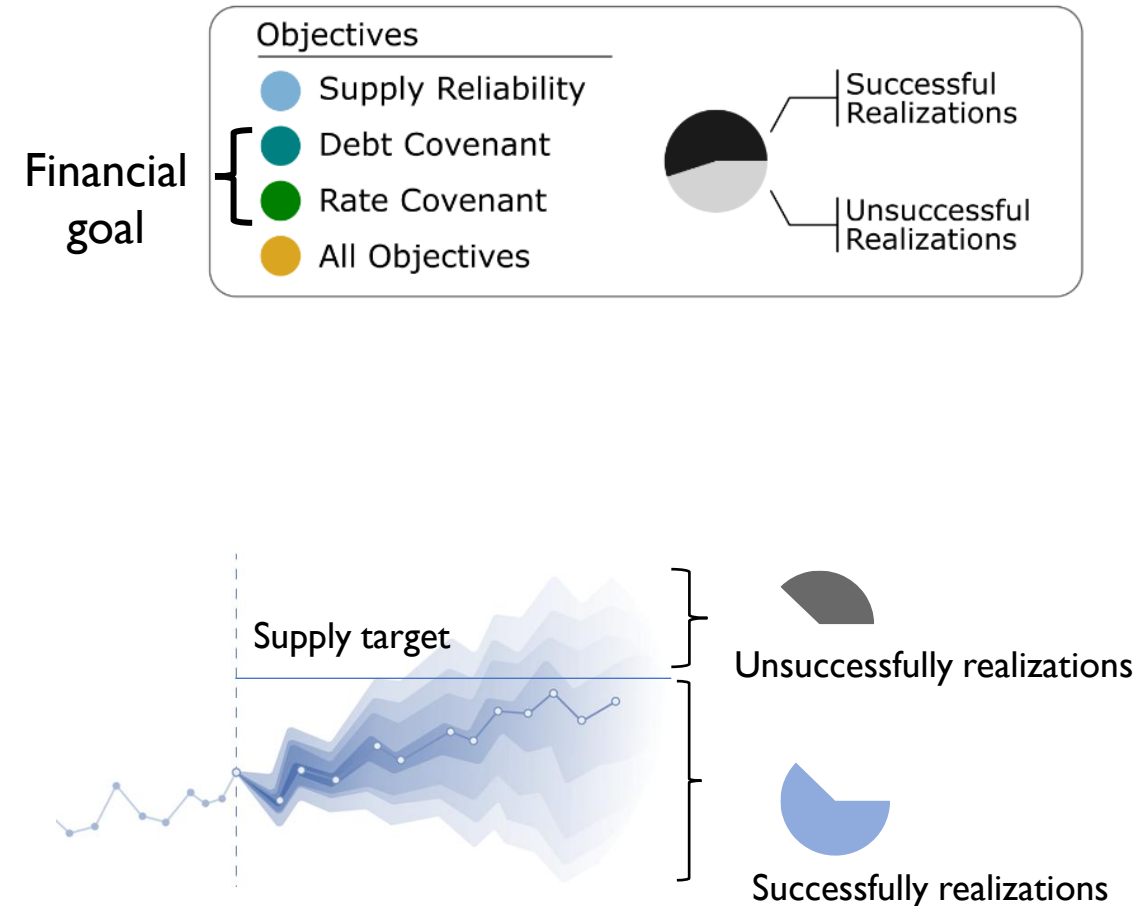


larger gray portion → higher probability of supply goal failure

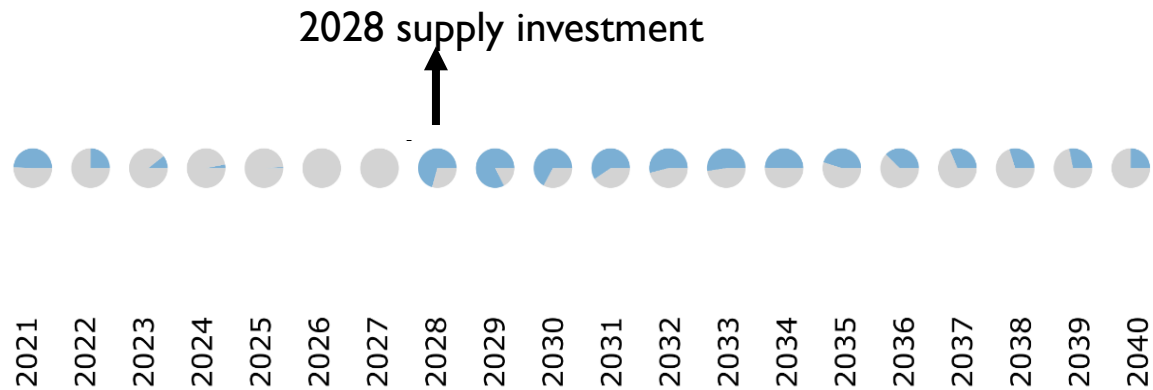
Where is our Goldilocks Zone?



Simulation period: 2021–2040



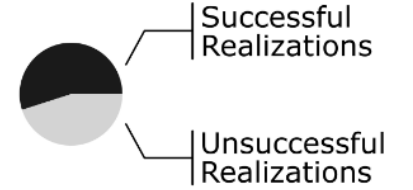
Where is our Goldilocks Zone?



Financial goal

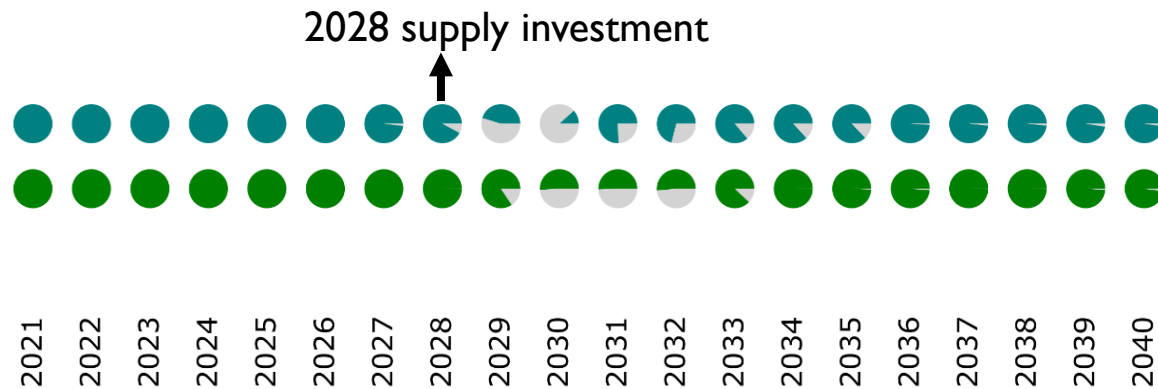
Objectives

- Supply Reliability
- Debt Covenant
- Rate Covenant
- All Objectives



- Supply reliability jumps in 2028 when the new projects come online.
- But reliability deteriorates later as demand keeps rising, especially in high-demand group.

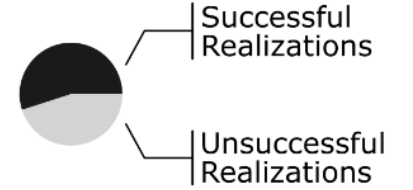
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Financial goal

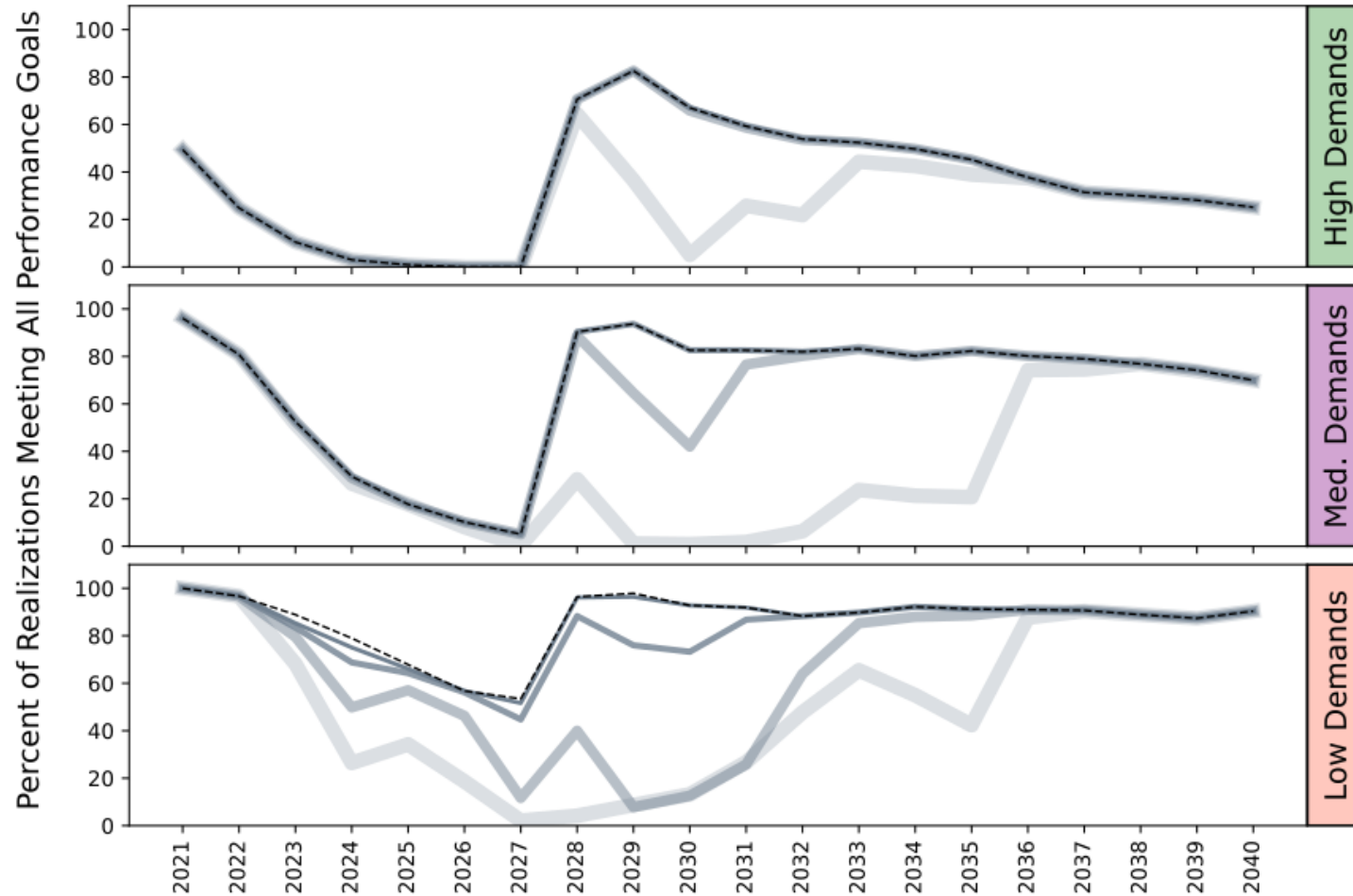
Objectives

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- Supply reliability jumps in 2028 when the new projects come online.
- But reliability deteriorates later as demand keeps rising, especially in high-demand group.
- Finance is weakest around 2028-2032 because new debt service peaks right after investment.
- low-demand group often fail financially; high-demand group more often fail on supply.

Rate policy comparison



Annual Rate Increase

Supply reliability goal met

$\leq 4\%$

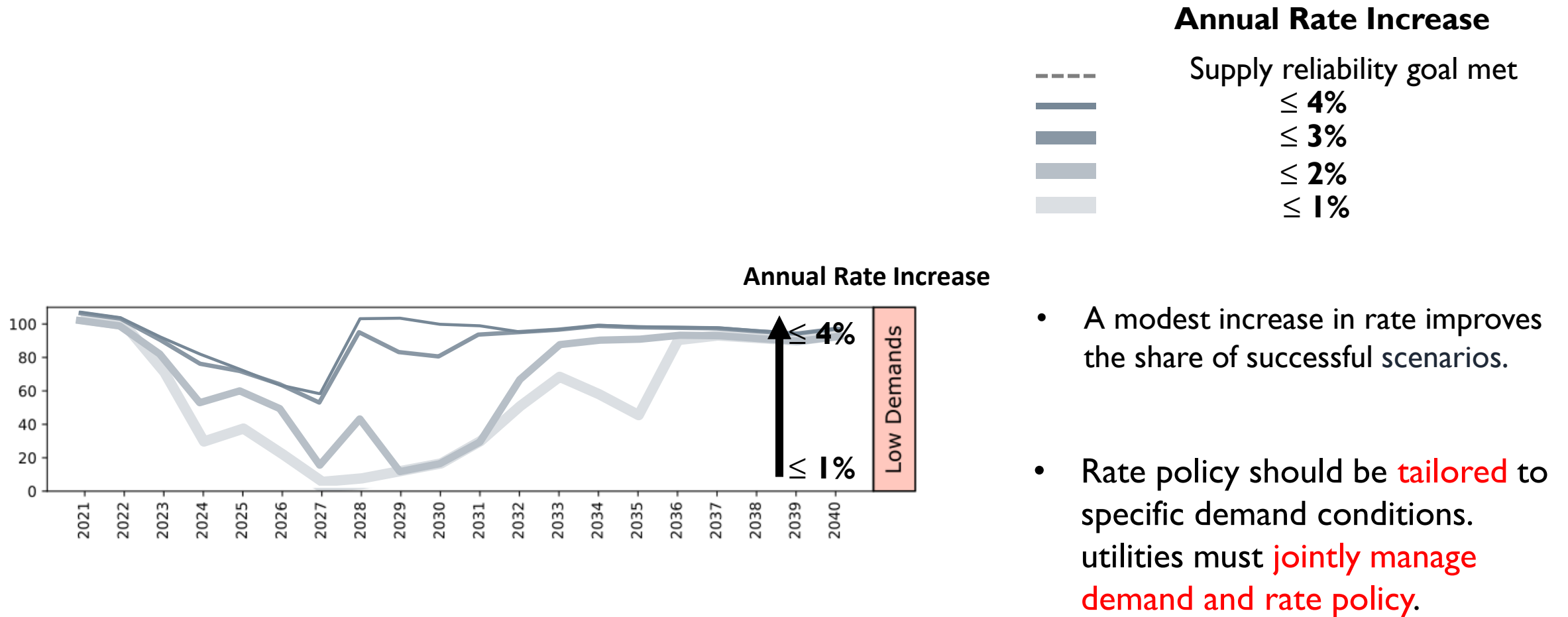
$\leq 3\%$

$\leq 2\%$

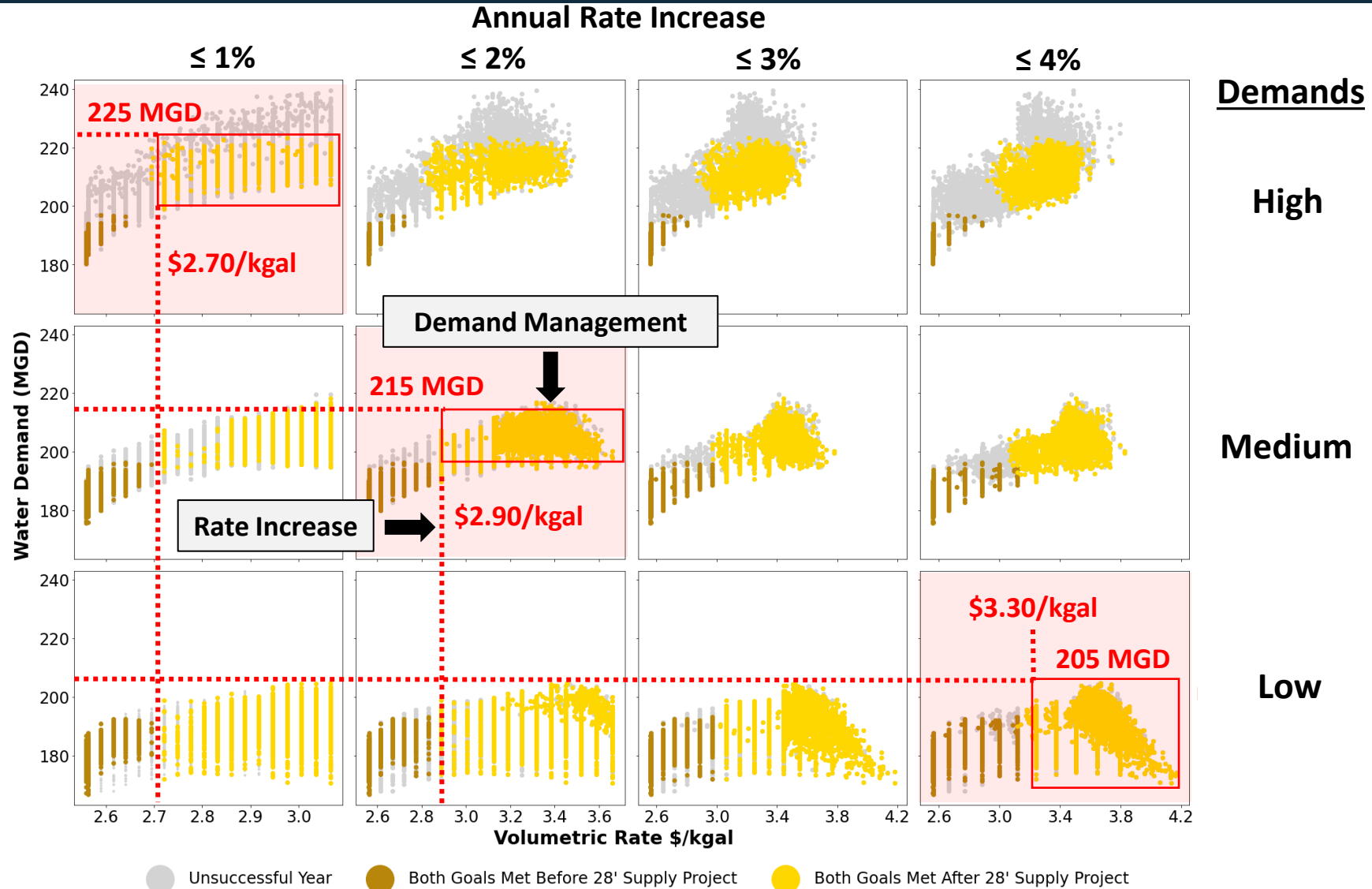
$\leq 1\%$

- A modest increase in rate improves the share of successful scenarios.

Rate policy comparison

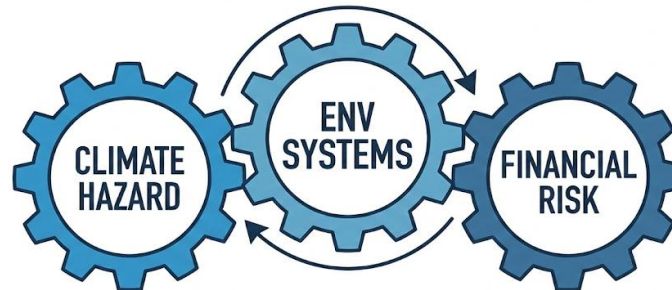


Where is our Goldilocks Zone?



Key Takeaway

- Utilities must balance supply reliability and financial stability in the face of growing uncertainty
- Climate and environmental risks are increasingly priced in financial markets.
- Better risk management requires integrated, system-aware tools that connect physical hazards to financial consequences and mitigation decisions.



Future Research Agenda

Area 1: Drought-related financial risks for US water utilities

Area 2: Hurricane risk propagation in Florida

Area 3: Wetland Mitigation and Downstream Flood Exposure



Future Research Agenda

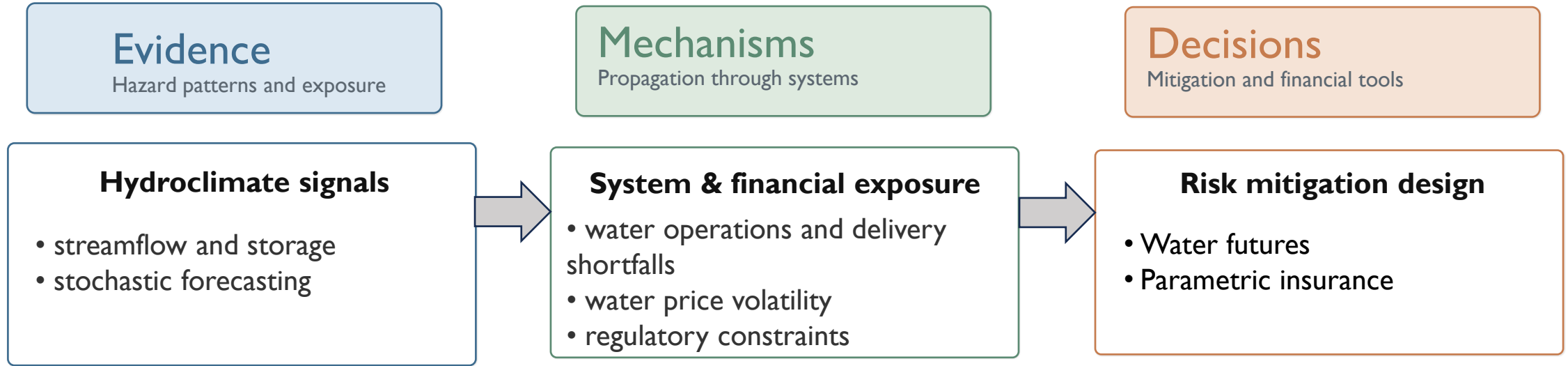
Area 1: Drought-related financial risks for US water utilities

Area 2: Hurricane risk propagation in Florida

Area 3: Wetland Mitigation and Downstream Flood Exposure



Drought as a Completed End-to-End Testbed



Completed in drought: hazard → operations → finance → mitigation

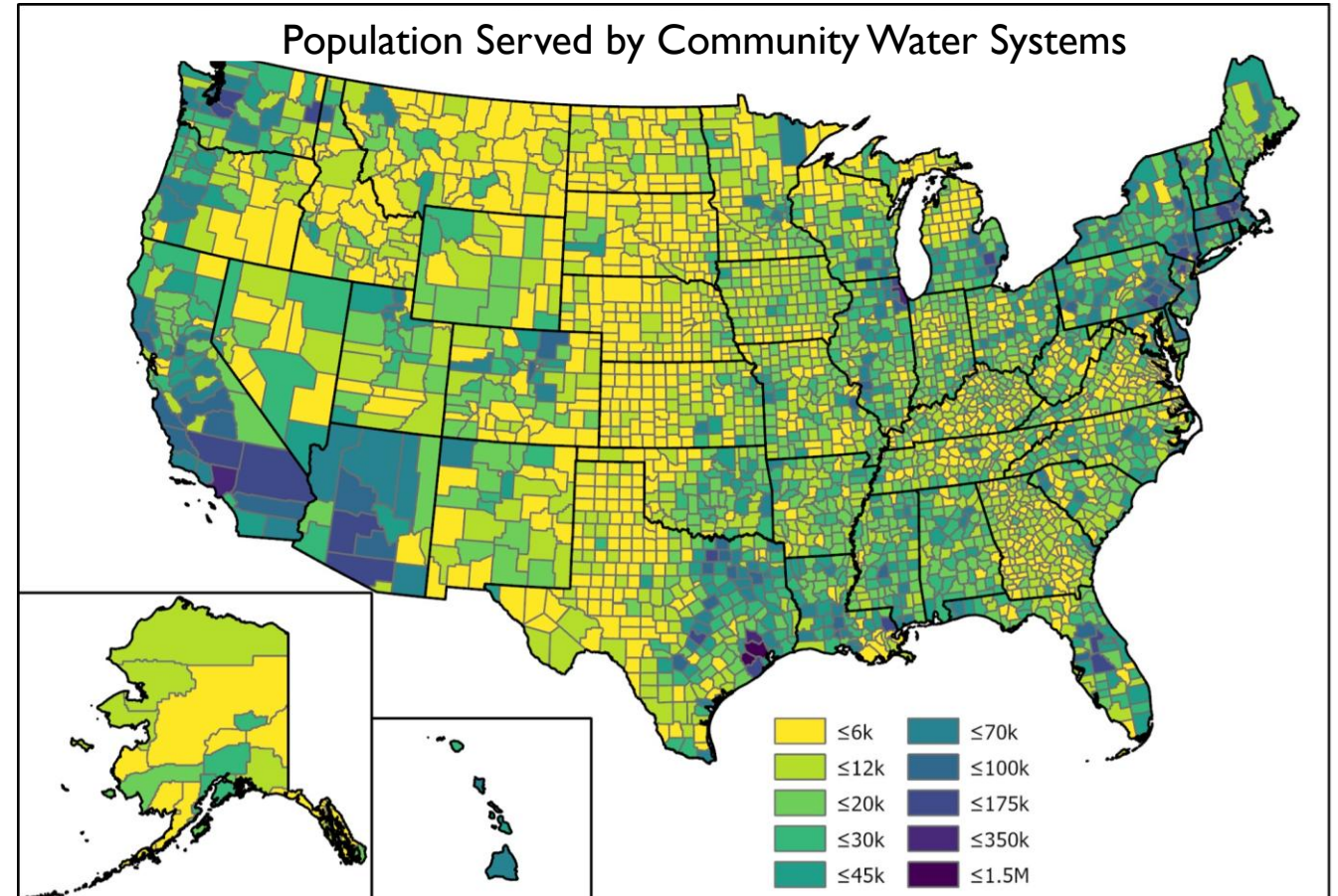
Next: scale the same framework to pooled utility risk management *across the U.S.*, especially for smaller systems that lack reserves and economies of scale.



Water Systems Lack Financial Tools to Manage Climate Risk

Small community water systems (<10,000 customers)

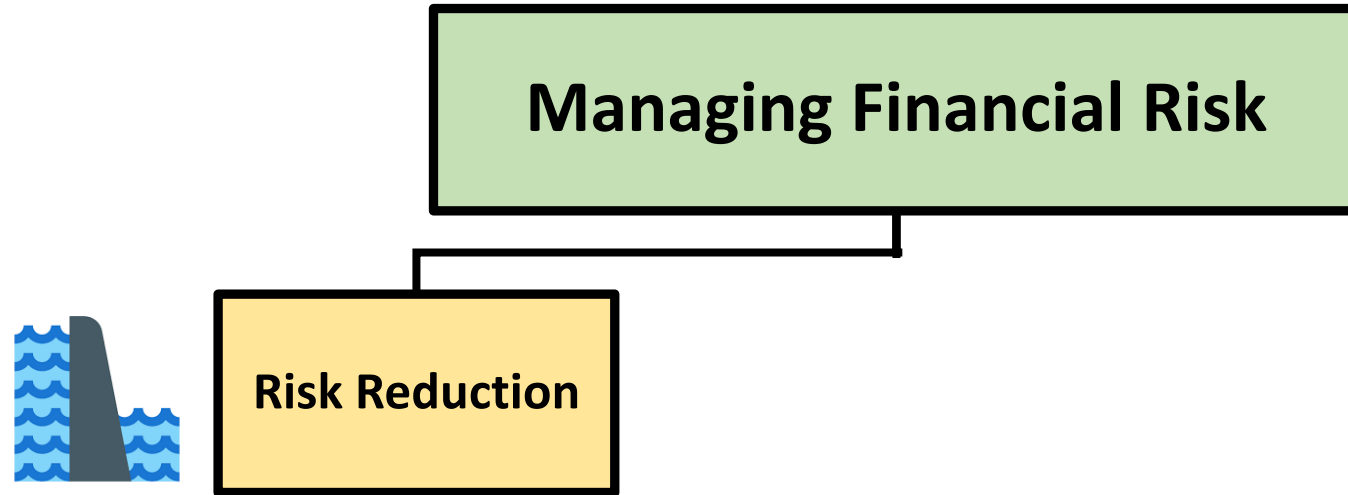
- ~92% of all U.S. systems.
- 53+ million people served
- Limited scale, reserves, and risk tools



Small systems face the same climate risks, with **fewer resources** to absorb financial shocks.



Different Tools for Managing Risk



Examples:

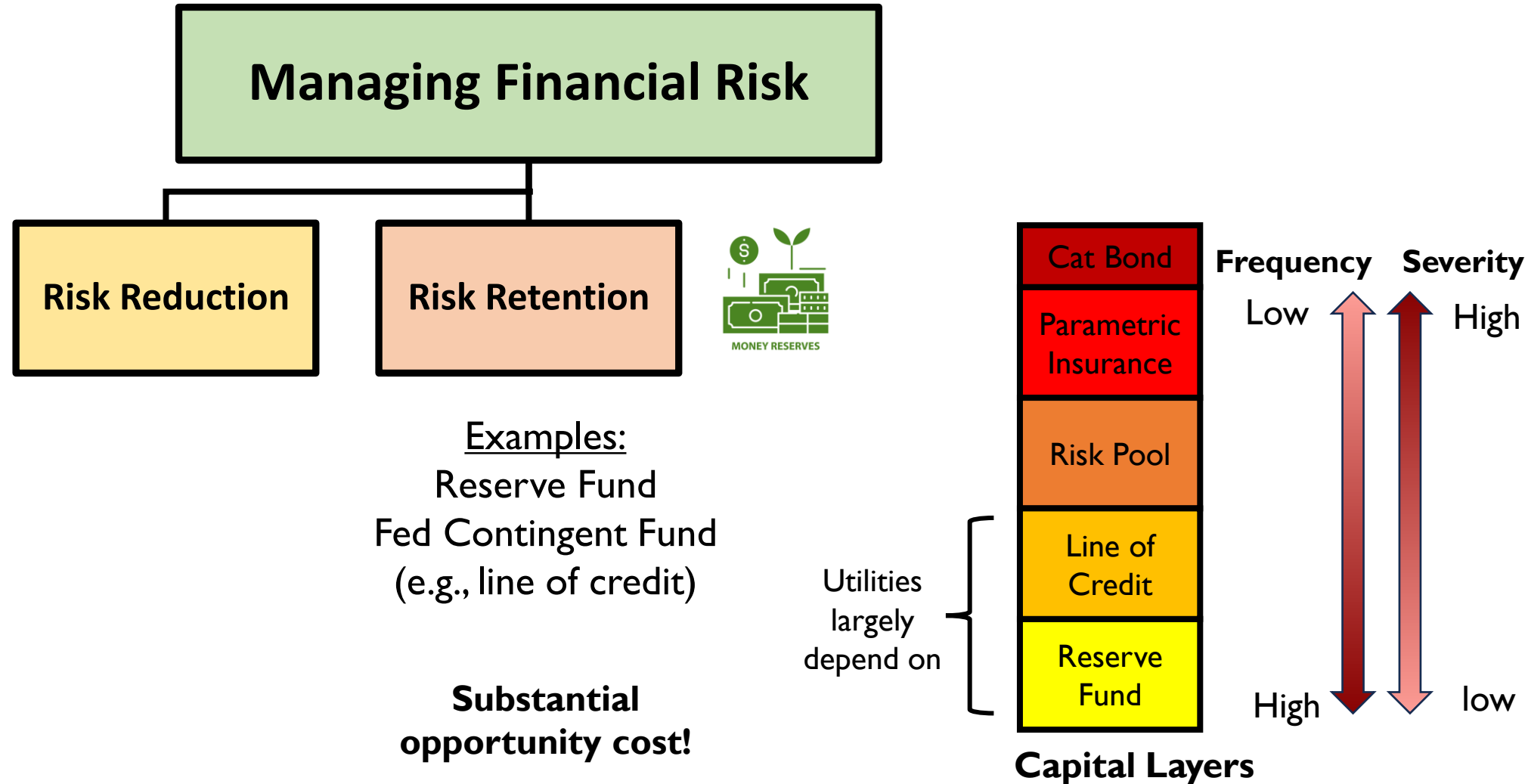
Infrastructure (e.g., reservoir);
Drought Pricing Scheme for
conservation



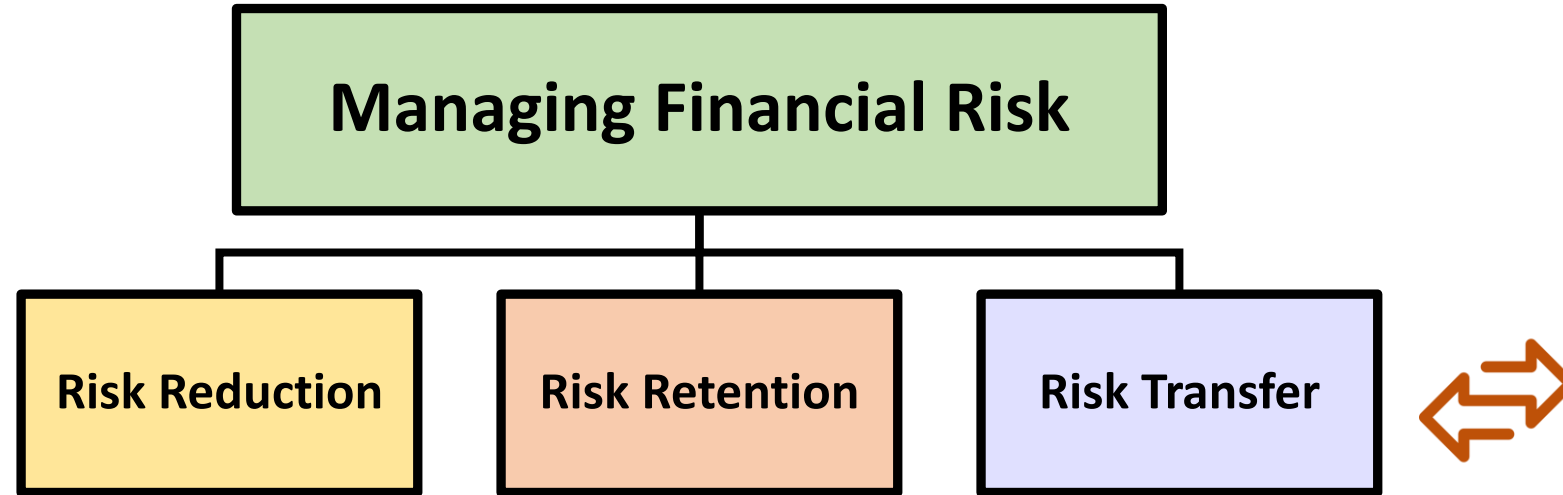
**Supply-side solution
is **costly!****



Different Tools for Managing Risk



Different Tools for Managing Risk



Examples:

Parametric/Index instruments
Insurance/Reinsurance
Risk Pools

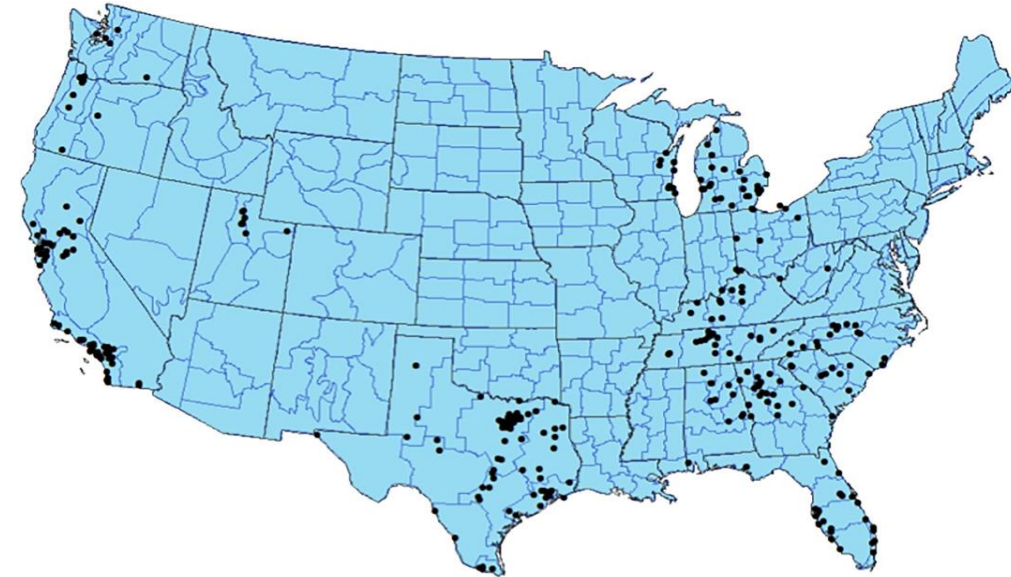
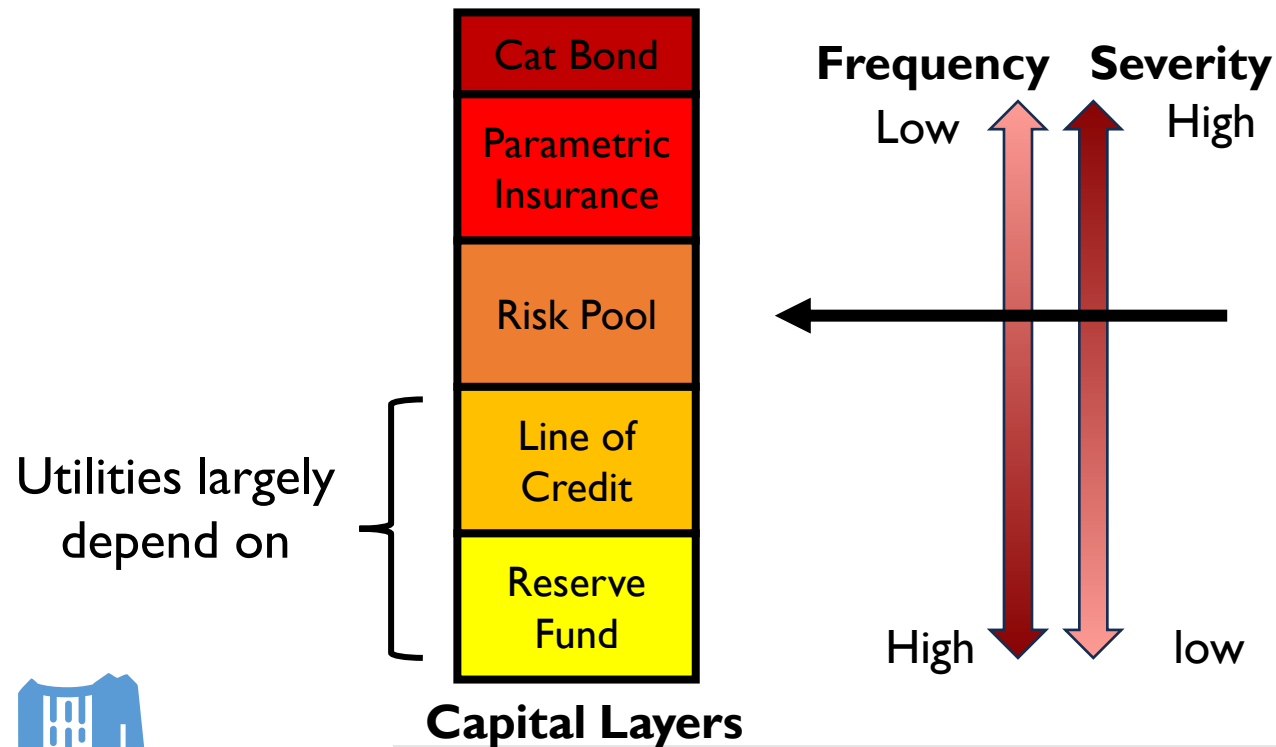


Extreme droughts require financial solutions that **act quickly, objectively, and at scale.**

Risk Pooling for Drought-Related Financial Risk

Risk Pooling Works When...

- Losses are driven by an observable index
- drought losses are **correlated (but not identical)** across utilities.



Different tools protect against different frequencies and severities of drought!



Parametric Insurance For Utilities *Tail Risk*

Parametric Insurance:

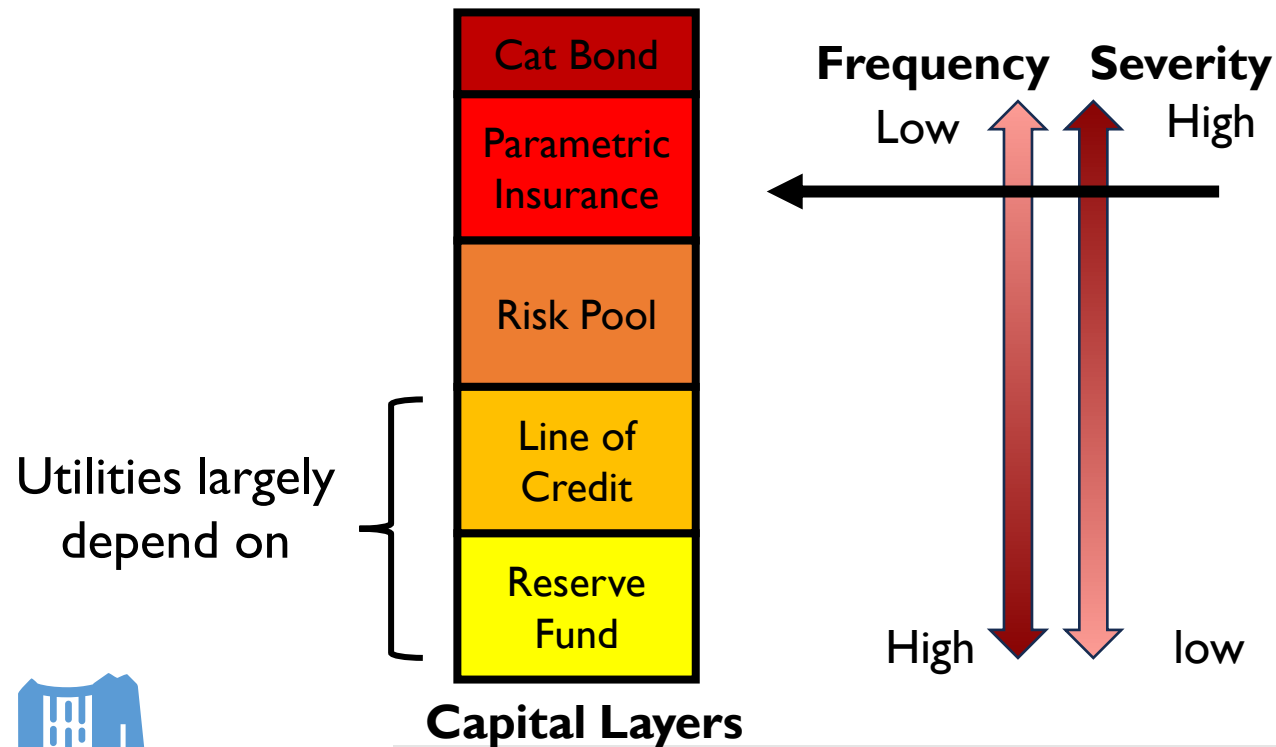
- Triggers can be based on **objective, non-manipulable indices**
- **Rapid payouts** stabilize cash flow
- Scalable



vs

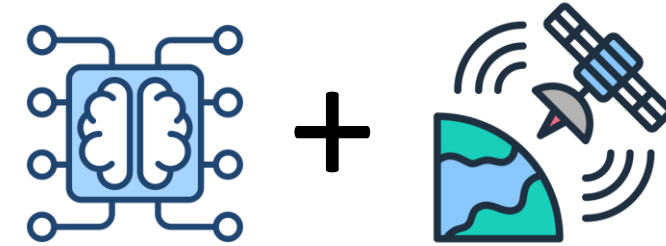
Indemnity-Based insurance:

- Loss verification required
- Payment **delays**
- **Limited** scalability



Advanced ML & Remote Sensing for Hybrid Parametric Triggers

- Basis risk: mismatch between payout and actual loss
- Use advanced remote sensing and ML to design multi-indicator triggers
- Combine hydrologic, climate, and storage signals to **better match realized losses**



Trigger type	Why it works	Limitation
Reservoir storage %	Directly tied to supply risk	Basin-specific
Snowpack level	Early-season predictor	Snow-dependent systems
Streamflow level	Hydrologic response indicator	Data latency
Palmer Drought Severity Index	Long-term moisture signal	Slow response
Precipitation anomaly	Anomaly detection	Weak linkage
Groundwater level	Captures buffered drought stress	Data availability



ML-based composite triggers can outperform single-index parametric contracts.

Future Research Agenda

Area 1: drought-related financial risks for US water utilities

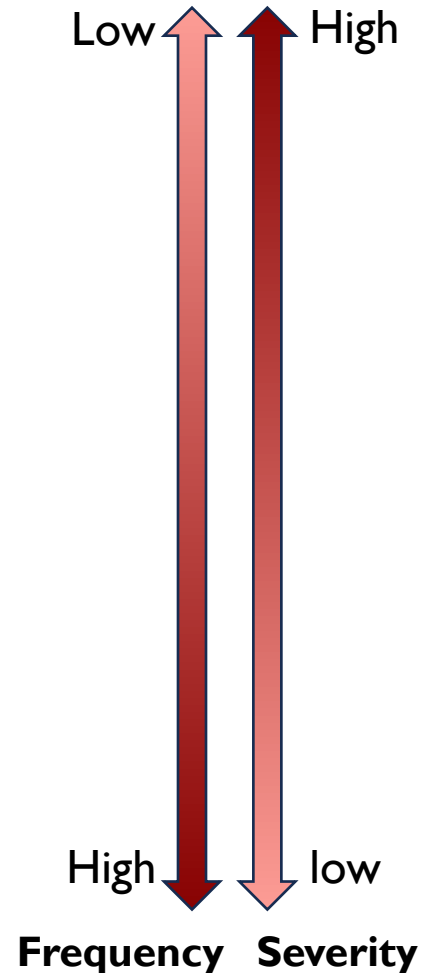
Area 2: Hurricane risk propagation in Florida

Area 3: Wetland Mitigation and Downstream Flood Exposure

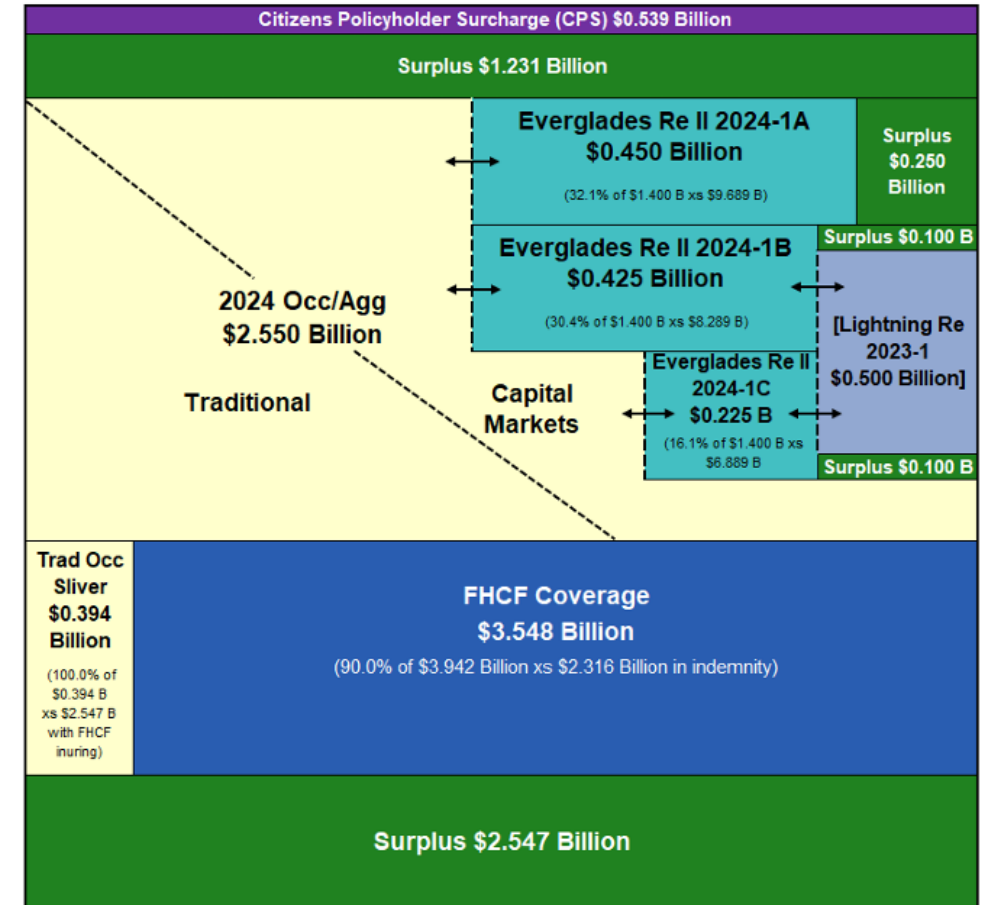


Why is it important to quantify (and price) risk?

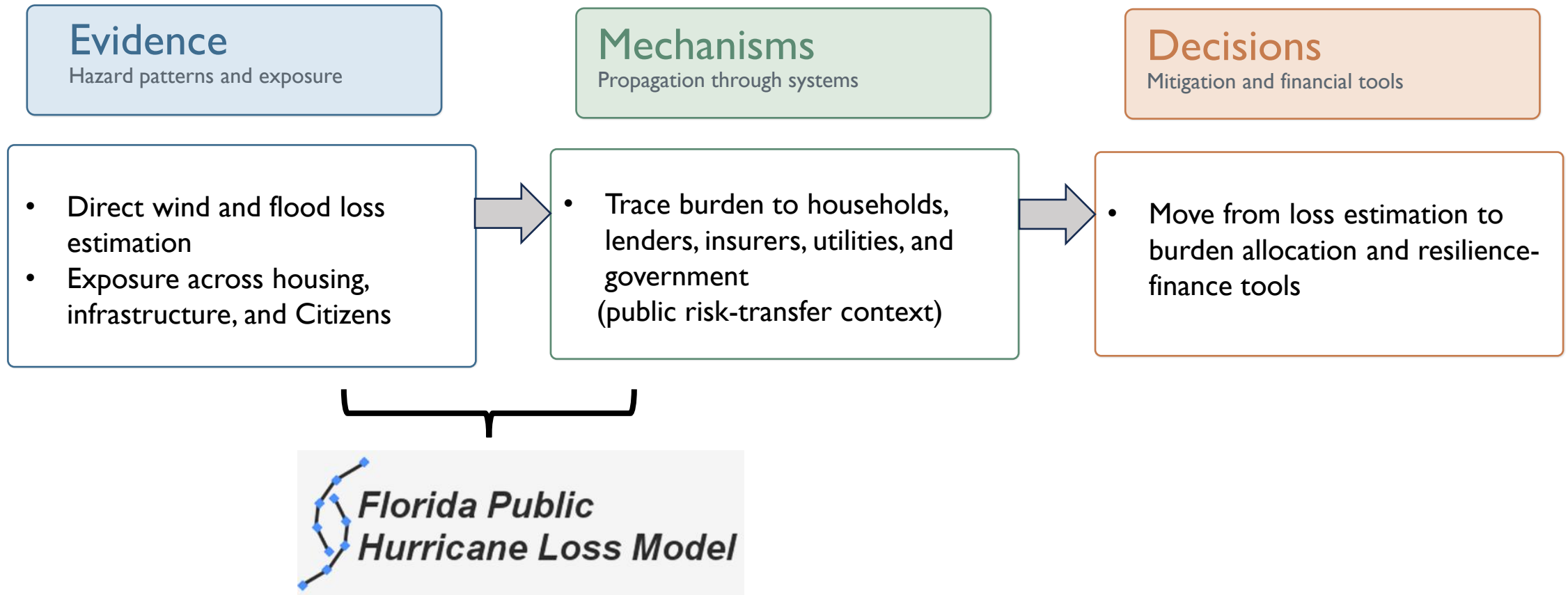
- Tail risk is expensive
- Florida relies on layered public–private risk transfer program
- Different layers absorb different losses
- We need to quantify the burden on households / insurers / lenders / government



Citizens FL - Hurricane Risk Transfer



Florida Hurricanes: From Evidence to Decisions



At FIU, I would build toward **risk propagation** and **decision support**.



Florida Hurricanes: From Evidence to Decisions



Evidence

Hazard patterns and exposure

Mechanisms

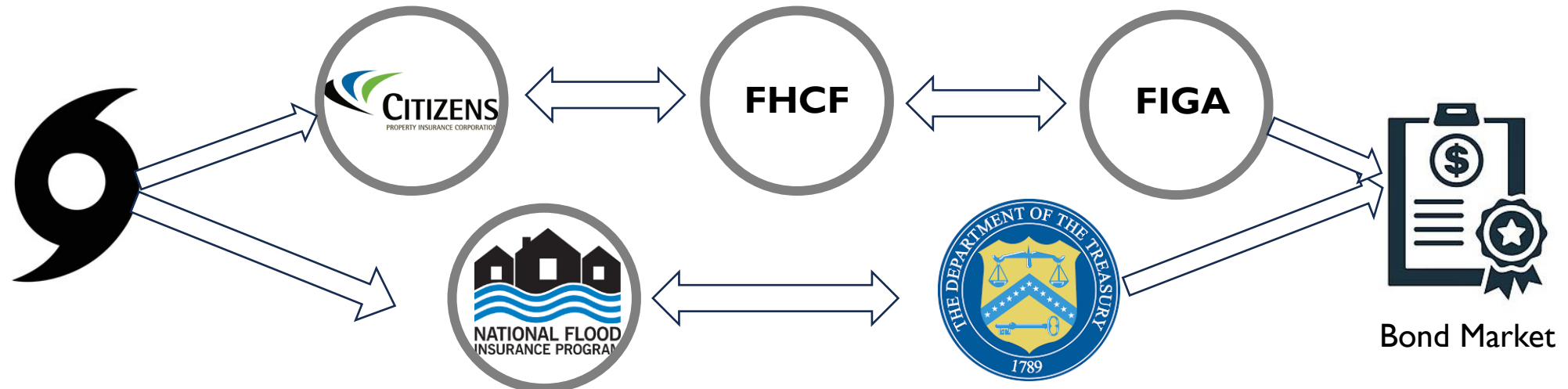
Propagation through systems

Decisions

Mitigation and financial tools

Core questions:

- Quantify the uninsured loss burden
- Trace spillovers across insurance and public finance systems
- Evaluate resilience tools and risk redistribution
- Stress-test system-wide performance under hurricane shocks



Future Research Agenda

Area 1: drought-related financial risks for US water utilities

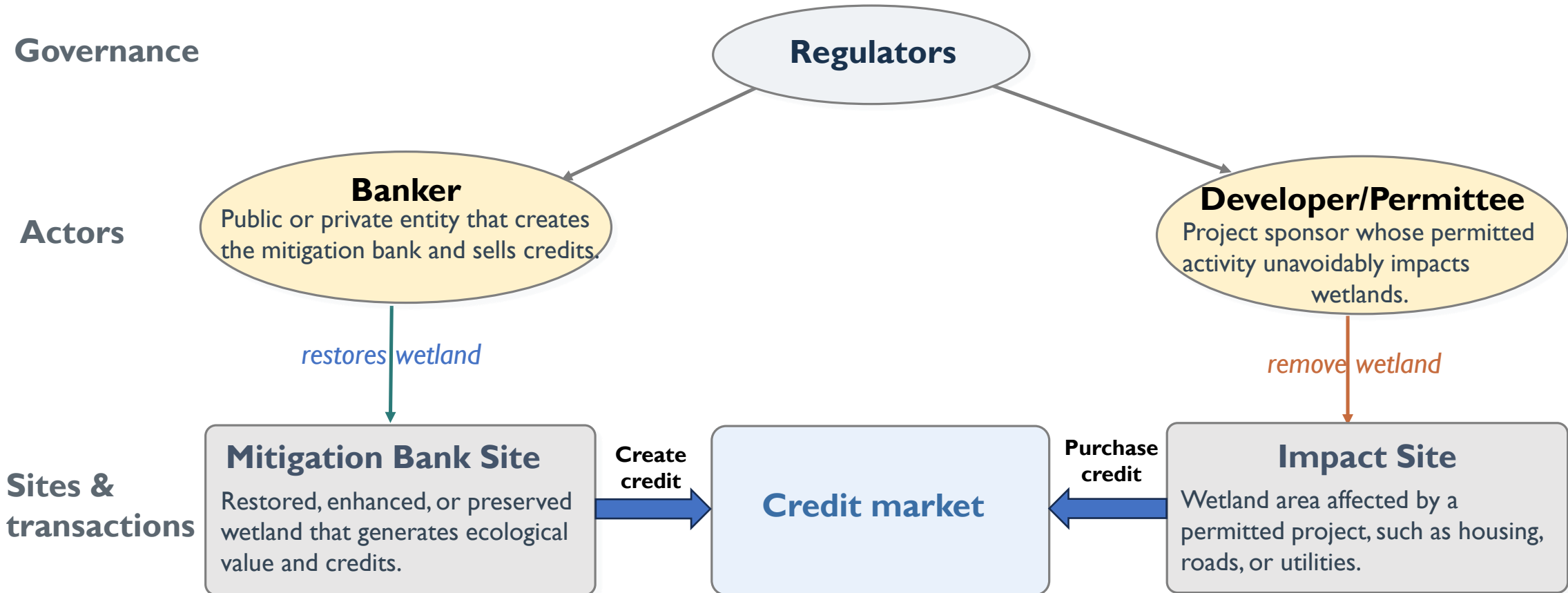
Area 2: wildfire financial risk management

Area 3: Wetland Mitigation and Downstream Flood Exposure



Wetland Mitigation Banking: Offsetting Wetland Losses

Wetlands are important because they function as natural infrastructure:
reducing flooding, supporting coastal resilience, improving water quality, and providing wildlife habitat.



Where the Current Credit System Falls Short

Today's mitigation metrics

- wetland acreage
- ecological value

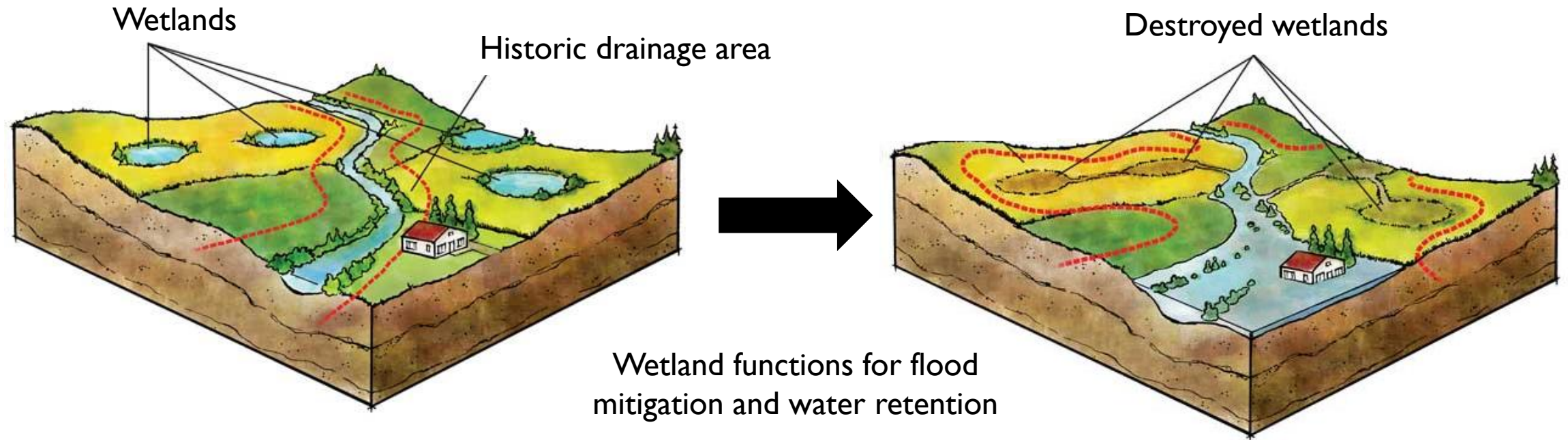
What is missing:

- downstream flood risk impacts
- who bears the added flood losses
- flood-mitigation value beyond legal jurisdiction

Current wetland mitigation banking compensates ecological loss, but not the **added downstream flood risk** created by wetland removal.



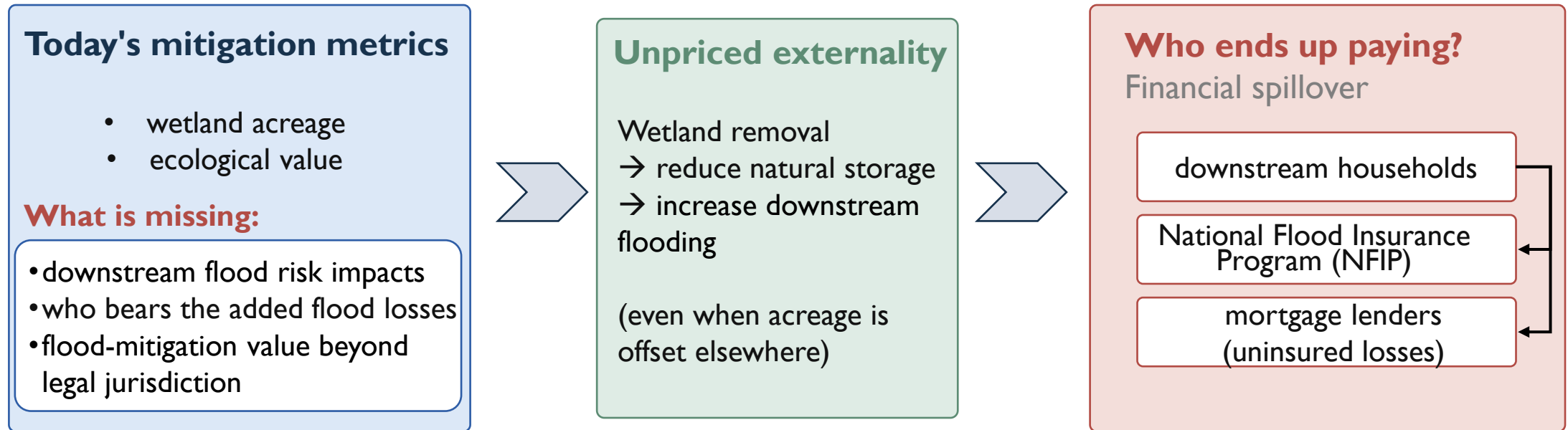
Where the Current Credit System Falls Short



- When wetlands are removed and restored elsewhere, **flood protection is not replaced** at the original site.
- This can increase downstream flood exposure, especially in **low-elevation coastal areas**.



Where the Current Credit System Falls Short



Current wetland mitigation banking compensates ecological loss, not by the **downstream flood risk** created → mismatch between **regulatory jurisdiction** and **actual hydrologic/economic value**.



Proposed Redesign

to account for flood attenuation when legal wetland protection does not fully capture hydrologic importance

1. Add flood-risk metrics to credits

- flood-storage capacity / avoided inundation
- peak-flow attenuation under storm scenarios
- site-specific downstream exposure weighting



2. financial-risk sharing

Compensation design should reflect who absorbs the added flood losses.

Wetland Mitigation
Banking Program



National Flood
Insurance Program

developers



downstream exposed
property owners



Thank you for listening!

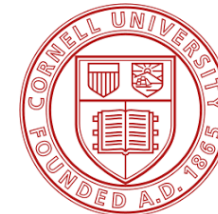
Q&A:

Managing Risk Under Climate Extremes:
How Hazards Propagate Through Natural, Engineered,
and Financial Systems

Contact:

✉ danli@unc.edu;

Personal website: <https://dlkency.github.io/>



Coupled models help us characterize risk

The **Natural** System:

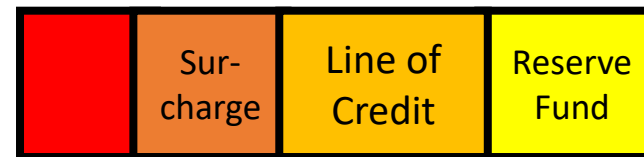
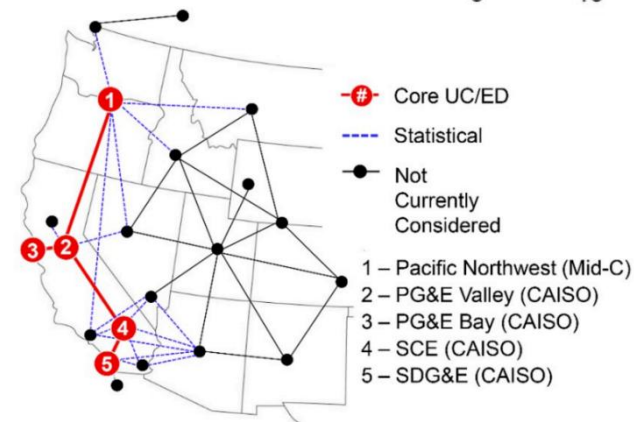
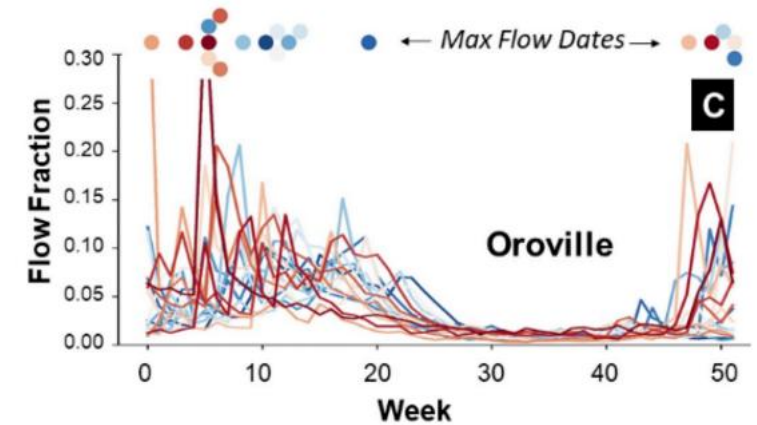
- *Hydrology*
- *Temperature/wind*
- *Wildfire*

The **Engineered** System:

- *Reservoir Operations*
- *Electric grid*

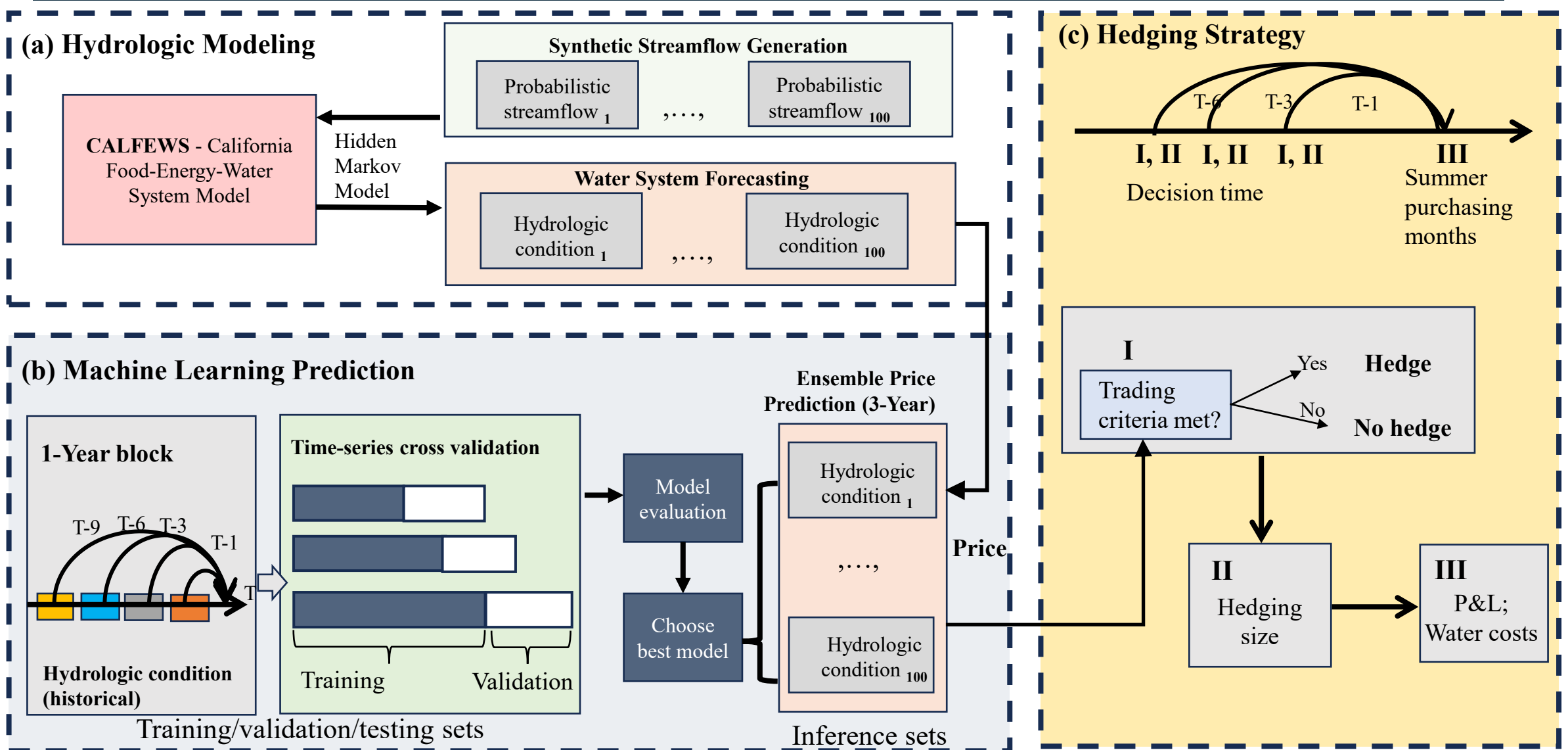
The **Financial** System:

- *Crop insurance*
- *Water market*
- *Municipal financing*

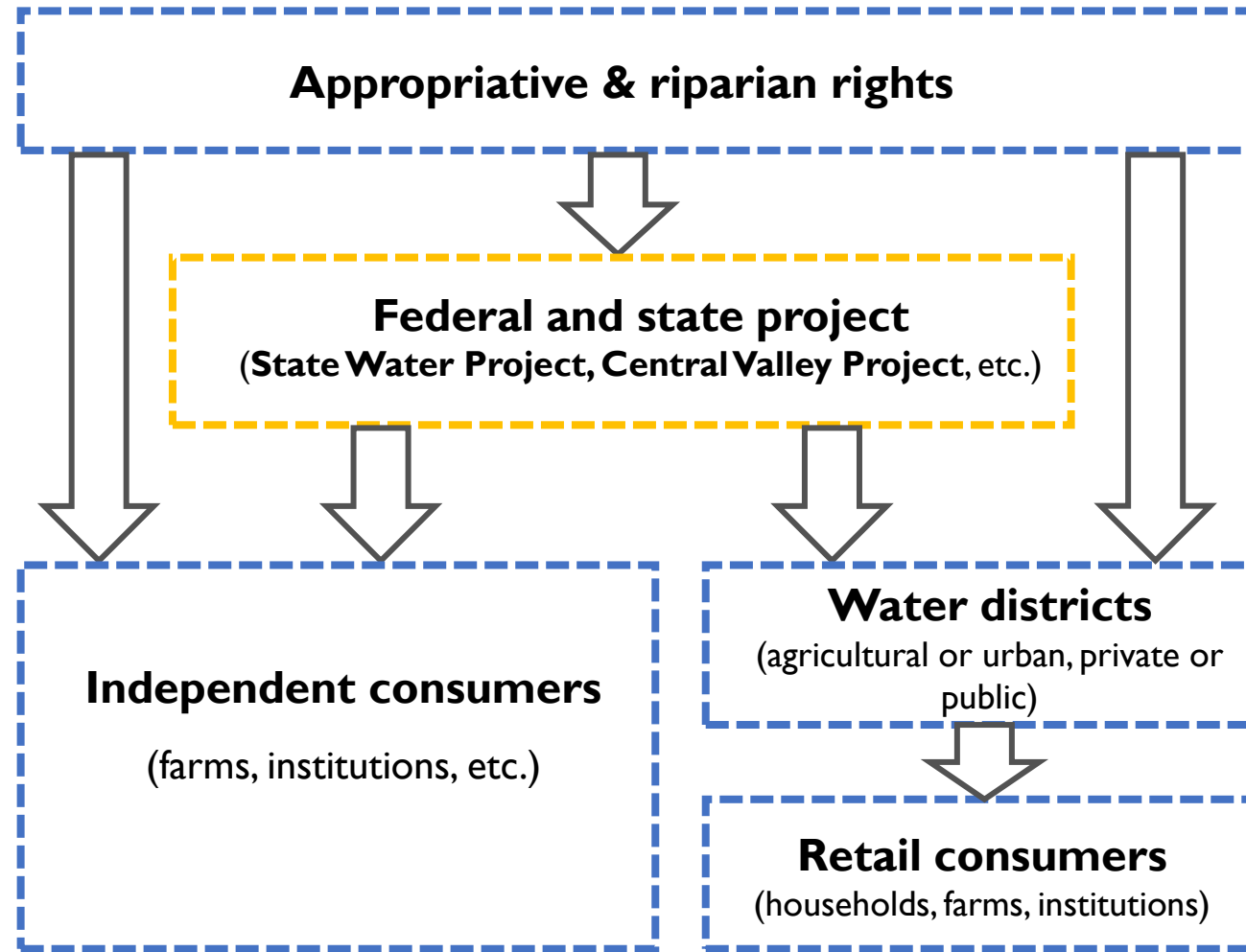


Method

A fully probabilistic, multi-stage framework links
hydrology → markets → financial risk → hedging decisions.



Structure of water allocation in California

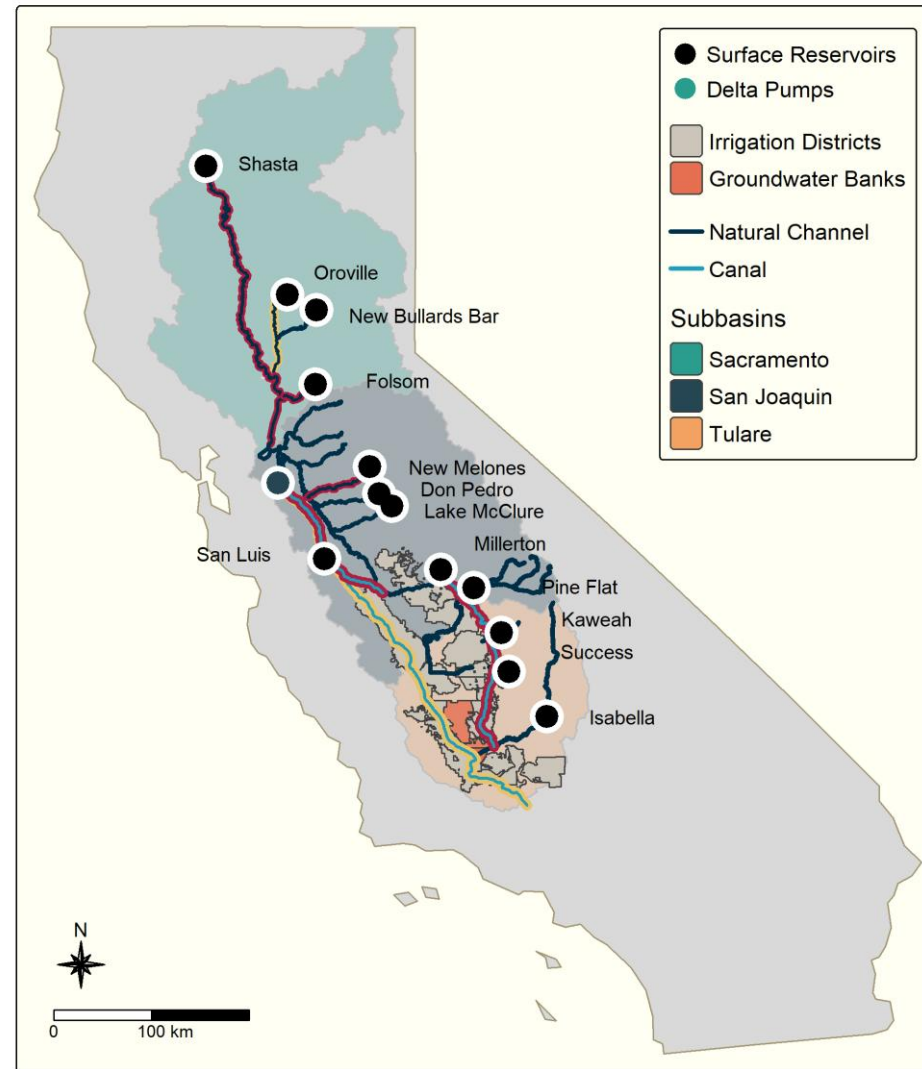


Allocation rules determine who loses water & who loses revenue, during drought.

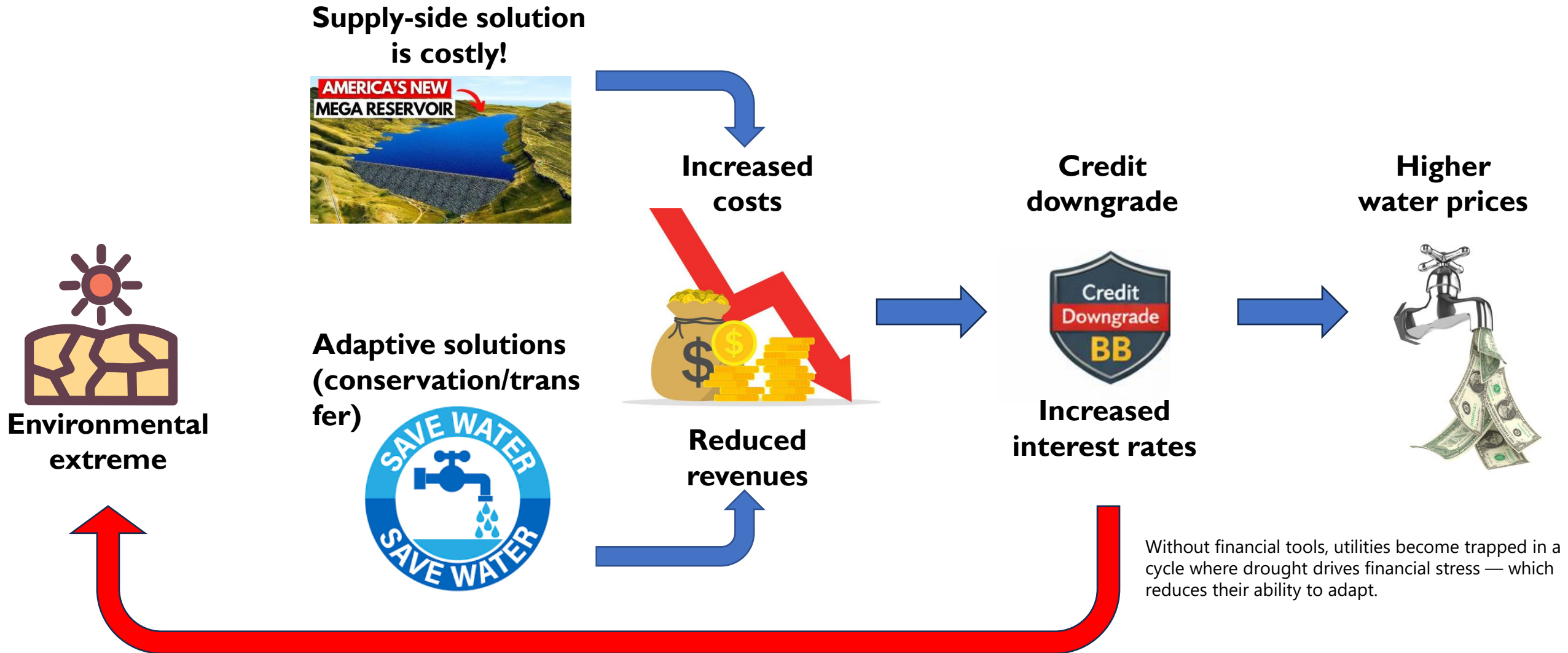
Addressing drought risk from a systems approach

- Daily timestep
- The California Food-Energy-Water System (CALFEWS) is a rules-based simulation model.
- CALFEWS represents CA's coordinated water resources operations across institutionally complex system, including:
 - Surface water reservoirs
 - Delta pumping stations
 - Groundwater banks
 - Irrigation district

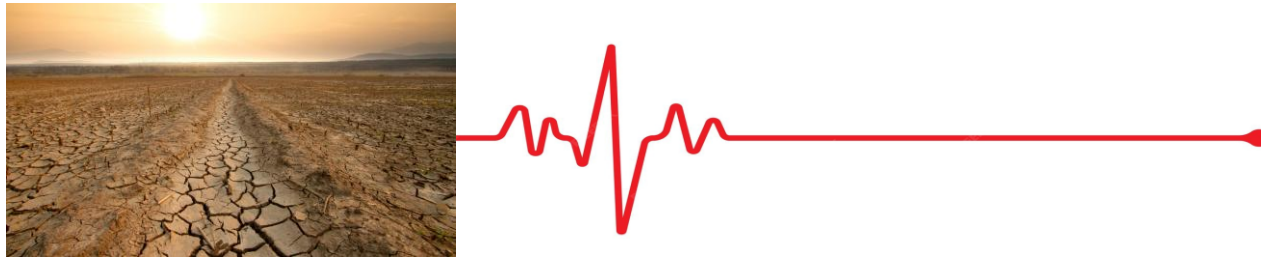
Zeff, Harrison B., et al. "California's food-energy-water system: An open source simulation model of adaptive surface and groundwater management in the Central Valley." *Environmental Modelling & Software* 141 (2021): 105052.



Drought issue is also a financial risk!

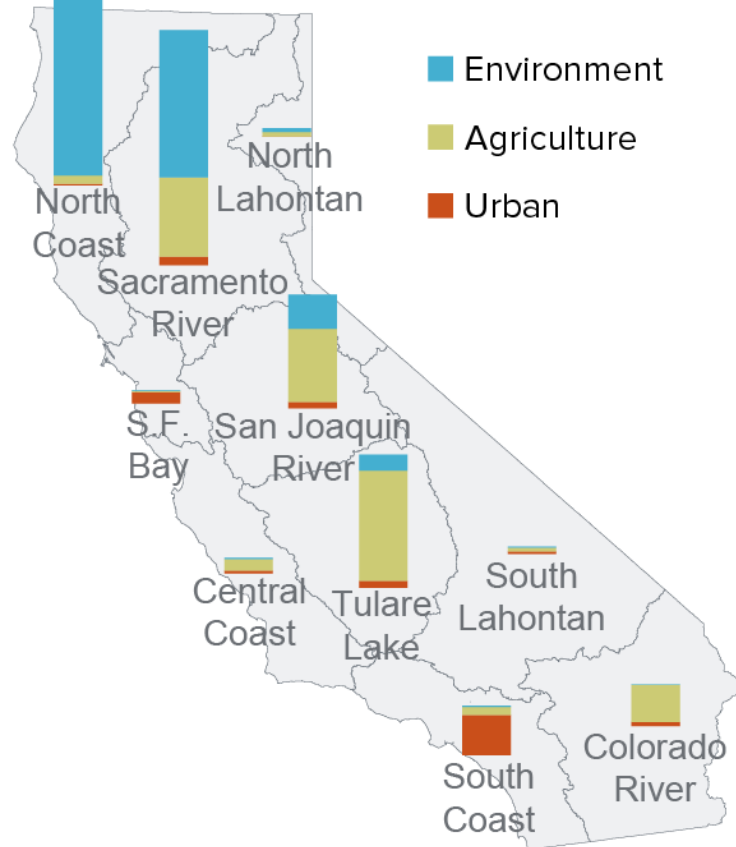


Why Does a Single Drought Year Have Long-Lasting Impacts?

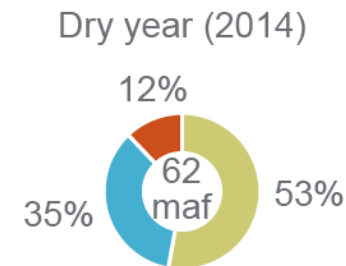
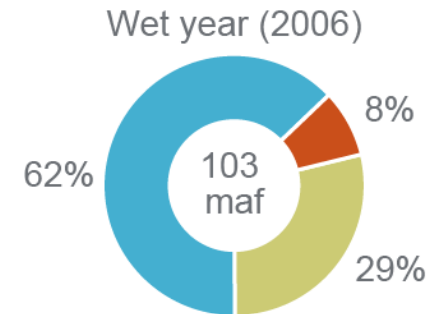


Managing Financial Risk in Water System

Average annual applied water use
(1998–2018)



Statewide applied water use,
millions of acre-feet (maf)

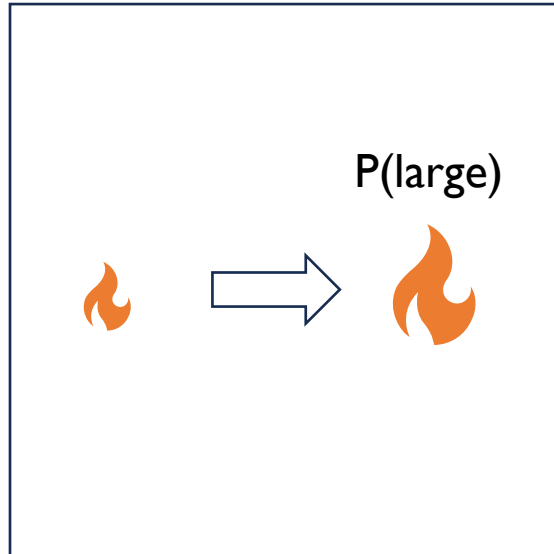


Cities, farms, and the environment compete for water

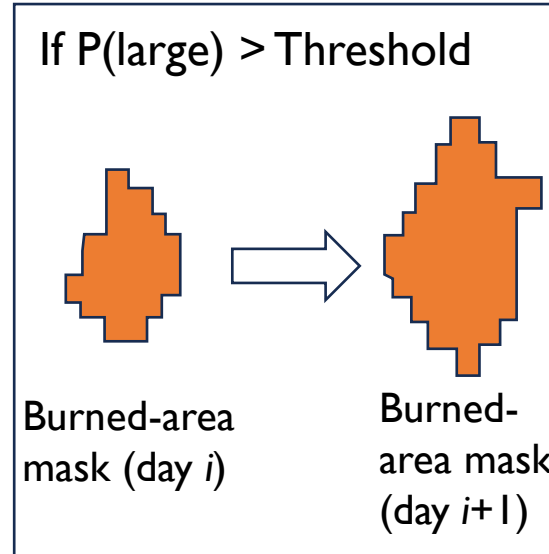
How we measure it

Multiple-step Modeling Framework: From Fire Probability to Economic Loss

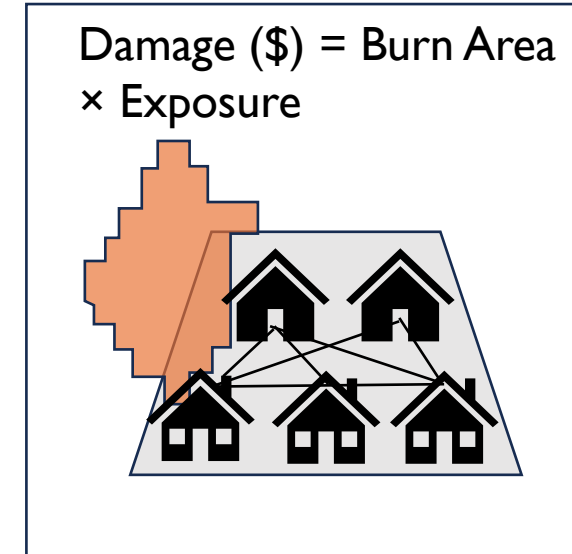
Model 1: Projecting Large Fires



Model 2: Fire Spread



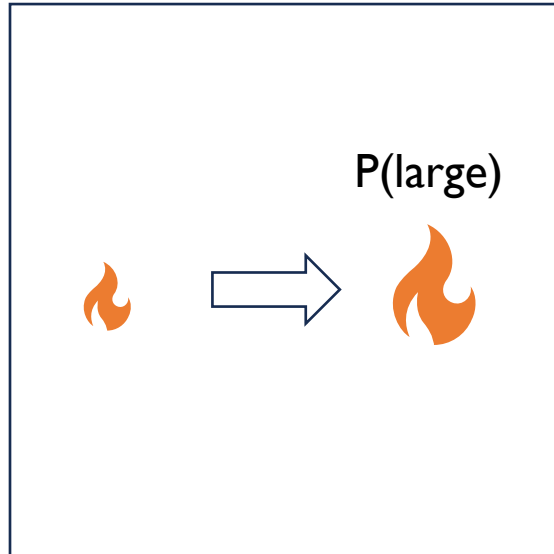
Model 3: Urban Conflagration



How we measure it

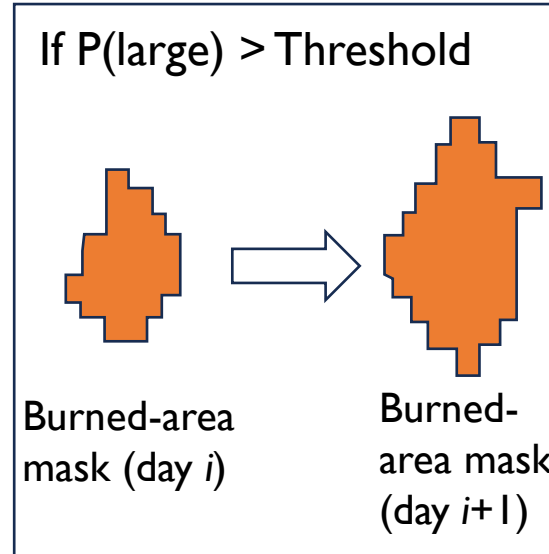
Multiple-step Modeling Framework: From Fire Probability to Economic Loss

Model 1: Projecting Large Fires



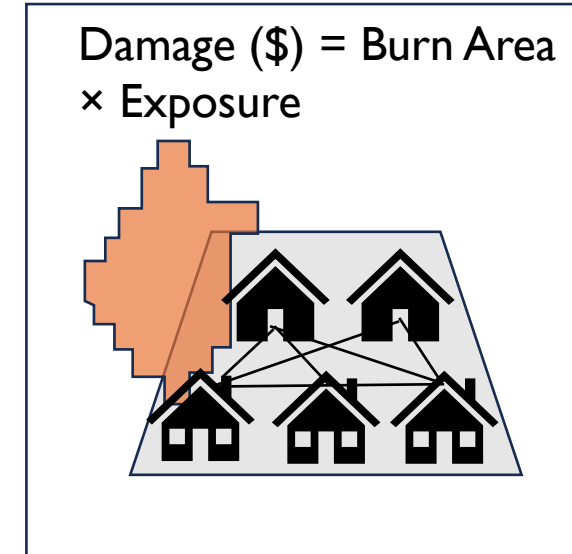
**What condition
grow large?**

Model 2: Fire Spread



**Speed & direction
of fire propagation**

Model 3: Urban Conflagration



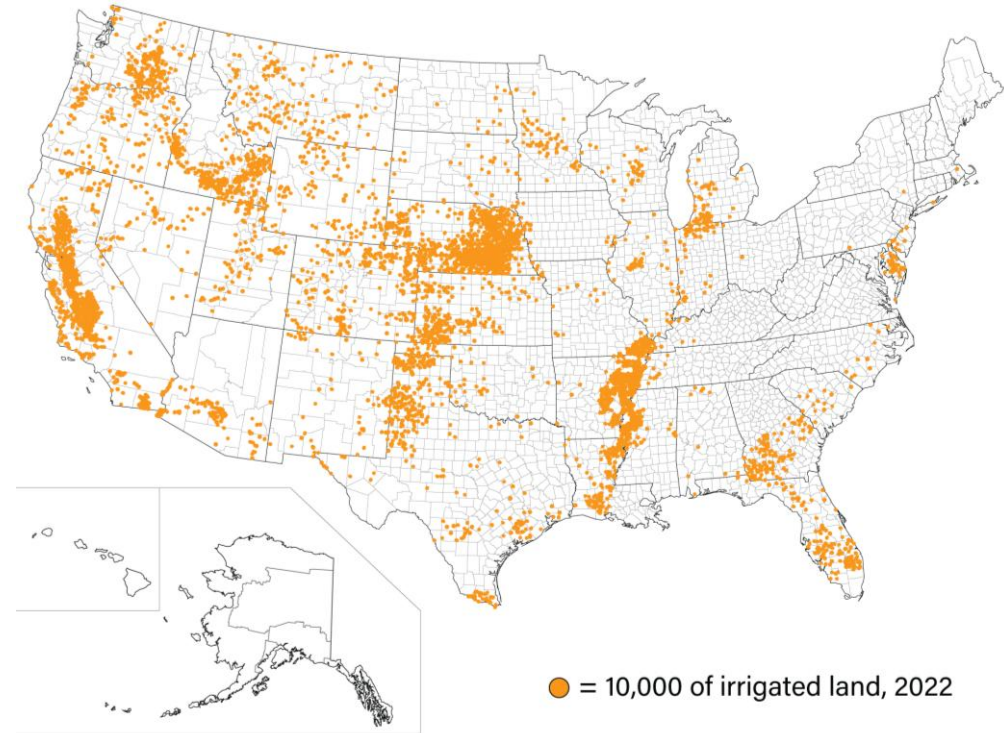
**Informs insurance
pricing & retrofit
programs**

**Integrated multi-stage models
for risk management and insurance**

Irrigation Districts Face Uninsured Drought Risk

U.S. acres of irrigated land by county

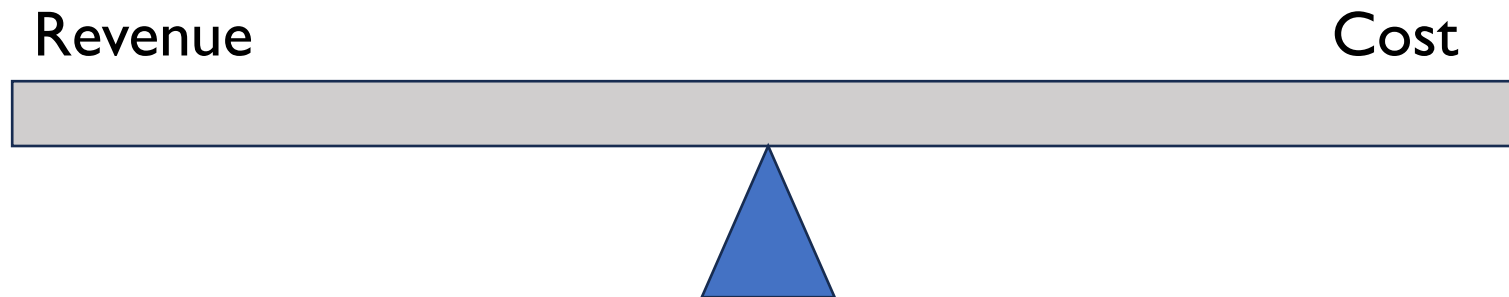
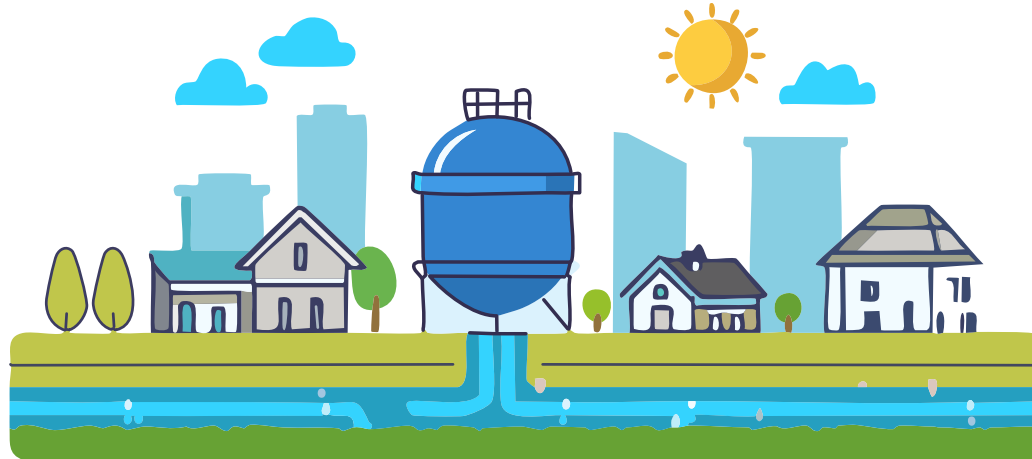
- Irrigation districts are **not eligible** for USDA crop insurance
- Raising volumetric water rates **shifts drought losses to farmers**
- Districts remain responsible for fixed infrastructure costs
- Increasing supply-side hydrologic volatility makes it **harder to maintain service quality**



A clear gap exists: districts bear drought-driven revenue losses with no formal financial protection.

Normal Water Year

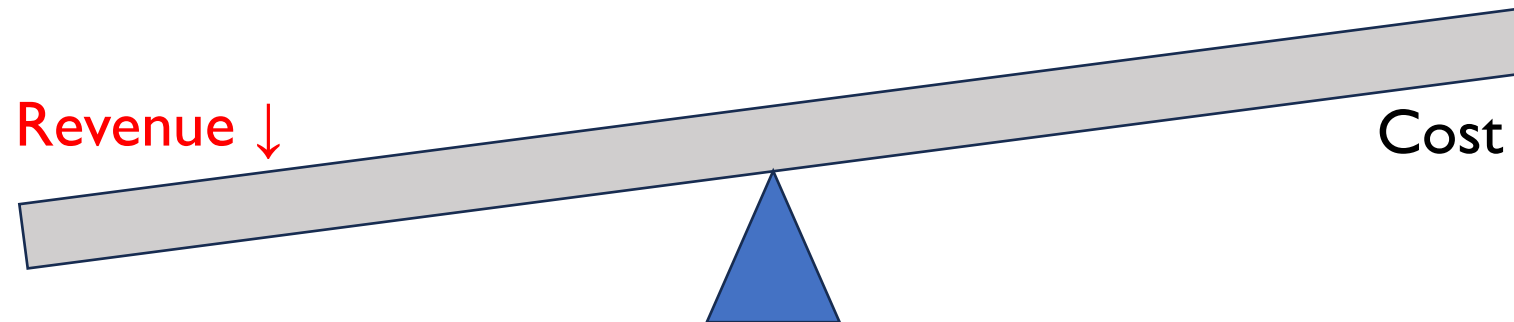
- Revenues align with demand
- Operating costs stable
- Maintenance and debt service follow planned schedules



In a normal year, water utilities function within a **stable financial environment**.

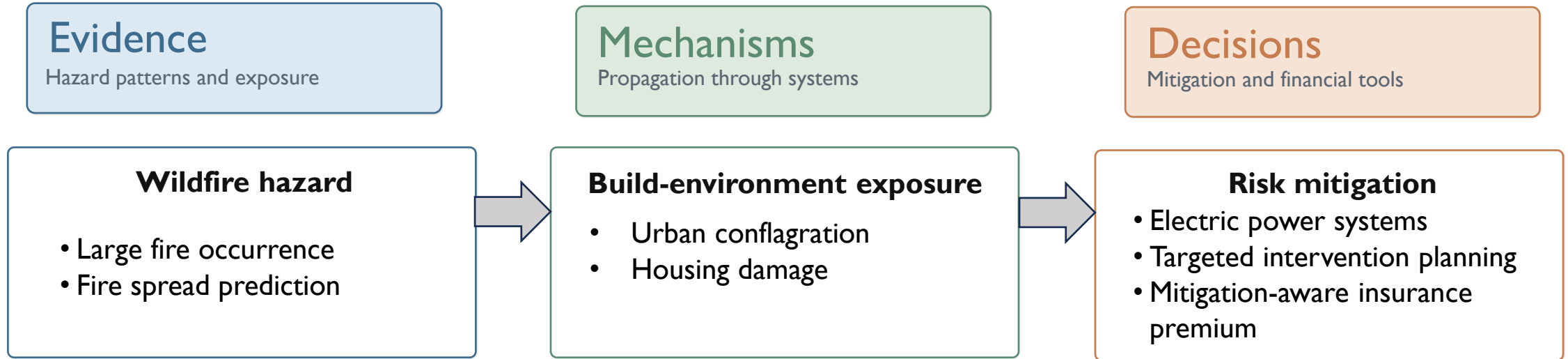
Drought Year

- **Revenues** ↓ as mandatory conservation reduces water use
- Costs remain unchanged



Physical shortages rapidly translate into **financial pressure**.

Future Research Agenda — Fire

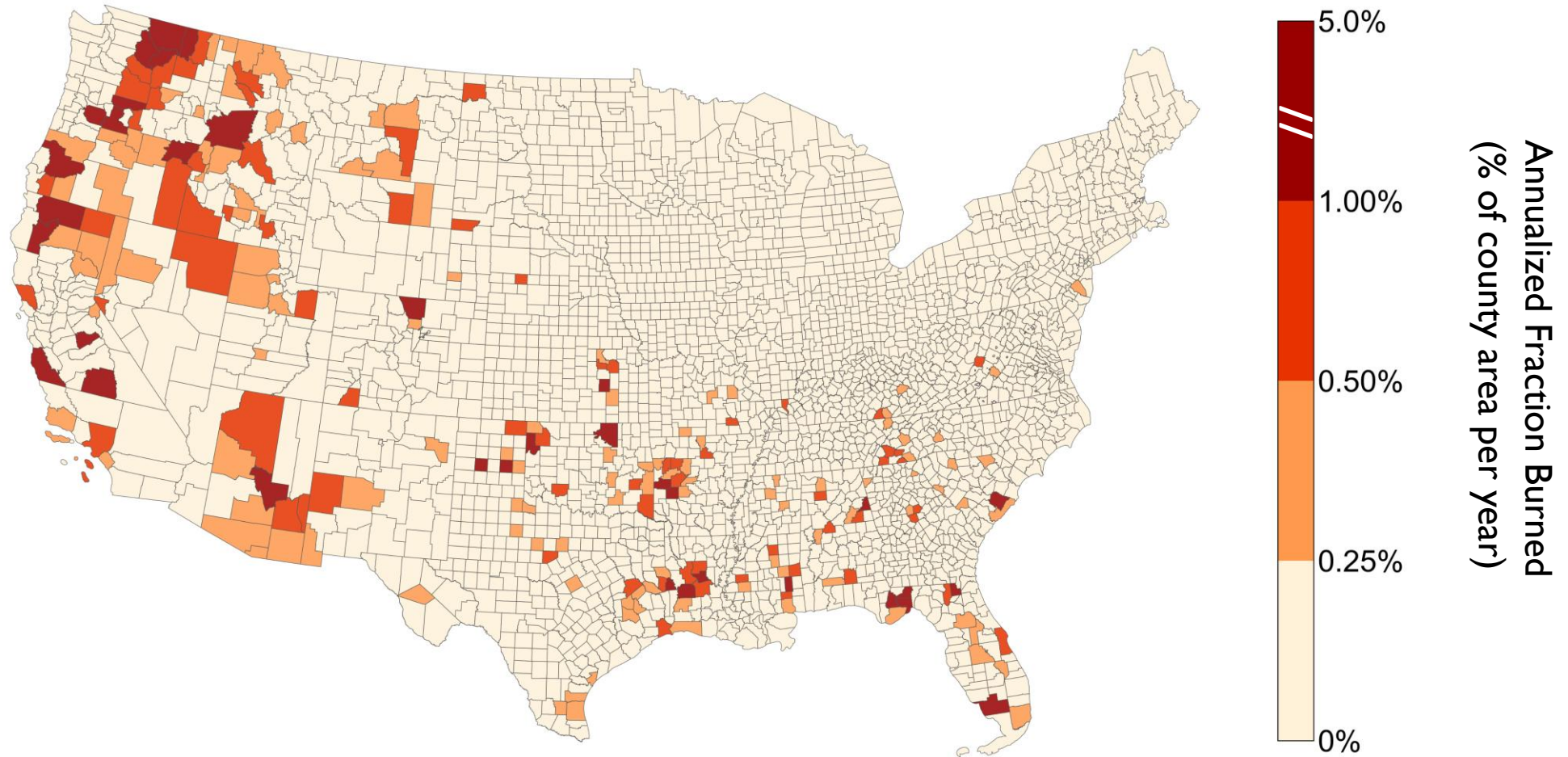


Current focus: the **front end** of the chain (multi-stage deep learning models for **fire occurrence**, **spatial spread**, and **structure-to-structure transmission**.)

Next: extend to infrastructure and community exposure, ultimately to decision-support tools for mitigation and resilience planning.

Where Fires Occur

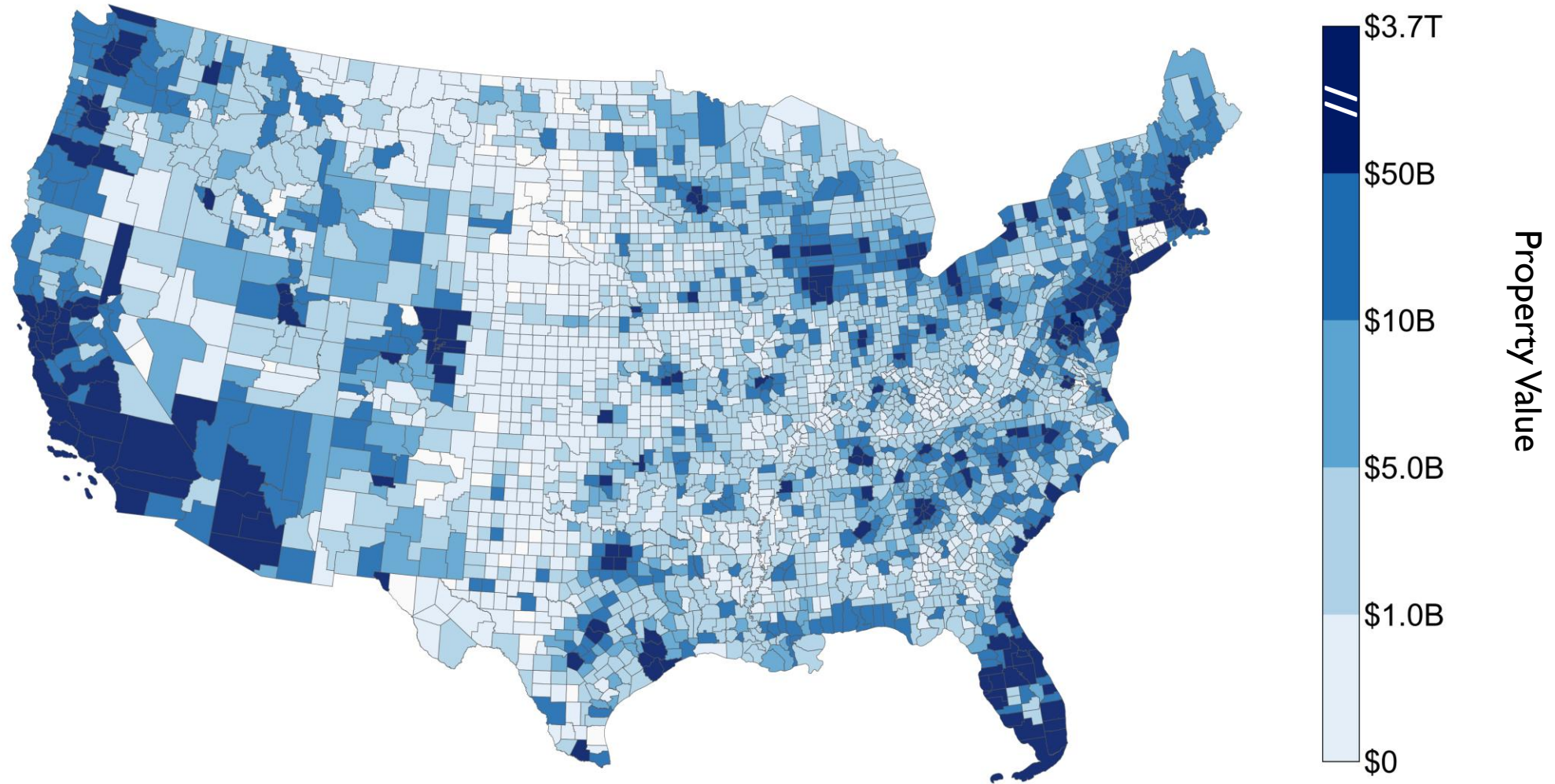
Annual Fraction Burn = Burned area / County Area



Source: US Forest Service Monitoring Trends in Burn Severity (MTBS),
burn areas 2015-2024

Where Assets Are

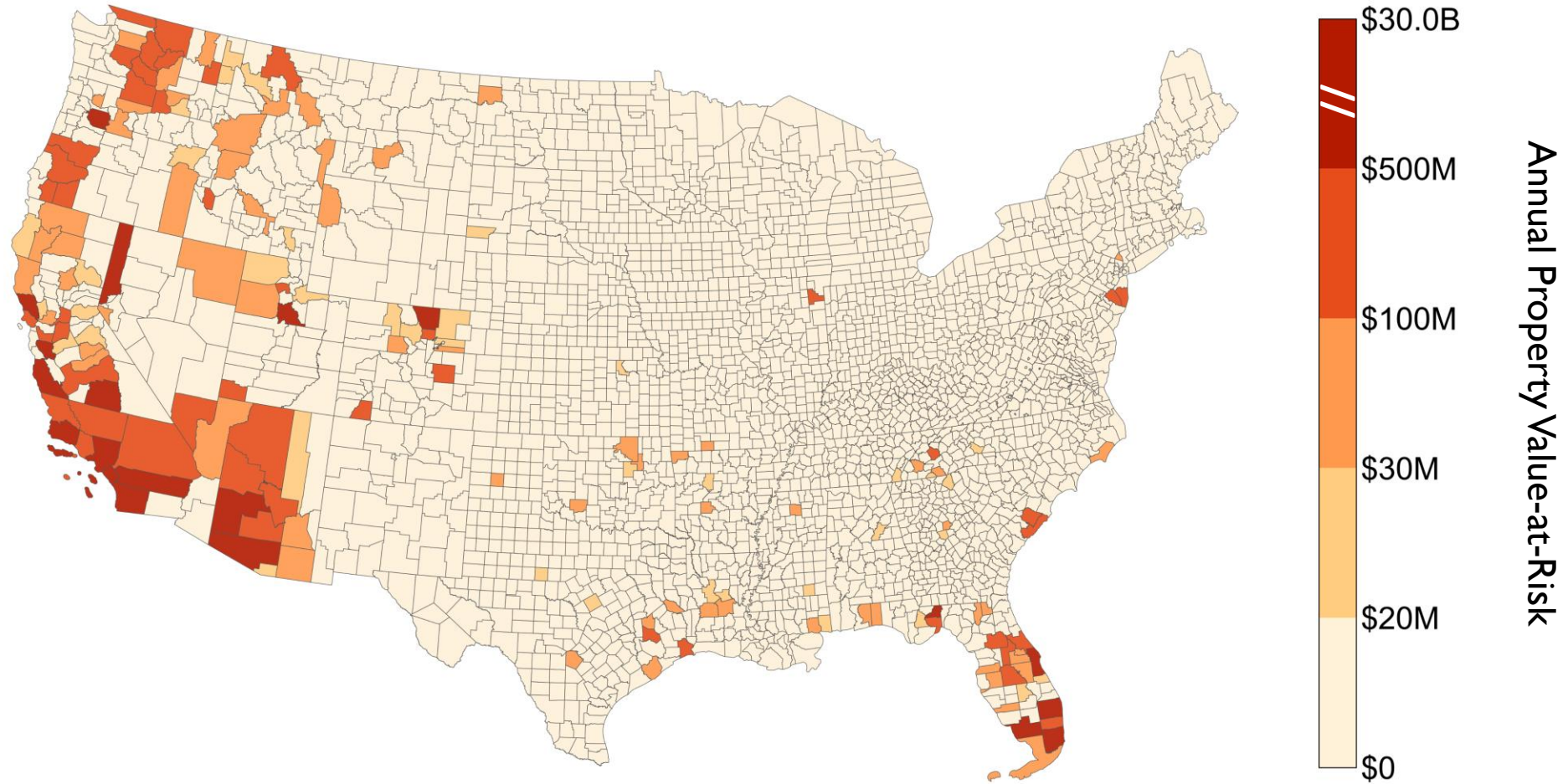
Total Property Value (Exposure) = Median Property Value × Housing Unit Count



Sources: Median property values are from the Zillow Home Value Index (December 2024), and housing unit counts are from the U.S. Census Bureau (2024 estimates).

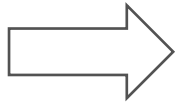
Property Value-at-Risk

Annual property value-at-risk = Annualized Fraction Burned $\times$ Total Property Value

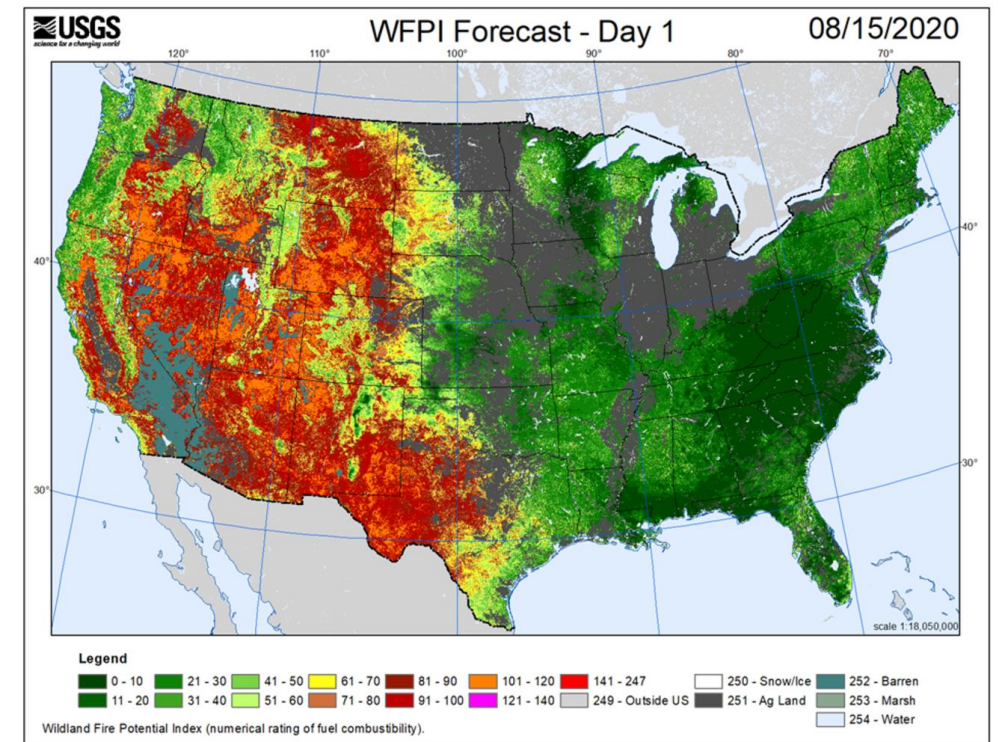


What People Usually Mean by “Wildfire”

Most of the modeling tools focus on **‘fuel-driven’ wildland fires**



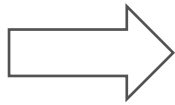
- Burn large forested areas
- Grow quite large
- Produce severe and widespread smoke impacts



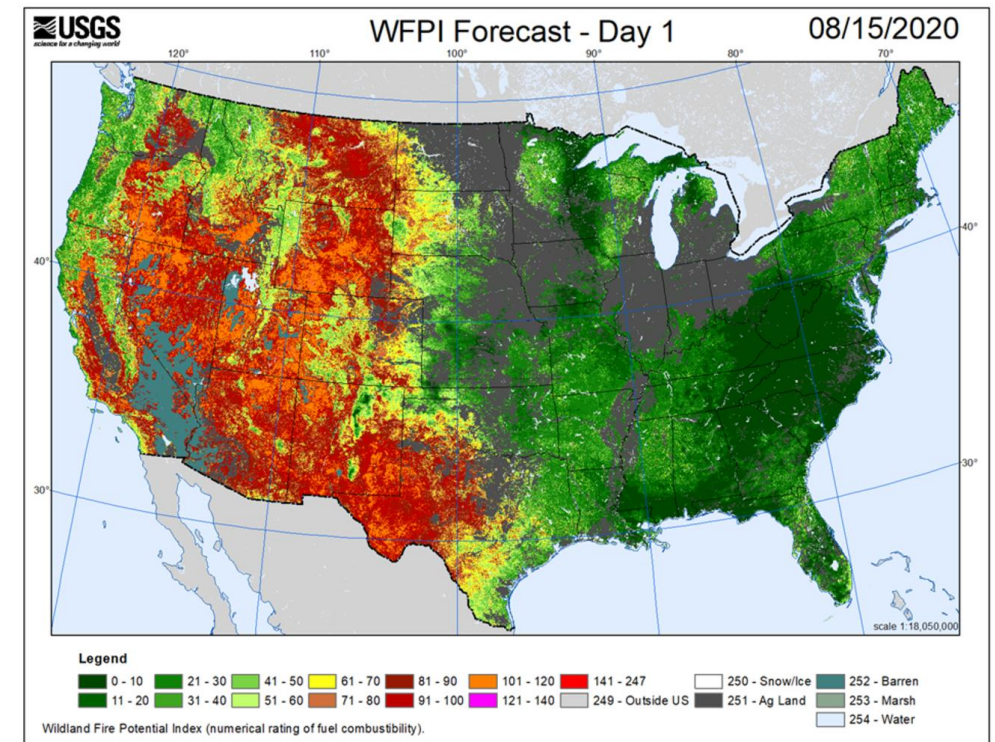
USGS Wildland Fire Potential Index (WFPI)

What People Usually Mean by “Wildfire”

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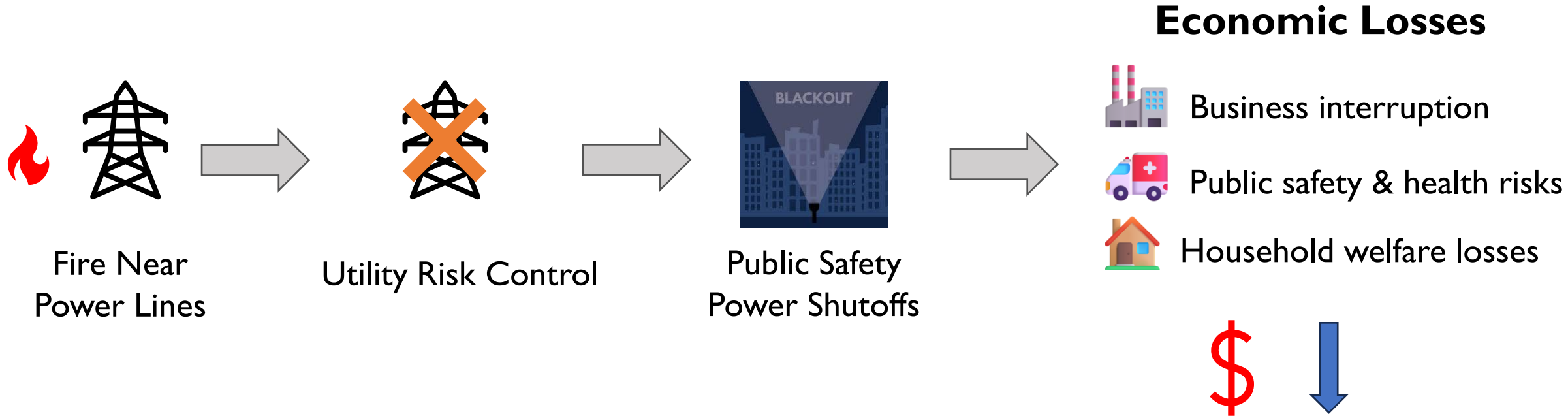


Existing models focus on fire behavior, not the **value of property or infrastructure at risk.**



USGS Wildland Fire Potential Index (WFPI)

Losses Are No Longer Limited to Direct Fire Damage



Losses Are No Longer Limited to Direct Fire Damage

